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Zeest: An Agentic Medical Copilot

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Abstract

Patient history, consultation notes and multimodal findings often require separate review, making clinical context difficult to maintain across encounters. Zeest is an agentic medical copilot that combines relational and semantic patient memory, consultation transcription, specialised clinical agents and retrieval of cited evidence. A reasoning agent produces a diagnosis and treatment draft, which undergoes critic review before the physician approves, modifies or rejects it. This work establishes the requirements, interface and database designs, architecture and comparative evaluation method; implementation and empirical evaluation remain future work.

Executive Summary

Zeest addresses the difficulty of reviewing patient information scattered across consultations, medical images, laboratory investigations and medication records. Physicians need relevant history alongside the current encounter, but preparing that context and documenting the consultation require additional effort. The intended application brings these tasks into one web workspace. Each physician accesses their own patient records, selects an encounter and reviews its historical context. Consultation audio becomes structured notes that can be checked and corrected before analysis. Supported imaging, laboratory information and medication context contribute identifiable findings rather than independent clinical decisions. Longitudinal memory combines explicit patient and encounter relationships with semantic retrieval, preserving the source of information across visits. Specialised agents organise the case, process supported inputs and retrieve relevant clinical guidance. The resulting diagnosis and treatment draft appears with cited evidence and critic findings so the physician can inspect its support and limitations. This workflow aims to make information continuity and recommendation review easier to follow while keeping clinical judgement with the physician. Its value requires assessment through implementation and controlled evaluation rather than assumptions about automation.

Approval, modification and rejection have different consequences in Zeest. Approval records the exact reviewed revision as a clinical decision; modification saves a pending revision, and rejection retains the review action without accepting the draft. Audit history preserves the physician, encounter, time and reviewed content, while patient ownership governs related information access. The current deliverable establishes the project vision, requirements, use cases, interface designs, database relationships, methodology and software architecture. It does not claim an implemented clinical system or measured improvements in patient outcomes. Development and evaluation require permitted de-identified or synthetic cases, suitable existing models and selected knowledge resources. Diagnostic models will be integrated rather than trained from scratch, and clinical coverage will follow their supported tasks. The evaluation method compares a single language model, a memory-augmented variant and the full system, with controlled contrasts for critic and multimodal contributions. These comparisons will determine which components provide useful support under common assessment conditions. Autonomous treatment, real clinical deployment and regulatory approval remain outside scope; every accepted decision requires physician oversight.

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Chapter 1 Introduction

Clinical decision support brings together patient information, documentation and analytical assistance. The following sections establish the background, purpose, intended readership and terminology needed to understand the work, followed by an outline of the remaining chapters.

1.1 Purpose of this Document

Zeest investigates how longitudinal memory, agent coordination, evidence retrieval and critic review can support clinical decision making. The central comparison examines a single language model, a memory-augmented variant and the full system, with controlled contrasts for critic and multimodal contributions. The requirements and design establish the physician workflow, component responsibilities and information relationships needed for these comparisons. Existing validated diagnostic models form the supporting groundwork; the research focuses on their integration with memory and review. Clinical coverage depends on suitable data and models, and evaluation uses de-identified or synthetic cases. The intended outcome is an assessable decision-support system whose recommendations remain subject to physician judgement. The design also distinguishes authorised clinical observations from generated drafts, identifies unresolved resource choices and establishes source-aware review. Implementation and comparative evaluation remain future work, so the described workflow sets requirements without presenting unmeasured outcomes or claiming clinical deployment. The assessment requires defined comparison conditions.

1.2 Intended Audience

The primary audience includes the project supervisor, academic evaluators and the student development team. Evaluators need a clear account of the clinical problem, research contribution and technical decisions, while developers need consistent requirements and component responsibilities. Physicians are also relevant readers because Zeest centres on their interpretation of patient information and review of generated recommendations. The descriptions of findings, evidence and approval controls should therefore remain understandable without requiring knowledge of agent orchestration. Researchers interested in clinical language models, memory and retrieval can examine how these capabilities are integrated. The writing connects technical responsibilities to user interactions so that each audience can follow the integrated system's intended behaviour. Interface designers can use the screen mappings to connect user tasks with requirements, while database and integration developers can check ownership and revision relationships. Clear distinctions between design, source evidence and measured outcomes help these

readers assess the same clinical workflow consistently.

1.3 Definitions, Acronyms, and Abbreviations

Key terms and abbreviations are defined below.

AI: Artificial Intelligence.

LLM: Large Language Model.

CDSS: Clinical Decision Support System.

RAG: Retrieval-Augmented Generation.

SDG: Sustainable Development Goal.

Encounter: An identifiable consultation or visit associated with a patient.

Longitudinal memory: Patient-specific information retained and retrieved across encounters.

Hybrid memory: Relational records combined with semantic vector representations.

Multimodal input: Clinical information represented through notes, images, laboratory information and medication context.

Agent: A specialised processing component with a defined responsibility.

Clinical draft: A generated diagnosis and treatment plan awaiting physician review.

Approved decision: Clinical content explicitly accepted by the physician for recording.

Audit history: Recorded physician actions associated with review and decision making.

Ablation study: A comparison of system variants used to examine component contributions.

1.4 Conclusion

Chapter 1 introduces the information problem and the purpose of Zeest. Chapter 2 defines the vision, objectives, scope and stakeholder needs. Chapter 3 examines the related research and applications that inform the technical approach. Chapter 4 specifies features, functional and non-functional requirements, use cases, resources, interface designs, data relationships and risks. Chapter 5 develops the methodology and comparative evaluation approach. Chapter 6 presents the high-level and low-level design of the integrated workflow. References identify the supporting sources, and the Appendix contains supplementary material. Together, these chapters connect the clinical motivation to a system design and an evaluation framework for investigating memory, reasoning, critic review and multimodal contributions. The resulting structure preserves the distinction between recorded observations, generated recommendations and accepted decisions. Source-aware records and review actions establish the connection between clinical assistance and accountability. Subsequent implementation must preserve these relationships before comparative evaluation can establish the value of each component.

Chapter 2 Project Vision

A project vision connects an identified problem with achievable objectives and clear boundaries. The following sections describe the problem domain, intended goals, scope, constraints and stakeholder needs that guide subsequent requirements and design decisions.

2.1 Problem Domain Overview

Clinical decision support assists physicians in interpreting patient information and considering possible diagnoses or treatment plans. Zeest operates within this domain by connecting the current consultation with longitudinal history and supported clinical inputs. The Patient Memory Agent retrieves relevant context, while the Clinical Case Agent organises it with transcribed notes and other findings. Imaging, Laboratory and Medication Safety agents analyse their respective inputs. Retrieved guidelines support clinical reasoning, followed by critic review and physician inspection. The application provides registration, login and case navigation around this workflow. These functions support access and review; responsibility for accepting a recommendation remains with the physician rather than the processing agents. Source-aware findings support inspection but do not establish diagnosis independently of the case. Relevant history must belong to the selected patient, and retrieved passages must remain visible to the reviewing physician. These boundaries distinguish clinical assistance from autonomous treatment and preserve accountability across consultations.

2.2 Problem Statement

Patient information is distributed across encounters, notes, imaging, laboratory findings and medication records, while many analytical tools process isolated inputs. Manual consultation documentation adds further preparation work and makes it difficult to maintain a coherent longitudinal view. This fragmentation can leave earlier observations disconnected from the current consultation, increase repeated information gathering and make generated recommendations difficult to assess. Without visible sources and review states, a physician may struggle to distinguish established observations from unsupported suggestions or identify the evidence behind a treatment draft. Zeest addresses this problem by connecting patient-specific history, structured consultation notes and specialist findings within one physician-controlled workflow. Hybrid memory retrieves relevant historical context, while cited clinical evidence and critic review support inspection of generated diagnosis and treatment drafts. The physician retains responsibility for approval, modification or rejection, and only explicitly approved content becomes a recorded clinical decision. Evaluation must establish the effectiveness of this integrated approach.

2.3 Problem Elaboration

Longitudinal continuity requires relevant retrieval rather than simply retaining every record. Historical observations must preserve their patient, encounter and source relationships so that they inform the correct case. Relational storage maintains explicit associations, while semantic retrieval identifies context related to the present information need. Consultation audio, images, laboratory information and medication context require different preparation and processing paths; missing or unusable inputs must remain distinguishable from completed findings. A generated draft introduces a further boundary between source information, retrieved guidance and physician judgement. Supporting passages need to remain inspectable, and critic findings must identify concerns without presenting them as conclusive clinical verification. Modification changes the saved draft without approving it, rejection excludes it from accepted decisions, and approval refers to the particular revision reviewed. Preserving these distinctions makes component comparisons meaningful because memory, specialist agents and critic review can be examined under consistent case conditions without assuming improvement in advance.

2.4 Goals and Objectives

The goal is to develop a physician-controlled medical copilot with longitudinal patient memory and coordinated multimodal reasoning. Consultation transcription supplies structured notes, specialist agents analyse supported inputs, and evidence retrieval grounds the generated diagnosis and treatment draft. Critic review and explicit physician decisions complete the connected clinical workflow. Table 2.1 lists the nine objectives covering this integration and its comparative evaluation. Memory and transcription establish the available information foundation; analysis, retrieval and reasoning prepare the recommendation; and approval controls preserve physician authority. The evaluation compares the single-language-model baseline, memory-augmented variant and full Zeest system to assess their contributions under consistent, clearly defined clinical assessment conditions. Each objective corresponds to identifiable requirements, component responsibilities and a verification activity. The comparison uses shared cases and an assessment process so changes in the available context do not become accidental differences in evaluation. Resource selection and clinical coverage remain explicit dependencies rather than additional objectives.

Table 2.1: Zeest objectives covering memory, clinical processing, physician review and comparative evaluation.

ID	Objective
O1	Develop hybrid longitudinal memory using relational storage and semantic retrieval.
O2	Convert consultation audio into structured clinical notes in real time.
O3	Integrate Patient Memory, Clinical Case, Imaging, Laboratory and Medication Safety agents.
O4	Integrate an existing validated pre-trained imaging classification model.
O5	Retrieve relevant clinical guidelines and present supporting citations.
O6	Generate a draft diagnosis for physician review within the clinical case workflow.
O7	Review generated recommendations through the Safety and Critic Agent before presentation.
O8	Provide approval, modification and rejection controls with a full audit history.
O9	Compare a single LLM, a memory-augmented variant and the full Zeest system.

2.5 Project Scope

Zeest includes physician registration and access, patient navigation, encounter review, consultation transcription and structured-note correction. Each physician accesses only their own patient records. Its workflow combines hybrid memory, specialised agents, imaging and laboratory inputs, medication context, evidence retrieval and critic review. Generated diagnoses and treatment plans require physician approval before recording. Development and evaluation use de-identified or synthetic cases, and clinical domain coverage depends on suitable data and validated models. Table 2.2 summarises the boundaries. Training diagnostic models from scratch, autonomous treatment, emergency and ICU systems, real-world deployment, regulatory approval, native mobile applications, patient-facing chat and the optional knowledge graph are excluded from the project scope. Supporting account functions do not introduce a separate administrative role or an external verification service. Patient ownership governs all related clinical operations, and draft revision links preserve the content reviewed. Clinical decisions remain separate from research outputs, even when both originate from the same encounter.

Table 2.2: Included capabilities and exclusions defining the boundaries of Zeest.

Included	Excluded
Physician registration, access and patient/encounter navigation	Separate administrator application
Consultation transcription and note review	Patient-facing chatbot and native mobile applications
Hybrid memory and specialised agents	Knowledge graph extension
Existing imaging models and clinical knowledge resources	Training diagnostic models from scratch
Evidence-supported drafts, critic review and physician decisions	Autonomous prescribing or treatment
De-identified/synthetic cases and supported data/model domains	Unsupported domains, emergency/ICU systems and real-world deployment
Controlled comparison of system variants	Regulatory approval and guaranteed clinical outcomes

2.6 Sustainable Development Goal (SDG)

Zeest aligns with Sustainable Development Goal 3, Good Health and Well-being, which promotes healthy lives and well-being for all ages. [1] Its intended contribution is to support physicians through clearer access to patient history, structured consultation documentation and reviewable clinical recommendations. Longitudinal memory connects earlier observations with the current encounter, while supporting evidence and critic findings help the responsible physician inspect the generated draft. Explicit approval keeps the clinical decision under professional judgement. Figure 2.1 shows the Good Health and Well-being goal. This alignment guides the focus on physician assistance; health-outcome improvements remain a matter for evaluation rather than an assumed result. The focus is assistance within a controlled clinical workflow rather than a claim that the integrated system already changes patient outcomes. Data permissions, supported model coverage and professional physician review limit the interpretation of this contribution. Comparative results are required before stronger statements about effectiveness can be made.



Figure 2.1: Good Health and Well-being.
Supporting physician review through longitudinal patient information and evidence-grounded clinical assistance.

2.7 Constraints

- Clinical coverage depends on permitted cases and validated existing models.
- Imaging inputs must match the selected model's task and supported labels.
- Transcription depends on supported languages and adequate audio quality.
- Evidence retrieval and medication review require suitable knowledge resources.
- Connected components must use compatible interfaces and representations.
- LangGraph and Google ADK responsibilities require a defined integration design.
- Performance limits depend on the selected hardware and processing services.

- Clinical decisions require explicit physician approval.
- Model formats, vector dimensions and service settings must be documented.

2.8 Business Opportunity

Zeest offers a coordinated workspace for consultation documentation, case preparation and recommendation review. Historical information, specialist findings and supporting guidelines can be considered within the same encounter of as disconnected outputs. The intended benefit is clearer access to clinical context and a more understandable review process. Integrating models and knowledge resources provides a foundation that can accommodate compatible capabilities without rebuilding the entire workflow. For researchers and developers, the opportunity lies in identifying which components provide useful support and which add complexity. Future adoption depends on evidence of that value and the requirements of the intended setting, rather than an assumed commercial deployment or pricing model. The design does not establish purchase costs, commercial demand or measured reductions in review time. These questions depend on implemented workflows and assessment in a setting. The immediate value is a traceable integration design that can be tested before any adoption claims are made.

2.9 Stakeholders Description/ User Characteristics

Physicians are the direct users and reviewers of Zeest. They need readable notes, clear patient context and an understandable relationship between findings, evidence and generated recommendations. Clinical knowledge supports their decisions, while basic familiarity with a web interface supports navigation and input tasks. Patients are affected stakeholders whose interests concern appropriate handling of their information and physician oversight of recorded decisions. The student team develops the integration and maintains consistent component responsibilities. The supervisor and evaluation panel guide and assess the problem, design and research approach. These groups share an interest in a coherent, accountable workflow while retaining distinct clinical, development and academic responsibilities. Patients do not operate a conversational interface within this scope. Their information remains associated with the responsible physician and relevant encounters. Researchers need permitted cases and consistent assessment procedures, while all stakeholders need the distinction between proposed assistance and a recorded professional decision to remain understandable.

2.9.1 Stakeholders Summary

Physicians are the direct users of Zeest and are responsible for interpreting findings and deciding whether to accept a clinical draft. Their interests include readable consultation notes, relevant history, visible

evidence and clear review controls. Patients are affected stakeholders whose information must remain associated with the correct physician and encounter; their role does not include a conversational interface in this scope. The student development team is responsible for translating the agreed requirements into compatible components, maintaining source relationships and preparing reproducible evaluation configurations. The supervisor and evaluation panel assess the clinical motivation, scope, technical rationale and evidence supporting the research contribution. These responsibilities are complementary: clinical users require understandable assistance, patients require appropriate information handling, and academic stakeholders require a justified design and controlled comparison method. Their common interest is an accountable workflow that preserves professional judgement while making the contribution of memory, specialised processing and review mechanisms assessable through future implementation and evaluation.

2.9.2 Key High-Level Goals and Problems of Stakeholders

Physicians need relevant history, reliable consultation notes and inspectable findings without losing the current patient context. Their main challenge is to assess information from several sources and determine whether it supports the generated draft. Patients need their information associated with the correct case and decisions reviewed by the responsible physician. The development team must connect processing components without creating ambiguous outputs or approval states. Academic reviewers need controlled comparisons that make the contributions of memory, agents, critic review and multimodal information assessable. Zeest addresses these needs through case continuity, identifiable source relationships and explicit review controls, with evaluation determining the usefulness of the integrated approach. The resulting vision connects clinical continuity with an accountable review boundary and a defined research comparison. It avoids assuming that more agents necessarily produce better recommendations. Resource permissions, supported tasks and measured component contributions determine whether the integrated design fulfils these stakeholder goals in practice.

2.10 Conclusion

Zeest's vision connects continuity of patient information with a physician-controlled clinical workflow. Longitudinal memory supplies relevant historical context, consultation transcription supports documentation, and specialised agents contribute findings from supported clinical inputs. Retrieved evidence and critic review accompany a diagnosis and treatment draft, while approval determines which reviewed revision becomes an accepted decision. These responsibilities define the integration objective without assuming that additional agents necessarily improve clinical recommendations. The scope also establishes

practical boundaries: permitted de-identified or synthetic cases, existing validated models, supported clinical domains and no autonomous treatment or real clinical deployment. Physicians need understandable information and control over decisions, patients need appropriate handling of their records, and the development and academic teams need consistent requirements and assessable comparisons. Resource selection and implementation remain dependencies for testing these goals. The resulting vision provides a clear basis for the requirements, methodology and design, with component value determined through controlled evaluation rather than architectural complexity alone.

Chapter 3 Literature Review / Related Work

Clinical decision support draws on medical language models, imaging analysis, evidence retrieval and agent coordination. The following review examines related applications, methods and resources, comparing their capabilities and limitations to identify the research gap.

3.1 Definitions, Acronyms, and Abbreviations

Additional abbreviations used in the literature review are listed below.

EMR: Electronic Medical Record.

EHR: Electronic Health Record.

NLP: Natural Language Processing.

CXR: Chest X-ray.

MIMIC: Medical Information Mart for Intensive Care.

ClinicalBERT: Clinical Bidirectional Encoder Representations from Transformers.

BioGPT: Biomedical Generative Pre-trained Transformer.

MedCPT: Medical Contrastive Pre-trained Transformer.

ReAct: Reasoning and Acting: a framework combining reasoning traces with tool actions.

ADK: Agent Development Kit.

3.2 Detailed Literature Review

This section reviews systems, models, datasets and methods related to Zeest. The topic is divided into clinical decision support, medical language models, imaging and multimodal learning, retrieval-based evidence support, agentic orchestration and clinical data resources. The purpose of this review is to identify what work already supports and where Zeest remains different. Thus, the literature is discussed according to the capability it contributes to the physician-controlled, longitudinal and multimodal clinical workflow. The comparison distinguishes clinical applications from representation models, specialist classifiers, retrieval methods and datasets. Each source is assessed for the task it demonstrates and the limits of transferring that capability into an encounter workflow. This distinction prevents medical question-answering performance or image classification findings from being presented as evidence of a clinical system. Source references identify the reviewed work, while critical assessment examines the additional ownership, provenance and review controls required by Zeest. Comparison distinguishes published capability from integration.

3.2.1 Clinical Decision Support and Electronic Records

3.2.1.1 Research Overview

Isabel DDx represents a differential diagnosis support application that helps clinicians consider possible diagnoses from entered features. [2] Its strength is that it addresses a real clinical reasoning task and can broaden the physician's diagnostic consideration. Its limitation for Zeest is that it is not described as a longitudinal patient-memory system with multimodal agents, audit-controlled approval and an ablation framework. UpToDate provides evidence-based clinical reference material for clinicians and is useful as a point of care knowledge resource. [3] Its reviewed reference capability motivates evidence access, while Zeest separately defines patient memory, specialist processing and revision-specific approval. OpenMRS demonstrates how an open source EMR can support patient record management, especially in resource-constrained settings. [4] Its relevance is strongest for patient and encounter organisation, while its limitation is that it is a record platform rather than an agentic reasoning and evidence-review workflow. Together, these works show useful clinical support functions without establishing the complete Zeest workflow.

3.2.1.2 Critical Assessment

The reviewed systems are strong in their individual areas. Isabel DDx supports differential diagnosis exploration, UpToDate gives clinicians access to curated medical reference content and OpenMRS shows how patient records can be organised in an open source EMR. Their documented contributions concern distinct parts of the target workflow. Differential diagnosis support does not automatically preserve encounter-specific memory, a reference system does not manage patient and encounter context, and an EMR does not by itself generate evidence-grounded drafts through specialist agents. Zeest additionally specifies revision-specific approval, modification, rejection and audit handling. Their limitations show why Zeest needs a combined method rather than one existing clinical tool. A useful assessment must distinguish the capability demonstrated by the source from the additional workflow Zeest requires. Integration still needs patient ownership, identifiable source relationships and explicit review states. These responsibilities cannot be inferred from benchmark performance or the presence of an existing software tool.

3.2.1.3 Relevance to Zeest

These systems show that Zeest should not treat clinical decision support as only a text-generation task. Differential diagnosis applications, medical references and EMR platforms each supply useful ideas for the project: possible diagnosis review, cited clinical knowledge and structured patient records. However,

the integration problem concerns the connection between these ideas. A physician needs to view the current consultation, retrieve relevant longitudinal history, inspect specialist findings and approve or reject the resulting draft. Existing reference or EMR systems do not automatically answer how a generated recommendation should be linked to a specific encounter, supporting evidence and audit action. Chapter 4 therefore specifies patient selection, encounter history, consultation note correction, evidence inspection and revision-specific approval as separate behaviours. The literature supports those boundaries by showing that information access and decision support are different responsibilities. Zeest's contribution is the integration of those responsibilities under physician control, not replacement of existing clinical judgement.

3.2.2 Medical and Biomedical Language Models

3.2.2.1 Research Overview

Med-PaLM 2 demonstrates that large language models can achieve strong performance on medical question answering benchmarks after medical alignment and evaluation. [5] GPT-4 medical challenge work shows broad medical reasoning capability across examination-style tasks, while documenting safety and reliability concerns that matter in clinical use. [6] These studies motivate the use of a powerful language model within Zeest, but their task setting is different from a longitudinal clinical workflow. Benchmark questions do not require storing a patient's encounter history, associating outputs with a physician action or preserving the difference between a draft and an approved clinical decision. The literature supports cautious use of language models as reasoning components. It does not support autonomous diagnosis, unreviewed treatment or the assumption that benchmark performance proves patient-level reliability. Zeest addresses this limitation by requiring retrieved evidence, critic review and explicit physician approval before any generated content becomes a recorded clinical decision. Component selection remains unresolved.

3.2.2.2 Critical Assessment

ClinicalBERT and BioGPT show another important direction: adapting language models to biomedical and clinical text. ClinicalBERT provides contextual embeddings trained on clinical notes, thereby contributing to patient-record representations and clinical NLP tasks. [7] BioGPT applies generative pretraining to biomedical text generation and mining, showing that domain-specific corpora can improve biomedical language processing. [8] Their strength is representation of specialised language, terminology and document patterns. Their limitation for Zeest is that they do not by themselves provide a physician workflow, multimodal processing path or approval-controlled memory update. They do not solve the problem of grounding a draft in patient-specific evidence. For Zeest, these works support

the general choice to use medical-language capability and semantic representations, but the embedding model and LLM remain unresolved. The methodology must treat model selection as a dependency and avoid claiming that any biomedical model has already been integrated.

3.2.2.3 Relevance to Zeest

These language-model studies are related to Zeest because the proposed system requires clinical text understanding, structured note use, reasoning over patient case context and generation of a reviewable draft. Their results support the decision to include language-model capability, but they also explain why Zeest requires additional controls. The proposed work adds longitudinal patient memory, cited evidence retrieval, critic physician review and physician approval around the generative component. In this way, Zeest uses medical language modelling as one part of a controlled clinical workflow rather than treating a model answer as an accepted clinical decision. The connection also creates verification targets: case inputs retain their originating sources, processing failures remain visible, and draft revisions preserve the evidence and critic findings presented to the responsible physician. Component value must be compared under common case definitions and reference procedures. Resource selection does not change the requirement for explicit approval before clinical decision recording.

3.2.3 Medical Imaging and Multimodal Learning

3.2.3.1 Research Overview

CheXNet used deep learning for pneumonia detection on chest radiographs and is frequently cited as evidence that imaging models can reach strong performance on a narrow radiology task. [9] CheXpert then provided a large chest radiograph dataset with uncertainty labels and an expert comparison benchmark across fourteen observations. [10] These works are relevant because Zeest intends to integrate a validated imaging classification model rather than train a new diagnostic model from scratch. Their limitation is equally important: a chest X-ray classifier produces findings for an imaging task, not a decision. It does not know the patient's longitudinal history, consultation, medication context or physician approval state. Therefore, Chapter 5 must place imaging analysis as one input to the clinical case rather than the method. The clinical domain depends on a dataset and validated model that can be used under the project constraints, so final resource selection remains part of the proposed methodology.

3.2.3.2 Critical Assessment

BioViL-T extends the imaging discussion by learning from temporal structure in biomedical vision-language processing. [11] It directly informs Zeest because Zeest is concerned with more than a single isolated input; the integrated system must reason over patient information across time and modalities.

However, a vision-language model still differs from the Zeest workflow. Zeest combines consultation notes, history retrieval, imaging outputs, laboratory information, medication context, evidence retrieval and critic review. A multimodal representation alone does not establish the audit trail or decision boundary required by Chapter 4. The review of imaging and vision-language research supports the need for careful integration. Imaging findings should retain their source and be displayed as specialist findings. They should not be silently merged into a generated conclusion without indicating how they relate to the encounter. This distinction motivates the text-only versus multimodal comparison in Chapter 5.

3.2.3.3 Relevance to Zeest

The imaging literature directly supports Zeest's plan to integrate an existing validated model instead of training a diagnostic model from scratch. It also supports the need for a multimodal methodology in which imaging findings are considered alongside notes, history, laboratory information and medication context. Zeest differs from isolated imaging systems by treating image analysis as a specialist contribution to a physician-reviewed case. The final draft remains subject to evidence retrieval, critic review and physician approval, so imaging output is not treated as an independent clinical decision. The connection also creates verification targets: inputs retain their sources, processing failures remain visible, and draft revisions preserve the evidence and critic findings presented to the physician. Component value must be compared under common case definitions and reference procedures.

3.2.4 Retrieval, Evidence Grounding and Medical Search

3.2.4.1 Research Overview

Retrieval-augmented generation connects a generator with retrieved documents so that generated answers can use external knowledge. [12] This directly informs Zeest because a draft should be accompanied by evidence that the physician can inspect. General RAG, however, does not guarantee that retrieved passages are clinically appropriate, patient-specific or sufficient for a diagnosis. MedRAG examines retrieval-augmented generation in medical question answering and provides a closer medical context for evaluating retrieval and generation together. [13] Its relevance lies in showing that retrieval choices can affect medical answer quality. Its limitation is that medical question answering still differs from a physician-facing encounter workflow with patient memory and approval actions. MedCPT contributes contrastive pre-training for biomedical retrieval using PubMed search logs, supporting the idea that medical retrieval benefits from domain-specific search representations. [14] Zeest can use these ideas only after selecting evidence sources and retrieval components.

3.2.4.2 Critical Assessment

The strength of RAG and medical retrieval work is that it gives a practical way to connect generated text with external sources. MedRAG and MedCPT are especially relevant because they move retrieval closer to biomedical and medical question answering. The limitation is that retrieval does not automatically make an answer correct. Retrieved passages may be incomplete, irrelevant or poorly matched to the patient case. A generator may overstate what the retrieved evidence supports. These weaknesses are important for Zeest because a clinical draft must be reviewable by a physician. The system must keep evidence visible, cite sources and allow critic review instead of hiding retrieval inside the generated answer. A useful assessment must distinguish the capability demonstrated by the source from the additional workflow Zeest requires. Integration still needs patient ownership, source relationships and explicit review states. These responsibilities cannot be inferred from benchmark performance or the presence of a software tool.

3.2.4.3 Relevance to Zeest

The evidence literature supports two design principles for the system. First, evidence must remain visible and citable rather than disappearing into a generated answer. Second, retrieval quality is a variable to be evaluated, not an assumed property of the architecture. Chapter 4 reflects these principles through requirements for guideline retrieval, citation display and supporting passages. Chapter 5 extends them into an evidence-retrieval procedure in which patient context and external guideline search are separated before the draft is produced. The physician should be able to inspect whether a cited passage supports the draft. The Safety and Critic Agent should review unsupported or contradictory content before presentation. This approach is stricter than ordinary RAG demonstrations because the final workflow concerns a draft and an audit record. The literature motivates RAG use, while the project constraints define the safeguards needed for Zeest.

3.2.5 Agentic Reasoning and Orchestration

3.2.5.1 Research Overview

ReAct shows how a language model can combine reasoning traces with actions, allowing an answer process to call tools, observe intermediate results and continue reasoning from those observations. [15] Agent orchestration frameworks, including ADK-style development patterns, extend this idea by separating responsibilities across specialised components rather than forcing one model interaction to perform every task. [16] These works are relevant because Zeest proposes separate agents for patient memory, case organisation, specialist analysis, retrieval, clinical reasoning and safety review. Their limitation

is that general agent frameworks do not define a clinical workflow by themselves. They do not decide what patient context is sufficient, how evidence should be displayed, or how a physician should approve, modify or reject a recommendation. Therefore, the literature supports the use of agentic decomposition as a design direction, while Chapter 5 must define the proposed medical sequence and its evaluation boundaries.

3.2.5.2 Critical Assessment

Agentic systems introduce risks that are especially important in a clinical report. More components can improve separation of responsibility, but they can create ambiguous handoffs, repeated context, failed tool calls or unsupported conclusions. Zeest's method must define what each stage receives and returns. The Patient Memory Agent retrieves history, the Clinical Case Agent organises the case, specialist agents produce typed findings, the Evidence and RAG Agent retrieves sources, the Clinical Reasoning Agent drafts the recommendation and the Safety and Critic Agent reviews it. The physician then approves, modifies or rejects the draft. This sequence is proposed, not executed. Its value will be examined through the ablation study rather than assumed in advance. ReAct and orchestration frameworks motivate the decomposition, while Zeest adds the patient-memory, evidence-inspection and physician-approval controls required for the medical setting. A useful assessment must distinguish the capability demonstrated by the source from the additional workflow Zeest requires.

3.2.5.3 Relevance to Zeest

This work supports Zeest's decision to divide responsibilities across specialised agents rather than using one prompt for the whole clinical task. The relationship is strongest in the idea that reasoning can be coordinated with tool use, retrieval and intermediate checks. Zeest extends that idea into a medical copilot workflow by adding patient memory, specialist clinical inputs, critic review and physician-controlled decision recording. The proposed research will compare this full agentic workflow against simpler variants so that the value of the decomposition can be assessed instead of assumed. The connection also creates verification targets: inputs retain their sources, processing failures remain visible, and draft revisions preserve the evidence and critic findings presented to the physician. Component value must be compared under common case definitions and reference procedures.

3.2.6 Clinical Datasets and Synthetic Cases

3.2.6.1 Research Overview

MIMIC-IV provides de-identified electronic health record data and can support Zeest's longitudinal-memory objective because it contains patient and encounter-level clinical information. [17] MIMIC-

CXR provides a de-identified chest radiograph dataset with associated reports, thereby contributing to imaging and multimodal case preparation. [18] These datasets support the idea that de-identified clinical data can be used for research, but their existence does not by itself grant access, determine the disease domain or authorize every use. Synthea offers a method and software mechanism for generating synthetic patients. [19] Its relevance is that synthetic cases can help develop workflows when real clinical data access is constrained. The limitation is that synthetic cases may not represent the full clinical complexity of real patients. Zeest's methodology distinguishes selected candidates, access status, preprocessing and final evaluation resources of treating all candidate datasets as already approved inputs. Resource permission and case suitability must be confirmed separately before incorporating candidates into comparative evaluation.

3.2.6.2 Critical Assessment

Across the dataset literature, the common lesson is that reproducible evaluation depends on case definition, permitted access, preprocessing, reference judgement and controlled comparisons. Zeest's planned evaluation therefore uses common case inputs across a single-LLM baseline, a memory-augmented variant and the full system. Any future results must come from those executed comparisons, not from claims imported from prior datasets or unrelated benchmark papers. A useful assessment must distinguish the capability demonstrated by the source from the additional clinical workflow Zeest requires. Integration still needs patient ownership, identifiable preserved source relationships and explicit physician review states. These responsibilities cannot be inferred from benchmark performance or the presence of an existing software tool. Dataset documentation describes available records rather than guaranteeing a usable evaluation case. Patient trajectories, supported modalities and source permissions require checking together. Reference judgements must be independent of generated answers, and configuration changes must remain identifiable when comparing system variants.

3.2.6.3 Relevance to Zeest

The dataset literature is related to Zeest because the project needs longitudinal cases, possible imaging inputs and safe development data. MIMIC-IV, MIMIC-CXR, CheXpert and Synthea provide candidate directions, but each has access, preprocessing and domain constraints. Zeest's proposed methodology therefore keeps a dataset and model register instead of claiming that every candidate resource is already selected. These resources will support the planned ablation study only after the final domain, permitted use and reference judgement process are confirmed. The connection also creates verification targets: inputs retain their sources, processing failures remain visible, and draft revisions preserve the evidence and critic findings presented to the responsible physician. Component value must be compared under

common case definitions and reference procedures. Resource selection does not change the requirement for explicit approval before clinical decision recording. The selected cases must preserve patient and encounter context and source relationships while distinguishing missing modalities from completed findings.

3.3 Literature Review Summary Table

Table 3.1 summarises the principal works reviewed for Zeest. The table mixes research papers, datasets and established clinical systems because the project is an R&D integration effort. Each item is included only for the capability it supports: clinical reference, record organisation, medical language modelling, imaging, retrieval, orchestration or evaluation data. The limitations column explains why the item does not already solve the Zeest problem. This distinction is necessary because Zeest combines several mature areas into one physician-controlled workflow. No row is treated as proof that Zeest has been implemented or evaluated. Instead, the table identifies the evidence base used to design the methodology and to justify the gaps addressed by the system. The comparison retains the separation between a reviewed capability and a physician workflow. Source evidence supports component decisions, while Zeest still requires ownership, revision links and explicit approval. The seventeen entries provide the evidence base for component selection.

Table 3.1: Summary of literature review.

This table compares the capabilities, relevance and limitations of the reviewed work.

Work	Method or capability	Relevance to Zeest	Limitation for Zeest
Isabel DDx [2]	Differential diagnosis support system.	Shows clinical value of diagnosis assistance.	Not a longitudinal multimodal agent workflow.
UpToDate [3]	Evidence-based clinical reference.	Motivates guideline and citation access.	Reference role does not establish Zeest's review controls.
OpenMRS [4]	Open source EMR platform.	Supports patient and encounter record thinking.	Not a reasoning, critic and RAG system.
Med-PaLM 2 [5]	Medical LLM alignment and evaluation.	Supports medical-language reasoning potential.	Benchmark QA is not patient-level workflow validation.
GPT-4 medical challenges [6]	Medical examination and benchmark analysis.	Motivates powerful LLM reasoning with caution.	Does not provide persistent memory or approval control.
ClinicalBERT [7]	Clinical text embeddings.	Supports clinical semantic representation.	Not a complete decision-support workflow.
BioGPT [8]	Biomedical generative pre-training.	Supports biomedical text generation and mining.	Does not ground drafts in patient evidence.
CheXNet [9]	Chest X-ray pneumonia classifier.	Candidate style for existing imaging model use.	Narrow imaging task without longitudinal context.
CheXpert [10]	Chest radiograph dataset and benchmark.	Candidate data/model domain for imaging support.	Access and final domain selection remain unresolved.
BioViL-T [11]	Biomedical vision-language temporal learning.	Supports multimodal and temporal motivation.	Does not define Zeest's physician approval workflow.
ReAct [15]	Reasoning and acting with tools.	Motivates stepwise agent interaction.	Not a clinical safety or memory architecture.
RAG [12]	Retrieval-augmented generation.	Foundation for cited evidence retrieval.	General method does not ensure clinical sufficiency.
MedRAG [13]	Medical retrieval-augmented generation benchmark.	Supports medical evidence retrieval evaluation.	Medical QA differs from encounter review.
MedCPT [14]	Biomedical retrieval pre-training.	Supports domain-aware evidence search.	Does not select Zeest's final retriever or corpus.
MIMIC-IV [17]	De-identified EHR dataset.	Candidate source for longitudinal cases.	Access, preprocessing and use conditions must be confirmed.
MIMIC-CXR [18]	De-identified chest radiographs and reports.	Candidate source for imaging and notes.	Does not by itself provide a complete evaluation protocol.
Synthea [19]	Synthetic patient generation.	Supports safe development case creation.	Synthetic cases may not capture real clinical complexity.

3.4 Conclusion

The reviewed literature supports the need for Zeest but clarifies its boundaries. Clinical support systems and EMR platforms show the importance of diagnosis assistance, evidence access and patient

records, yet they do not combine these functions into a longitudinal agentic workflow. Medical LLMs and biomedical language models provide useful reasoning and representation capabilities, but their benchmark performance does not remove the need for evidence, critic review and physician approval. Imaging and vision-language work supports the integration of validated models, while showing why imaging findings must remain tied to their source and task. RAG and medical retrieval research motivate cited evidence retrieval, but retrieval quality must be tested rather than assumed. Dataset work provides possible resources for evaluation, subject to access and domain constraints. The resulting gap is the controlled integration of patient memory, multimodal findings, evidence retrieval, critic review and physician decision logging. Comparisons establish the value of this integration.

Chapter 4 Software Requirement Specifications

Software requirements define observable functions, expected qualities and the conditions under which a system operates. The following sections organise features, requirements, use cases, interfaces, data relationships and risks into a consistent specification for development and verification.

4.1 List of Features

- Physician registration, login and logout.
- Physician-specific patient listing, selection and information review.
- Encounter history review and encounter selection.
- Real-time consultation transcription into structured notes.
- Physician review and correction of consultation notes.
- Imaging, laboratory and medication input handling.
- Longitudinal patient history retrieval through hybrid memory.
- Specialised imaging, laboratory and medication analysis.
- Clinical guideline retrieval with citations and supporting passages.
- Draft diagnosis and treatment plan generation.
- Safety and consistency review through the critic agent.
- Physician approval, modification and rejection of clinical drafts.
- Recording of explicitly approved clinical decisions.
- Approved decision and audit history review.
- Comparative evaluation of baseline, memory and full-system variants.

4.2 Functional Requirements

Tables 4.1 through 4.5 specify clinical functions; Table 4.6 specifies comparative evaluation.

Table 4.1: Physician access and patient navigation.

ID	User/actor	Required behaviour
FR01	Physician	The system shall allow a physician to sign up using their name, email and password.
FR02	System	The system shall validate registration fields and reject invalid or duplicate email addresses.
FR03	System	The system shall create a valid account and direct the physician to Log In.
FR04	System	The system shall associate each patient record with its responsible physician.
FR05	System	The system shall reject access or processing of another physician's patient records.
FR06	Physician	The system shall allow the physician to log in using a registered account.
FR07	Physician	The system shall allow the physician to log out and end the current session.
FR08	System	The system shall reject invalid login credentials without establishing a session.
FR09	Physician	The system shall allow the physician to view only the patients belonging to the logged-in physician.
FR10	Physician	The system shall allow the physician to select a patient from the patient list.
FR11	Physician	The system shall display the selected patient's information.
FR12	Physician	The system shall display the selected patient's encounter history.
FR13	Physician	The system shall allow the physician to open a selected encounter for consultation and review.

Table 4.2: Consultation recording and clinical inputs.

ID	User/actor	Required behaviour
FR14	Physician	The system shall allow the physician to start consultation audio capture.
FR15	Physician	The system shall allow the physician to stop consultation audio capture.
FR16	System	The system shall convert consultation audio into structured clinical notes.
FR17	Physician	The system shall display the structured consultation notes for physician review.
FR18	Physician	The system shall allow the physician to correct and save structured consultation notes.
FR19	Physician	The system shall allow the physician to supply an imaging input for the selected encounter.
FR20	Physician	The system shall allow the physician to supply laboratory information for the selected encounter.
FR21	Physician	The system shall allow the physician to supply medication context for the selected encounter.
FR22	System	The system shall reject unsupported or unusable clinical inputs with an explanatory message.

Table 4.3: Memory and specialised case processing.

ID	User/actor	Required behaviour
FR23	System	The system shall retain patient information across encounters using relational records and semantic representations.
FR24	System	The system shall retrieve relevant history for the selected patient and current encounter.
FR25	System	The system shall organise current inputs and retrieved history into the clinical case.
FR26	System	The system shall analyse supported imaging inputs using an existing validated pre-trained classification model.
FR27	System	The system shall process supplied laboratory information through the Laboratory Agent.
FR28	System	The system shall assess medication context through the Medication Safety Agent and an established knowledge resource.
FR29	Physician	The system shall display imaging, laboratory and medication findings with their input sources.
FR30	System	The system shall display the processing status or failure of a requested clinical analysis.

Table 4.4: Draft generation, evidence and critic review.

ID	User/actor	Required behaviour
FR31	Physician	The system shall allow the physician to request generation of a clinical draft for the selected encounter.
FR32	System	The system shall retrieve relevant clinical guidelines through the Evidence and RAG Agent.
FR33	System	The system shall generate a draft diagnosis from the prepared case and retrieved evidence.
FR34	System	The system shall generate a draft treatment plan for physician review.
FR35	System	The system shall submit the clinical draft to the Safety and Critic Agent before physician presentation.
FR36	Physician	The system shall display guideline citations and supporting passages alongside the clinical draft.
FR37	Physician	The system shall display critic findings alongside the clinical draft.

Table 4.5: Physician decisions and audit history.

ID	User/actor	Required behaviour
FR38	Physician	The system shall allow the physician to explicitly approve a selected clinical draft.
FR39	Physician	The system shall allow the physician to edit and save a revised clinical draft.
FR40	Physician	The system shall allow the physician to reject a selected clinical draft.
FR41	System	The system shall record the explicitly approved draft revision as a clinical decision.
FR42	System	The system shall record physician review actions with the actor, time, draft revision and encounter context.
FR43	Physician	The system shall allow the physician to inspect the encounter's audit history.
FR44	Physician	The system shall allow the physician to view approved decisions for their selected patient and encounter.

Table 4.6: Research requirements for comparing the contribution of system components.

ID	Required behaviour
RR01	The evaluation framework shall support a single-LLM baseline, a memory-augmented variant and the full Zeest system.
RR02	The evaluation framework shall use the same case inputs and reference process for comparisons between variants.
RR03	The evaluation framework shall record the configuration and output associated with each evaluated variant.
RR04	The evaluation framework shall support controlled comparisons of critic review and multimodal input contributions.

4.3 Quality Attributes

4.3.1 Usability

The physician interface must make patient review, consultation capture and clinical decisions easy to follow. Patient and encounter context should remain visible so that the physician can identify which case is being reviewed. Navigation must use consistent labels for patient history, consultation and review actions. Structured notes need a readable presentation and a clear means of correcting transcription errors before processing. Specialist findings and retrieved evidence should be distinguishable from the generated diagnosis and treatment draft. The physician must understand whether a draft is pending, modified, rejected or approved. Approval, modification and rejection controls require clear labels because they produce different outcomes. Messages should explain unusable inputs and failed operations without suggesting that analysis succeeded. The interface should make the next available action understandable without showing internal framework labels or requirement codes as application content. Empty states should indicate what information is unavailable and which user action can supply it.

4.3.2 Reliability and Integrity

Zeest must preserve the association between each patient, encounter, input and recorded review action. Information retrieved from earlier encounters must retain its source and must not become associated with a different patient during case preparation. Processing components need to return identifiable results or failures so that incomplete analysis cannot be mistaken for completed findings. A generated draft must remain separate from an approved clinical decision, including when the physician edits or rejects it. Approval must refer to the saved revision that the physician reviewed, and the recorded content must match that revision. Audit entries need to preserve the actor, action and context. These relationships make the outcome of each operation understandable and protect the continuity of the case. Failed requests must leave an understandable state without creating a partial accepted decision. Repeated approval attempts must not create duplicate decisions. Saved outcomes must remain distinguishable

from pending operations and rejected requests.

4.3.3 Security and Confidentiality

Clinical operations require a valid authenticated physician session, and each physician can access only their own patient records. Ownership checks apply to patient details, encounters, clinical inputs, drafts, approved decisions and audit history. A request for another physician's record must not reveal its information or permit processing. Registration validates account information, login establishes access and logout ends the session. All stored passwords require salted hashing rather than plaintext storage. [20] Development and evaluation use de-identified or synthetic cases. Account restrictions and data-handling practices address different aspects of confidentiality, and both apply across consultation documentation, specialised analysis and physician review rather than only on the login screen. Ownership must be enforced by application services as well as by the visible patient list. Changing a patient or encounter identifier in a request must not bypass these restrictions. Error responses should avoid exposing another physician's record, and clinical data permissions must be checked separately.

4.3.4 Performance

The consultation workflow requires timely transcription, history retrieval and clinical processing so that the physician can follow the progress of the encounter. Audio transcription should proceed incrementally without waiting until the whole consultation has finished. Other processing stages must expose their state so that a pending operation is distinguishable from a completed result or a failure. Performance depends on the selected language model, imaging model, knowledge resources and execution environment. It must be assessed through measurements, including transcription delay and the time required for retrieval and analysis. The application must avoid presenting incomplete output as a final result merely to appear responsive. Numerical acceptance limits require an agreed environment and workload before evaluation. Measurements must record the workload, service configuration and stage boundaries so comparisons remain interpretable. Pending tasks should not prevent the physician from understanding the encounter state. Performance checks must therefore identify each stage and its completion state.

4.3.5 Maintainability and Reusability

The components of Zeest need responsibilities and consistent interfaces so that the integration can be understood and maintained. Transcription, memory retrieval, specialised analysis, evidence retrieval and reasoning should exchange identifiable inputs, outputs and failure information. A problem in one stage must be distinguishable from a failure elsewhere in the workflow. Existing diagnostic models and knowledge resources form the supporting groundwork, while the integration connects their results

to patient context and physician review. Maintaining that connection requires changes to preserve the patient and encounter associations used by downstream components. Reusability concerns retaining these interfaces when a compatible supporting component is replaced. It does not require every model or knowledge resource to be interchangeable without adaptation. Versioned contracts and source identifiers help isolate the effect of an integration change. Each component should expose its resource restrictions and errors through the same application boundary. Documented contracts provide the basis for targeted integration checks.

4.3.6 Extensibility

Zeest must accommodate compatible clinical data and diagnostic models within its established workflow. Extending clinical coverage depends on suitable case information and a validated model for the task, rather than on the language model's ability to discuss a condition. A new input or model must supply findings that can be associated with the correct patient, encounter and source. The Clinical Case Agent must be able to organise those findings alongside the historical context. Evidence retrieval, clinical reasoning and critic review must continue to operate on identifiable information, and physician approval must remain necessary before a decision is recorded. Extensibility concerns adding compatible capabilities while preserving the relationships and controls already required by the system. New coverage must preserve evidence visibility and explicit failure states. Supporting a broader topic in generated text alone does not establish validated clinical analysis. Compatibility requires documented inputs, outputs and preserved decision boundaries for every extension.

4.3.7 Traceability

The physician must be able to relate a displayed finding or recommendation to the information used in producing it. Clinical inputs need identifiable patient, encounter and source associations, while retrieved guidelines require citations and supporting passages. Specialist findings must remain linked to their input sources so that the physician can distinguish an imaging result from a laboratory or medication finding. The generated diagnosis and treatment draft must retain its supporting evidence and critic findings during review. Physician actions require identifiable context, including the reviewed revision and the content accepted through approval. An audit entry records what action occurred; it does not itself establish that a draft became an approved decision. Source relationships must remain available after note corrections or draft modifications. Findings and citations refer to the revision used, while the approval action identifies the content accepted. Source identifiers must survive retrieval, presentation and later inspection of the reviewed case.

4.4 Non-Functional Requirements

Tables 4.7 and 4.8 define the constraints supporting these attributes.

Table 4.7: Non-functional requirements for interaction, traceability and component quality.

ID	Concern	Required constraint
NFR01	Performance	The system shall support incremental consultation transcription without waiting for the entire consultation to end.
NFR02	Usability	The interface shall use consistent navigation and terminology across patient, consultation and review views.
NFR03	Decision integrity	The system shall exclude pending and rejected drafts from approved clinical decisions.
NFR04	Traceability	Displayed findings, citations and review actions shall retain their source and encounter relationships.
NFR05	Confidentiality	Development and evaluation case information shall be de-identified or synthetic.
NFR06	Maintainability	Processing components shall expose defined input, output and failure contracts.
NFR07	Extensibility	A new clinical domain shall require compatible case data and a validated existing diagnostic model.
NFR08	Access control	Protected clinical operations shall require authentication and ownership of the selected patient record.

Table 4.8: Non-functional requirements for access, persistence, performance and decision integrity.

ID	Concern	Required constraint
NFR09	Credential protection	Stored passwords shall use salted password hashing rather than plaintext storage.
NFR10	Case integrity	Processing and retrieval shall preserve physician ownership and patient/encounter associations.
NFR11	Failure handling	Failed processing or record persistence shall not display a success outcome or create a falsely approved decision.
NFR12	Revision integrity	Approval shall identify the exact saved draft revision and its reviewed content.
NFR13	Timing observability	Transcription delay, history retrieval time and analysis duration shall be separately measurable.
NFR14	Capacity specification	The supported concurrent-user limit shall be declared for the agreed execution environment.
NFR15	Review clarity	The interface shall visibly distinguish pending, approved and rejected clinical drafts.
NFR16	Reuse	Replacing a specialist integration shall preserve its agreed interface and the physician-approval boundary.

4.5 Assumptions

- Physicians register before accessing protected patient records.
- Each patient belongs to a responsible physician and has identified encounters.
- Physicians can interpret findings and review clinical drafts.
- An audio device and recording permission are available for consultation capture.

- Clinical inputs match the formats supported by the selected components.
- Required processing services and knowledge resources are accessible.
- Permitted de-identified or synthetic cases cover the selected clinical domain.

4.6 Use Cases

Figure 4.1 shows actor participation and included or conditional behaviour. Tables 4.9 through 4.30 describe the complete identified workflow.

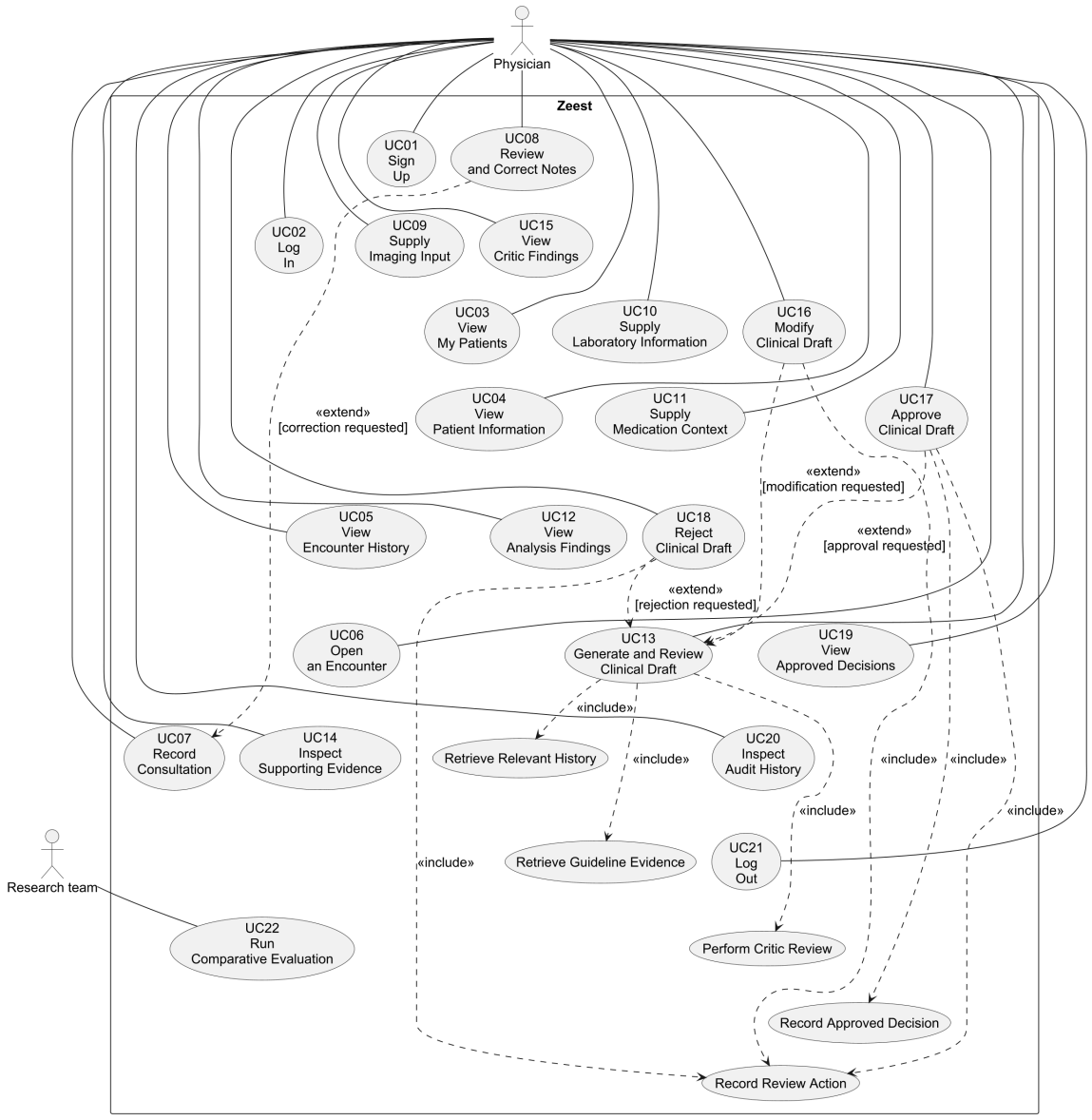


Figure 4.1: Zeest use cases inside one system boundary, with Physician and Research team actors outside. Dashed arrows distinguish mandatory included processing from conditional review actions.

Table 4.9: UC01: Sign Up.

Name		UC01: Sign Up	
Actors		Physician	
Summary		The physician creates an account using their registration information.	
Pre-Conditions		The physician is not logged in and registration is available.	
Post-Conditions		An account is created and the login page is displayed.	
Special Requirements		Valid email address; required name and password; securely stored password.	
Basic Flow			
Actor Action		System Response	
1	Selects Sign Up.	2	Displays the registration form.
3	Enters name, email and password and submits.	4	Validates the information, creates the account and displays Log In.
Alternative Flow			
3	Submits incomplete or invalid information.	4-A	Identifies the invalid or missing fields without creating an account.
3	Uses an email already registered.	4-B	Explains that the email is already registered without creating another account.

Table 4.10: UC02: Log In.

Name		UC02: Log In	
Actors		Physician	
Summary		The physician accesses their account and patient records.	
Pre-Conditions		The physician has a registered account and is not logged in.	
Post-Conditions		An authenticated session is established.	
Special Requirements		Valid registered email and password.	
Basic Flow			
Actor Action		System Response	
1	Selects Log In.	2	Displays the login form.
3	Enters email and password and submits.	4	Authenticates the account and opens its patient list.
Alternative Flow			
3	Enters invalid credentials.	4-A	Displays an invalid-credentials message without establishing a session.

Table 4.11: UC03: View My Patients.

Name		UC03: View My Patients	
Actors		Physician	
Summary		The physician views and selects from their patient records.	
Pre-Conditions		The physician is logged in.	
Post-Conditions		The physician's patient list is displayed.	
Special Requirements		Only records belonging to the logged-in physician are accessible.	
Basic Flow			
Actor Action		System Response	
1	Opens My Patients.	2	Retrieves and displays only the physician's patient records.
3	Selects a patient.	4	Retains the selected patient and provides access to their information.
Alternative Flow			
1	Opens a list without assigned patients.	2-A	Displays an empty patient-list message.

Table 4.12: UC04: View Patient Information.

Name		UC04: View Patient Information	
Actors		Physician	
Summary		The physician reviews the selected patient's available information.	
Pre-Conditions		The physician is logged in and one of their own patients is selected.	
Post-Conditions		Patient information is displayed.	
Special Requirements		Physician ownership must be checked before displaying the record.	
Basic Flow			
Actor Action		System Response	
1	Opens the selected patient.	2	Checks ownership and displays patient information.
3	Reviews the information.	4	Keeps the selected patient context for subsequent actions.
Alternative Flow			
1	Requests another physician's patient record.	2-A	Denies access without displaying the record.

Table 4.13: UC05: View Encounter History.

Name		UC05: View Encounter History	
Actors		Physician	
Summary		The physician reviews the selected patient's previous encounters.	
Pre-Conditions		The physician is logged in and one of their own patients is selected.	
Post-Conditions		Historical encounter information is displayed.	
Special Requirements		History must retain its patient, encounter and source relationships.	
Basic Flow			
Actor Action		System Response	
1	Selects Encounter History.	2	Displays the patient's previous encounters.
3	Selects a previous encounter.	4	Displays available notes and clinical information with their sources.
Alternative Flow			
1	Opens history with no previous encounters.	2-A	Displays an empty-history message.

Table 4.14: UC06: Open an Encounter.

Name		UC06: Open an Encounter	
Actors		Physician	
Summary		The physician opens the encounter used for consultation and review.	
Pre-Conditions		The physician is logged in and one of their own patients is selected. An encounter record exists.	
Post-Conditions		The encounter becomes the context for clinical actions.	
Special Requirements		The encounter must belong to the selected patient and physician.	
Basic Flow			
Actor Action		System Response	
1	Selects an encounter.	2	Checks its patient and physician associations.
3	Opens the encounter.	4	Displays its consultation workspace and existing inputs.
Alternative Flow			
1	Selects an unavailable or unauthorised encounter.	2-A	Rejects the selection without exposing its information.

Table 4.15: UC07: Record Consultation.

Name		UC07: Record Consultation	
Actors		Physician	
Summary		The physician captures audio and obtains structured consultation notes.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open.	
Post-Conditions		Captured consultation content is presented as structured notes.	
Special Requirements		Working microphone, recording permission and supported transcription service.	
Basic Flow			
Actor Action		System Response	
1	Selects Start Recording.	2	Checks audio access and begins capture and transcription.
3	Conducts the consultation.	4	Transcribes incoming audio into structured notes.
5	Selects Stop Recording.	6	Stops capture and presents the resulting notes.
Alternative Flow			
1	Cannot grant microphone permission.	2-A	Reports the access problem without recording.
3	Continues when transcription fails.	4-A	Reports the failure and identifies incomplete notes.

Table 4.16: UC08: Review and Correct Notes.

Name		UC08: Review and Correct Notes	
Actors		Physician	
Summary		The physician reviews and saves corrections to consultation notes.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. Structured notes are available.	
Post-Conditions		The reviewed notes are retained for the encounter.	
Special Requirements		Corrections must preserve the encounter association.	
Basic Flow			
Actor Action		System Response	
1	Opens the notes.	2	Displays the available structured notes.
3	Edits the notes and selects Save.	4	Validates and saves the corrected content.
Alternative Flow			
3	Submits unusable note content.	4-A	Requests correction and retains the prior saved content.

Table 4.17: UC09: Supply Imaging Input.

Name		UC09: Supply Imaging Input	
Actors		Physician	
Summary		The physician supplies imaging information for the selected encounter.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open.	
Post-Conditions		The supplied input is retained and submitted to its specialist agent.	
Special Requirements		Input must meet the selected imaging model's supported format and task.	
Basic Flow			
Actor Action		System Response	
1	Opens the imaging input control.	2	Displays the supported input requirements.
3	Provides the input and submits.	4	Validates the input and submits it for specialised processing.
Alternative Flow			
3	Submits missing or unsupported input.	4-A	Explains the validation problem without treating the input as processed.

Table 4.18: UC10: Supply Laboratory Information.

Name		UC10: Supply Laboratory Information	
Actors		Physician	
Summary		The physician supplies laboratory information for the selected encounter.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open.	
Post-Conditions		The supplied input is retained and submitted to its specialist agent.	
Special Requirements		Laboratory values must retain their supplied units and source context.	
Basic Flow			
Actor Action		System Response	
1	Opens the laboratory input control.	2	Displays the supported input requirements.
3	Provides the input and submits.	4	Validates the input and submits it for specialised processing.
Alternative Flow			
3	Submits missing or unsupported input.	4-A	Explains the validation problem without treating the input as processed.

Table 4.19: UC11: Supply Medication Context.

Name		UC11: Supply Medication Context	
Actors		Physician	
Summary		The physician supplies medication information for the selected encounter.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open.	
Post-Conditions		The supplied input is retained and submitted to its specialist agent.	
Special Requirements		Medication information must remain linked to the selected patient and encounter.	
Basic Flow			
Actor Action		System Response	
1	Opens the medication input control.	2	Displays the supported input requirements.
3	Provides the input and submits.	4	Validates the input and submits it for specialised processing.
Alternative Flow			
3	Submits missing or unsupported input.	4-A	Explains the validation problem without treating the input as processed.

Table 4.20: UC12: View Analysis Findings.

Name		UC12: View Analysis Findings	
Actors		Physician	
Summary		The physician inspects specialist findings for the encounter.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. Clinical inputs have been submitted.	
Post-Conditions		Available findings and processing states are displayed.	
Special Requirements		Findings must identify their source; failures must not appear as successful results.	
Basic Flow			
Actor Action		System Response	
1	Opens Analysis Findings.	2	Displays imaging, laboratory and medication results with processing states.
3	Selects a finding.	4	Displays its source and available analysis information.
Alternative Flow			
1	Requests findings while processing is incomplete.	2-A	Displays the pending or failed state rather than a completed result.

Table 4.21: UC13: Generate and Review Clinical Draft.

Name		UC13: Generate and Review Clinical Draft	
Actors		Physician	
Summary		The physician requests a diagnosis and treatment draft for review.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. Case information is available.	
Post-Conditions		An evidence-supported draft and critic findings are available for physician review.	
Special Requirements		Relevant history, evidence retrieval and critic review precede presentation.	
Basic Flow			
Actor Action		System Response	
1	Selects Generate Draft.	2	Prepares the case with inputs, specialist findings and relevant history.
3	Waits for the requested draft.	4	Retrieves evidence, generates the draft, performs critic review and presents the output.
5	Reviews the presented draft, evidence and critic findings.	6	Keeps the draft pending while the physician chooses a review action.
Alternative Flow			
1	Requests generation without usable case information.	2-A	Explains the missing inputs.
3	Waits during a processing failure.	4-A	Reports the failed stage without presenting an unreviewed output as completed.

Table 4.22: UC14: Inspect Supporting Evidence.

Name		UC14: Inspect Supporting Evidence	
Actors		Physician	
Summary		The physician reviews citations and passages supporting the draft.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. A generated draft is available.	
Post-Conditions		Available supporting evidence is displayed.	
Special Requirements		Citations and passages must retain their source and draft relationships.	
Basic Flow			
Actor Action		System Response	
1	Opens Supporting Evidence.	2	Displays citations and retrieved passages.
3	Selects an evidence item.	4	Displays its source reference and supporting content.
Alternative Flow			
1	Reviews a draft with missing supporting evidence.	2-A	Identifies the missing support without inventing citations.

Table 4.23: UC15: View Critic Findings.

Name		UC15: View Critic Findings	
Actors		Physician	
Summary		The physician reviews the safety and consistency assessment of the draft.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. A generated draft is available. Critic review has been performed.	
Post-Conditions		Critic findings are displayed alongside the draft.	
Special Requirements		The review must identify the draft it assessed and retain physician decision authority.	
Basic Flow			
Actor Action		System Response	
1	Opens Critic Findings.	2	Displays the available safety and consistency findings.
3	Reviews a finding.	4	Keeps the finding associated with its assessed draft.
Alternative Flow			
1	Requests an unavailable critic review.	2-A	Displays its unavailable state without claiming a successful review.

Table 4.24: UC16: Modify Clinical Draft.

Name		UC16: Modify Clinical Draft	
Actors		Physician	
Summary		The physician edits and saves a generated clinical draft.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. A generated draft is available. The draft is pending review.	
Post-Conditions		A revised draft is saved and remains pending approval.	
Special Requirements		Saving a modification must not record an approved decision.	
Basic Flow			
Actor Action		System Response	
1	Selects Modify.	2	Displays editable draft content.
3	Edits the content and selects Save.	4	Stores the revised draft and modification action without approval.
Alternative Flow			
3	Cancels the edit.	4-A	Retains the previous saved revision.

Table 4.25: UC17: Approve Clinical Draft.

Name		UC17: Approve Clinical Draft	
Actors		Physician	
Summary		The physician explicitly accepts the reviewed draft revision.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. A generated draft is available. The displayed revision is pending review.	
Post-Conditions		The accepted content is recorded as an approved clinical decision.	
Special Requirements		Approval must identify the exact saved revision and responsible physician.	
Basic Flow			
Actor Action		System Response	
1	Selects Approve.	2	Displays the revision and content to confirm.
3	Confirms approval.	4	Records the accepted decision and corresponding audit action.
Alternative Flow			
3	Confirms after the revision has changed.	4-A	Requests review of the current saved revision before approval.
3	Cancels confirmation.	4-B	Leaves the draft pending.

Table 4.26: UC18: Reject Clinical Draft.

Name		UC18: Reject Clinical Draft	
Actors		Physician	
Summary		The physician rejects a pending clinical draft.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open. A generated draft is available. The draft is pending review.	
Post-Conditions		The draft is marked rejected and the action is logged.	
Special Requirements		Rejected content must not become an approved decision.	
Basic Flow			
Actor Action		System Response	
1	Selects Reject.	2	Requests confirmation for the selected draft.
3	Confirms rejection.	4	Records rejection and its audit action without recording a clinical decision.
Alternative Flow			
3	Cancels rejection.	4-A	Leaves the existing draft state unchanged.

Table 4.27: UC19: View Approved Decisions.

Name		UC19: View Approved Decisions	
Actors		Physician	
Summary		The physician reviews accepted decisions for their selected patient.	
Pre-Conditions		The physician is logged in and one of their own patients is selected.	
Post-Conditions		Approved decision content and its encounter context are displayed.	
Special Requirements		Only explicitly approved decisions belonging to the physician's patient may be displayed.	
Basic Flow			
Actor Action		System Response	
1	Opens Approved Decisions.	2	Displays the patient's approved decisions grouped by encounter.
3	Selects a decision.	4	Displays accepted content and its approval context.
Alternative Flow			
1	Opens a patient with no approved decisions.	2-A	Displays an empty approved-decisions message.

Table 4.28: UC20: Inspect Audit History.

Name		UC20: Inspect Audit History	
Actors		Physician	
Summary		The physician reviews the encounter's recorded decision actions.	
Pre-Conditions		The physician is logged in and an encounter for one of their own patients is open.	
Post-Conditions		Review actions are displayed without changing them.	
Special Requirements		Actions must retain time, physician identity, draft revision and encounter context.	
Basic Flow			
Actor Action		System Response	
1	Opens Audit History.	2	Displays recorded review actions.
3	Selects an action.	4	Displays its revision and decision context.
Alternative Flow			
1	Opens an encounter without review actions.	2-A	Displays an empty audit-history message.

Table 4.29: UC21: Log Out.

Name		UC21: Log Out	
Actors		Physician	
Summary		The physician ends the application session.	
Pre-Conditions		The physician is logged in.	
Post-Conditions		The session ends and protected records require login.	
Special Requirements		Protected patient information must not remain accessible through the ended session.	
Basic Flow			
Actor Action		System Response	
1	Selects Log Out.	2	Terminates the session and displays Log In.
Alternative Flow			
1	Selects Log Out after the session has expired.	2-A	Displays Log In without restoring access.

Table 4.30: UC22: Run Comparative Evaluation.

Name		UC22: Run Comparative Evaluation	
Actors		Research team	
Summary		The research team compares the defined system variants.	
Pre-Conditions		Permitted cases, reference assessments and configurations are available.	
Post-Conditions		Outputs are recorded with their cases and configurations for comparison.	
Special Requirements		The baseline, memory variant and full system must use consistent assessment conditions.	
Basic Flow			
Actor Action		System Response	
1	Selects cases and configurations.	2	Checks the required inputs and comparison settings.
3	Starts evaluation.	4	Runs the variants and records outputs and configuration details.
Alternative Flow			
1	Selects unavailable cases or incomplete settings.	2-A	Identifies missing prerequisites without fabricating results.

4.7 Hardware and Software Requirements

Tables 4.31 and 4.32 identify the supporting resources.

4.7.1 Hardware Requirements

A development computer must support the application services, storage and the tools used to integrate the processing components. The physician interface needs a browser-capable device and an audio input source for consultation capture. Microphone access must be available when the recording workflow is used, and connectivity must support whichever processing services are selected. Storage must retain the permitted case inputs, relational records and associated semantic representations without separating them from their identifying context. Compute and memory requirements depend on the imaging and transcription models, the language-model access arrangement and the size of the case collection. Table 4.31 records these resource needs with the configuration values that remain to be established. Numerical compute, storage and concurrent-user limits must be established for the selected workload rather than assumed from a candidate model. Resource failures should be identifiable during integration checks. No resource specification alone establishes measured capacity for the integrated clinical processing workflow.

Table 4.31: Hardware resources supporting the Zeest workflow.

Resource	Purpose	Specification
Development/processing computer	Application services and selected local processing	Processor and RAM: TBD after model selection.
Physician device	Web interface and review actions	Browser-capable device; supported configurations: TBD.
Audio input	Consultation capture	Working audio source and capture permission.
Storage	Clinical inputs, records, audit actions and vectors	Capacity and retention configuration: TBD.
Graphics acceleration	Local inference where required	GPU requirement depends on the selected model.
Connectivity	Access to selected processing and evidence services	Connection requirements depend on service arrangement.

4.7.2 Software Requirements

FastAPI provides the backend integration layer, while Next.js supports the physician-facing web interface. LangGraph and Google ADK are the named orchestration technologies for coordinating the agent responsibilities, with their exact division to be agreed in the detailed design. Gemini supplies the language-model capability used within the reasoning workflow, and PyTorch supports integration of the imaging model. PostgreSQL stores patient, encounter and review information, with pgvector providing the semantic representation layer alongside relational records. A transcription component and clinical knowledge resources the dependencies needed for consultation capture, guideline retrieval and medication review. Table 4.32 connects these technologies to their responsibilities. Component versions, model identifiers, representations and access conditions must be recorded when the environment is selected. Component versions and permitted resource use must be recorded before comparative runs. Framework choices do not establish completed integration or guaranteed compatibility. Each selected dependency requires a recorded configuration and an identifiable integration failure path.

Table 4.32: Software technologies and processing dependencies used in the Zeest design.

Technology/resource	Responsibility	Configuration
FastAPI	Backend integration and application interfaces	Version and interface contracts: TBD.
Next.js	Physician-facing web application	Version: TBD.
LangGraph; Google ADK	Agent orchestration	Responsibility division: TBD.
Gemini	Language-model reasoning	Model and access configuration: TBD.
PyTorch; existing imaging model	Supported image classification	Validated model and label set: TBD.
PostgreSQL; pgvector	Relational records and semantic memory	Embedding model, dimension and versions: TBD.
Transcription component	Audio-to-notes processing	Model, languages and input formats: TBD.
Guidelines; medication knowledge resource	Evidence retrieval and medication review	Selected resources and use conditions: TBD.
MIMIC-IV; MIMIC-CXR; PubMed; Synthea; imaging candidates	Clinical/evidence data and synthetic case strategy	Permitted access and final cases/domain: TBD.

4.8 Graphical User Interface

The physician screens support registration, access to the physician's own patients, encounter review, consultation capture and clinical decisions. Related functions share pages or tabs rather than requiring a separate page for every use case. The research team uses a separate comparative-evaluation screen. Tables 4.33 and 4.34 identify each screen, user and covered use case. Figures 4.2, 4.3 and 4.4 show their navigation. Existing accounts begin at Login; registration is available for new accounts. Review branches to modification, approval or rejection, while accepted decisions and audit actions remain distinguishable. Patient and encounter context is maintained throughout clinical work. Screen captions identify the intended user and covered functionality. These images are interface mockups rather than evidence of executed authentication, recording, analysis or decisions. Empty states and illustrative references show layout only. The primary screen flow and the physician-system swimlane provide complementary views, while the research flow separates evaluation work from clinical review. Implementation testing remains future work.

Table 4.33: Physician access, patient and input screens mapped to users and use cases.

Screen	Page / state	User	Use cases
G01	Sign Up (Figure 4.5)	Physician	UC01
G02	Log In (Figure 4.6)	Physician	UC02, UC21
G03	My Patients (Figure 4.7)	Physician	UC03, UC21
G04	Patient Details (Figure 4.8)	Physician	UC04–UC06
G05	Consultation (Figure 4.9)	Physician	UC07–UC08
G06	Imaging Input and Findings (Figure 4.10)	Physician	UC09, UC12–UC13
G07	Laboratory Input and Findings (Figure 4.11)	Physician	UC10, UC12–UC13
G08	Medication Input and Findings (Figure 4.12)	Physician	UC11, UC12–UC13

Table 4.34: Review, decision, audit and research screens mapped to users and use cases.

Screen	Page / state	User	Use cases
G09	Clinical Review and Evidence (Figure 4.13)	Physician	UC13–UC14
G10	Critic Findings (Figure 4.14)	Physician	UC15
G11	Modify Clinical Draft (Figure 4.15)	Physician	UC16
G12	Approval Confirmation (Figure 4.16)	Physician	UC17
G13	Rejection Confirmation (Figure 4.17)	Physician	UC18
G14	Approved Decisions (Figure 4.18)	Physician	UC19
G15	Audit History (Figure 4.19)	Physician	UC20
G16	Comparative Evaluation (Figure 4.20)	Research team	UC22

4.9 Database Design

The logical design separates physician accounts, patient ownership, clinical information, generated drafts and accepted decisions. Each patient has one responsible physician, and encounters inherit that ownership through the patient. Clinical case inputs retain saved revisions, allowing findings and draft preparation to identify their exact source content. Draft revisions keep diagnosis and treatment text distinct from approval actions and accepted decisions. Retrieved evidence and critic reviews refer to the assessed revision. Semantic representations originate from an input revision or an approved decision. Separate evaluation records connect permitted cases, run settings and recorded outputs. These relationships preserve continuity, source traceability and physician authority across the defined clinical workflow. The schema is a logical design rather than an implemented database. Patient ownership and revision consistency require explicit validation alongside foreign keys. Model-dependent vector dimensions remain configuration choices, and evaluation outputs represent records to be populated after actual runs rather than supplied numerical findings.

4.9.1 ER Diagram

Figure 4.21 presents the complete conceptual model with entities, relationship diamonds and attribute ovals together. Dashed regions group accounts and clinical information, drafts and supporting evidence, physician review and memory, and comparative evaluation. Blue rectangles identify entities, green ovals identify attributes, and amber diamonds identify named relationships. Primary keys are underlined and foreign keys are marked FK. Cardinalities express ownership and record associations, while revision keys preserve the exact content used or reviewed. Figures 4.22, 4.23, 4.24 and 4.25 enlarge the same regions; cross-region relationships remain visible in the complete model. The grouped enlargements retain the same entity and relationship names as the overview, allowing attributes and cardinalities to be read at print size. Login and registration are account operations rather than separate persistent clinical entities. Evaluation links remain separate from clinical decisions, so a comparative output cannot be-

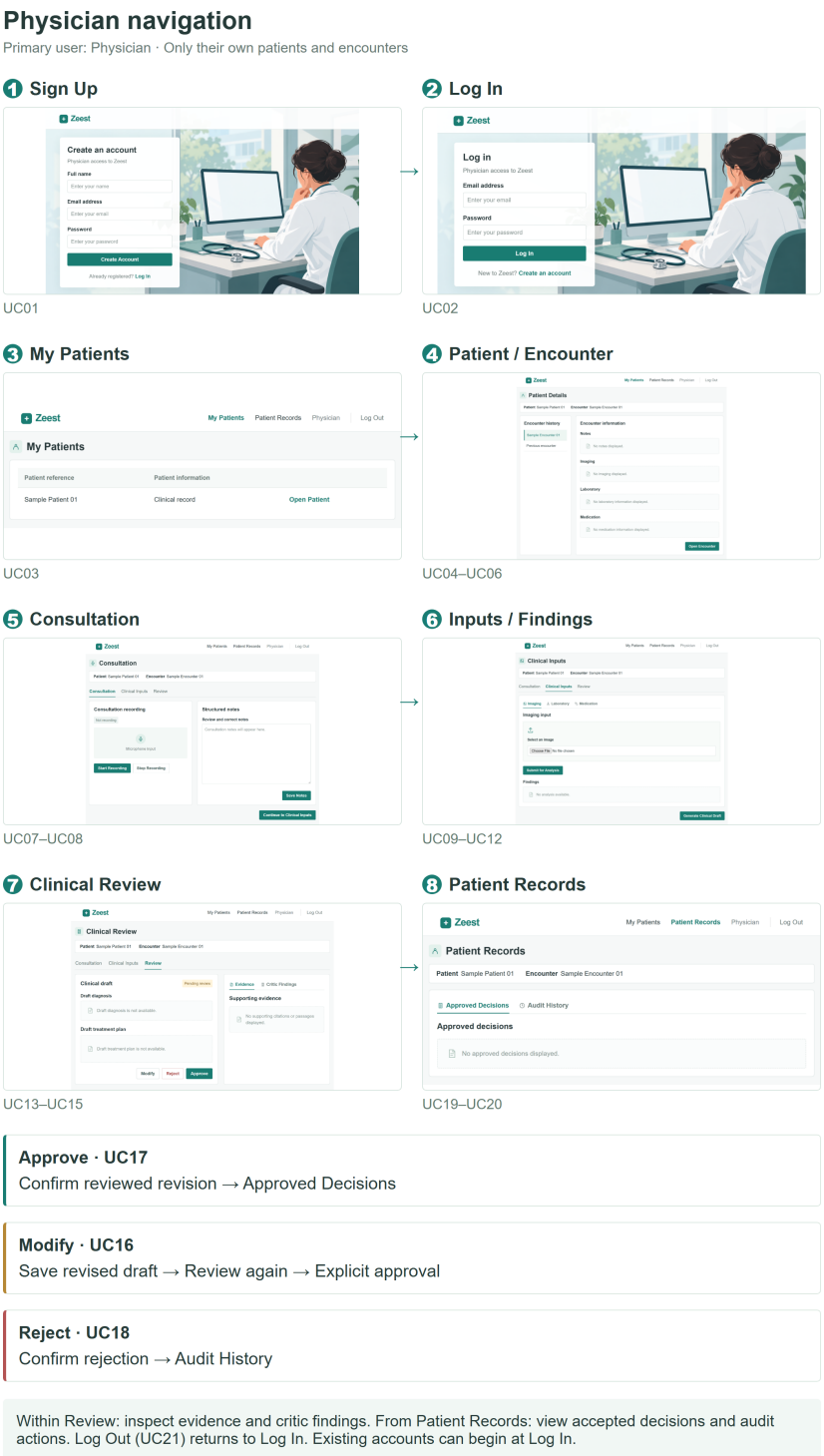


Figure 4.2: Primary physician navigation through the numbered screen thumbnails. Review branches to modification, approval or rejection; Patient Records exposes approved decisions and audit history. Logout returns to Login.

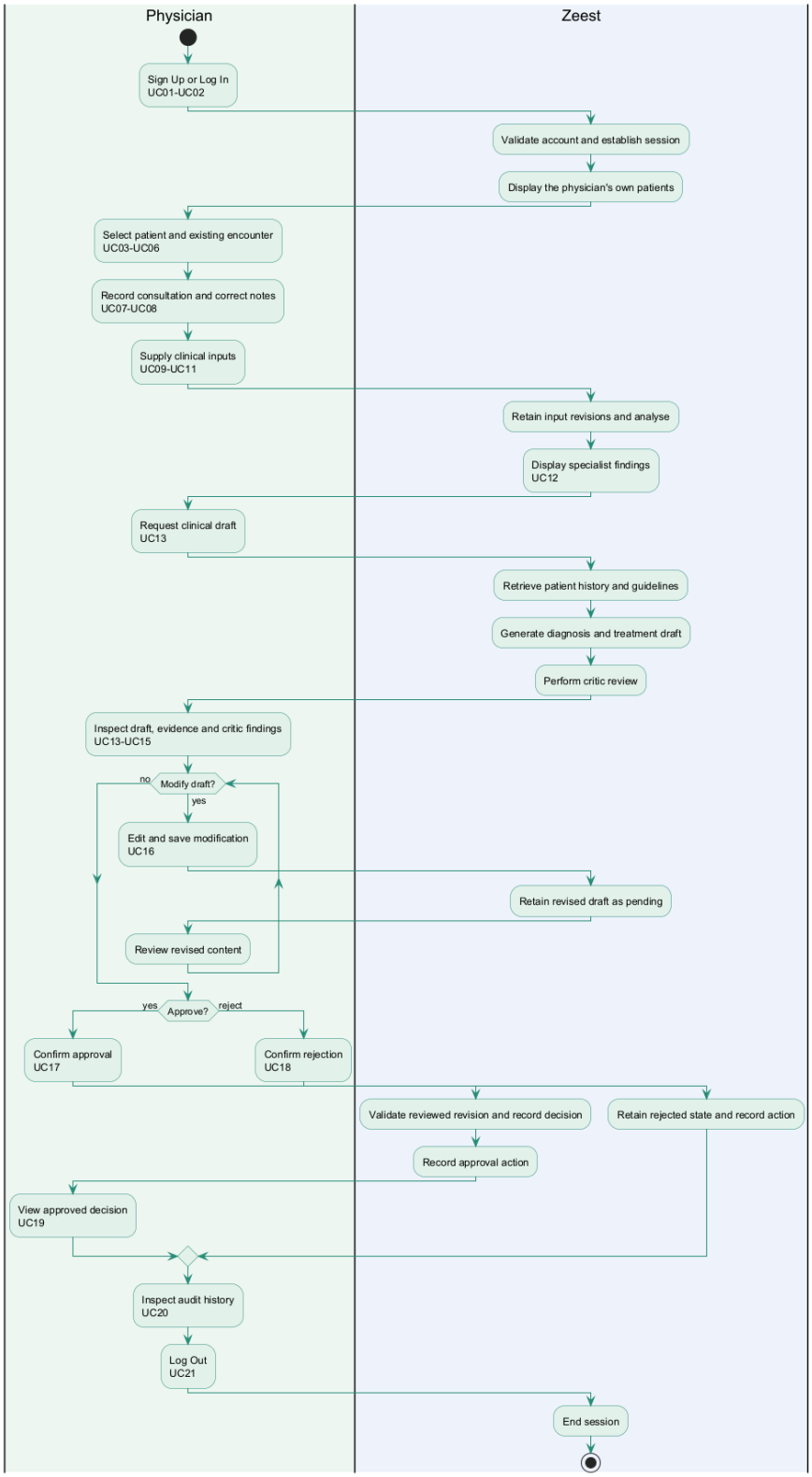


Figure 4.3: Physician–Zeest swimlane navigation retained alongside the screen-thumbnail flow. Physician actions appear in the left lane and system processing in the right, including ownership checks, clinical analysis, draft modification, explicit approval or rejection, audit review and logout.



Figure 4.4: Research-team navigation for UC22: select permitted cases, configure variants, validate and run evaluation, then inspect outputs associated with each case and configuration.

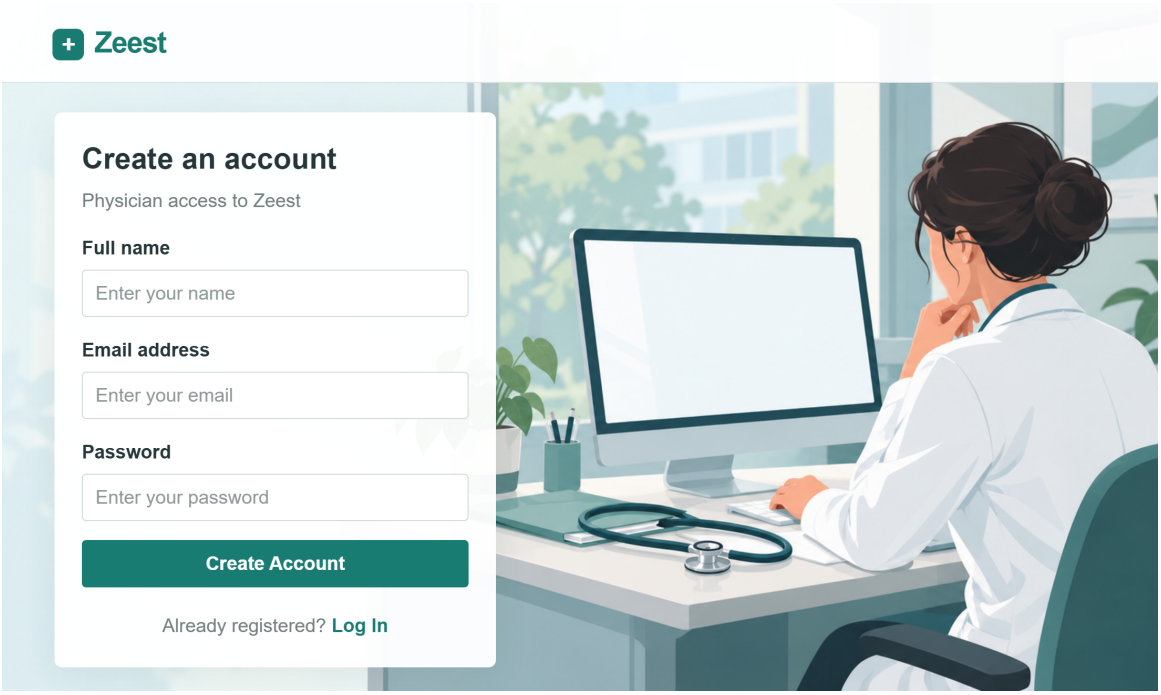


Figure 4.5: Physician registration form with name, email and password fields (UC01).

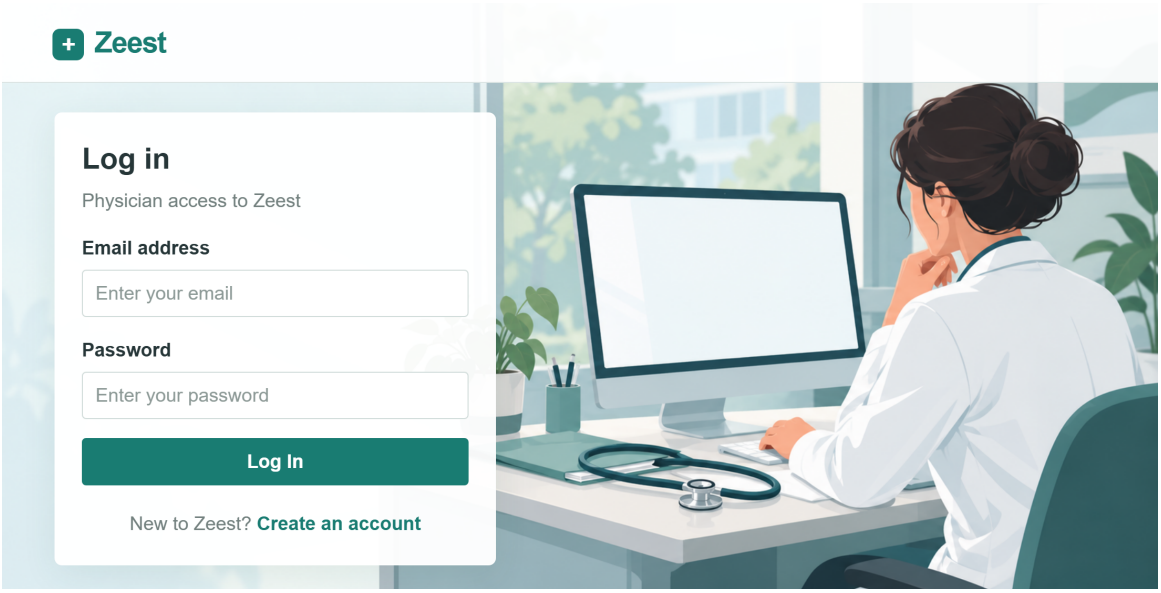


Figure 4.6: Physician login form and account-access controls (UC02, UC21).

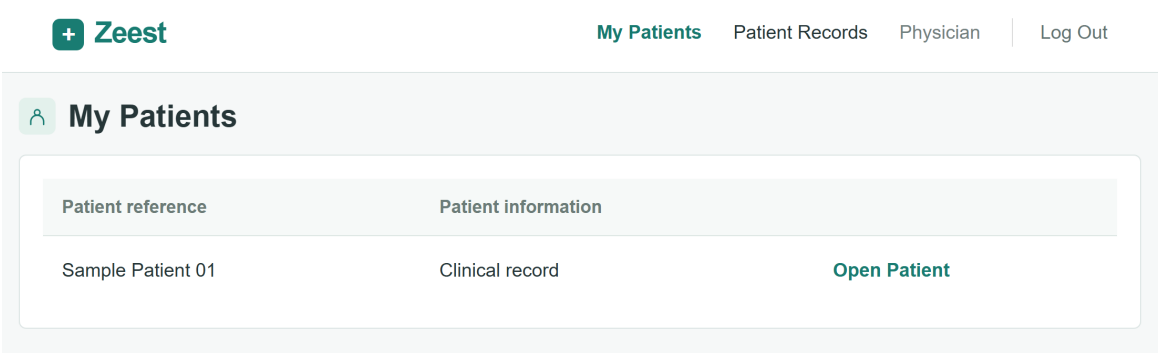


Figure 4.7: Physician patient list and selection controls (UC03, UC21).

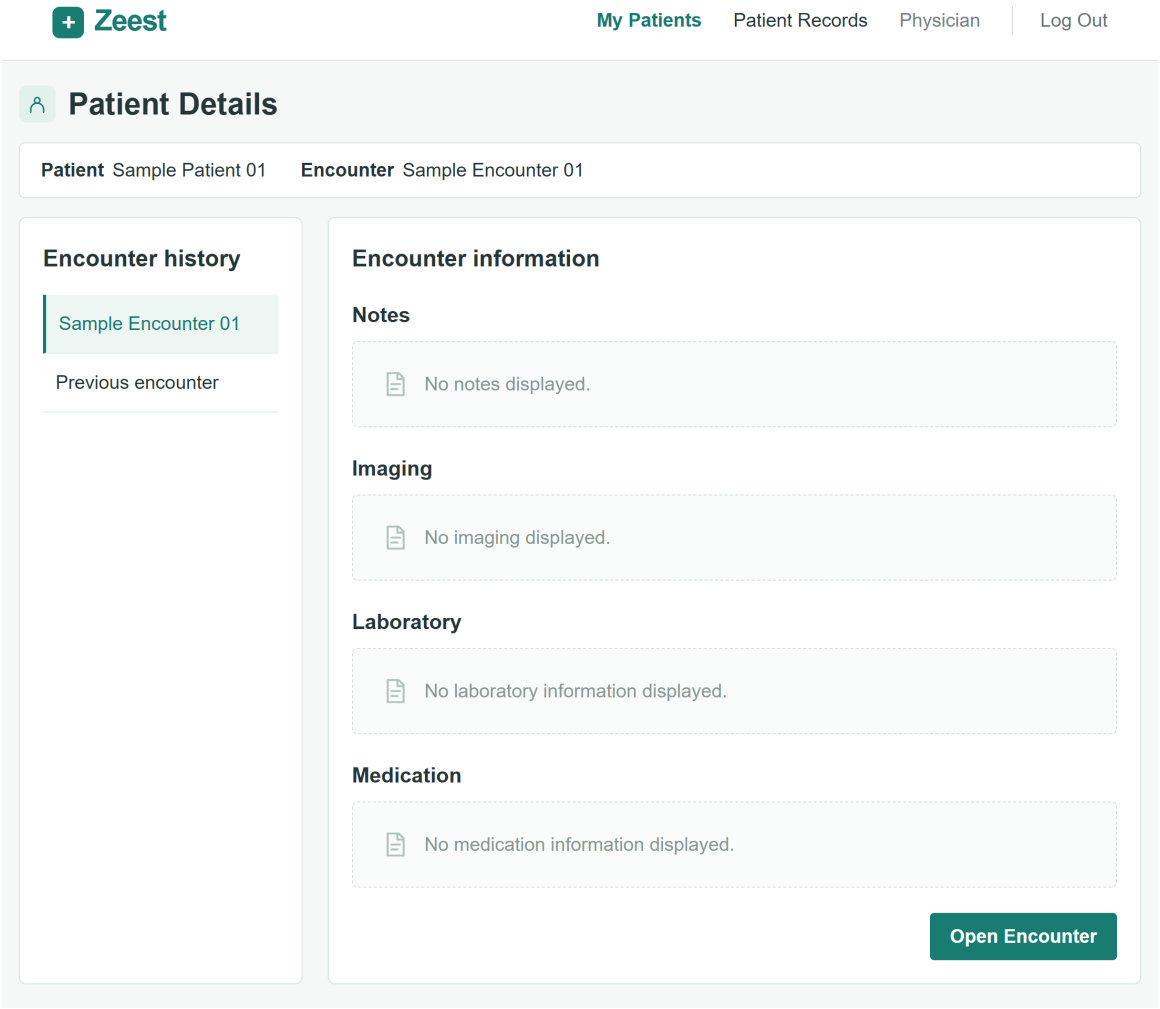


Figure 4.8: Patient context, history and encounter selection (Physician, UC04–UC06).

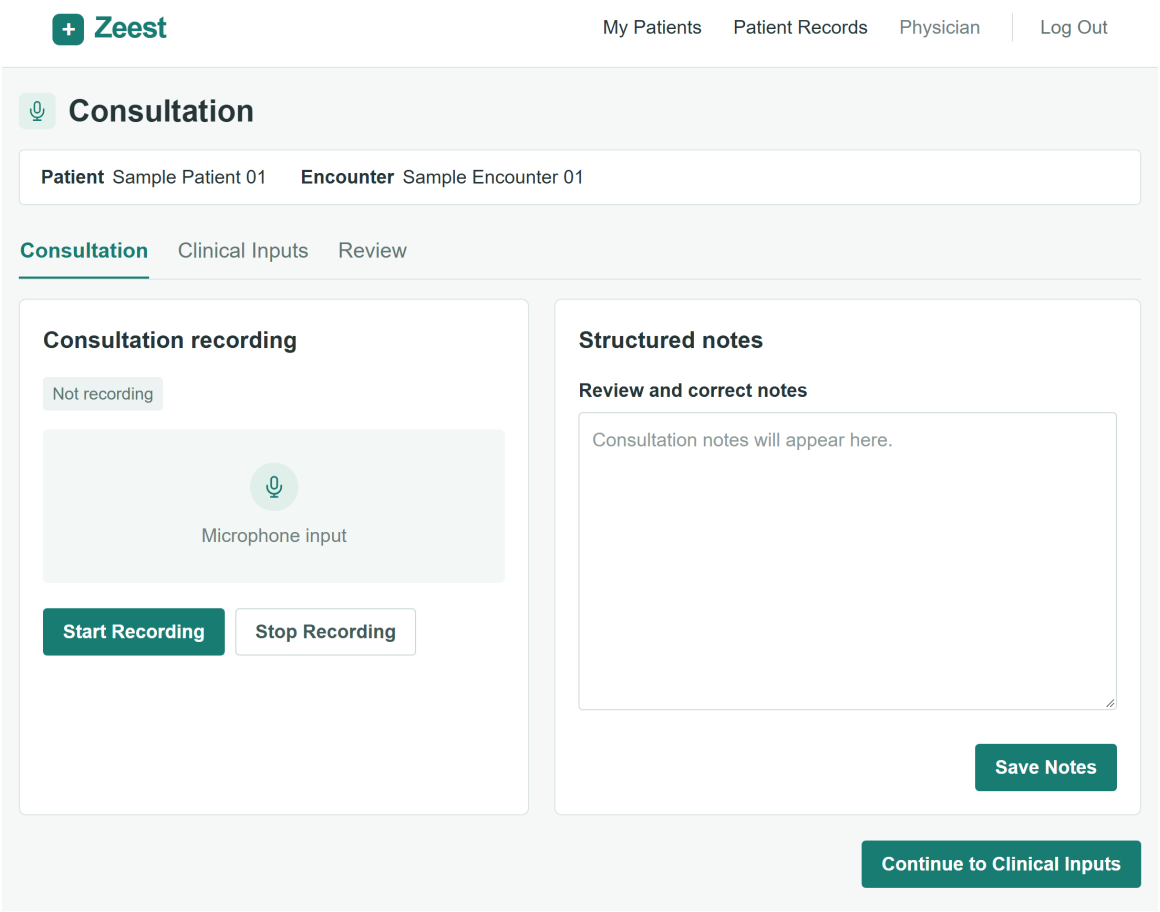


Figure 4.9: Consultation recording controls and editable notes (Physician, UC07–UC08).

come an approved patient record merely because it was saved. Region enlargements preserve the same underlying schema.

4.9.2 Data Dictionary

Tables 4.35 onward define the entities, attributes, data types, constraints and meanings used by the logical schema. Primary keys identify records, composite keys preserve associations, and nullable fields accommodate absent values only where their processing state permits. Each patient's physician association governs access to encounters, case inputs, drafts, actions and decisions. Corrections create input revisions rather than replacing saved content; draft evidence, recorded critic findings and source links identify the exact revision they support. Approval validates the reviewed revision and physician action together, records matching accepted content, and permits at most one decision for that revision. Modified or rejected drafts remain outside accepted decisions. Semantic memory links authorised input revisions or approved decisions to their vector representations, with model-dependent dimensions configured after model selection. Evaluation fields describe case definitions, configurations and outputs; numerical measurements are populated only after an executed run. These relationships maintain source

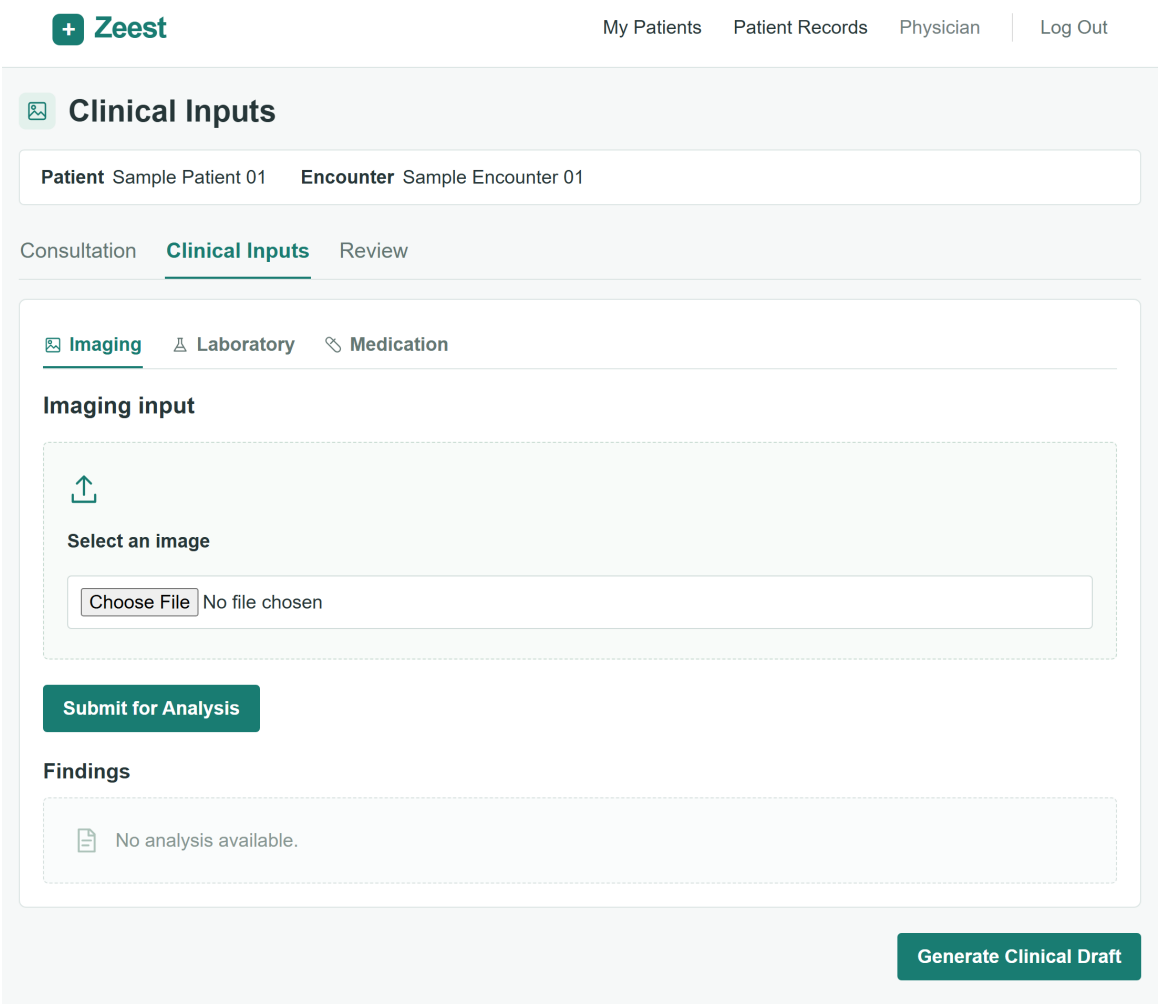


Figure 4.10: Imaging upload and findings area (Physician, UC09, UC12).

traceability without inventing missing clinical information.

Table 4.35: Physician attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`physician_id`	UUID	PK; not null	Unique physician account.
`full_name`	text	Required; not null	Name supplied at registration.
`email`	text	Unique; not null	Normalised registration and login email.
`password_hash`	text	Required; not null	Salted hash; plaintext passwords are never stored.
`created_at`	timestamptz	Required; not null	Account creation time.

Table 4.36: Patient attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`patient_id`	UUID	PK; not null	Unique patient record.
`physician_id`	UUID	FK; not null	Physician who owns the record.
`patient_ref`	text	Unique; not null	De-identified or synthetic reference.
`context_text`	text	Optional; nullable	Available patient context.

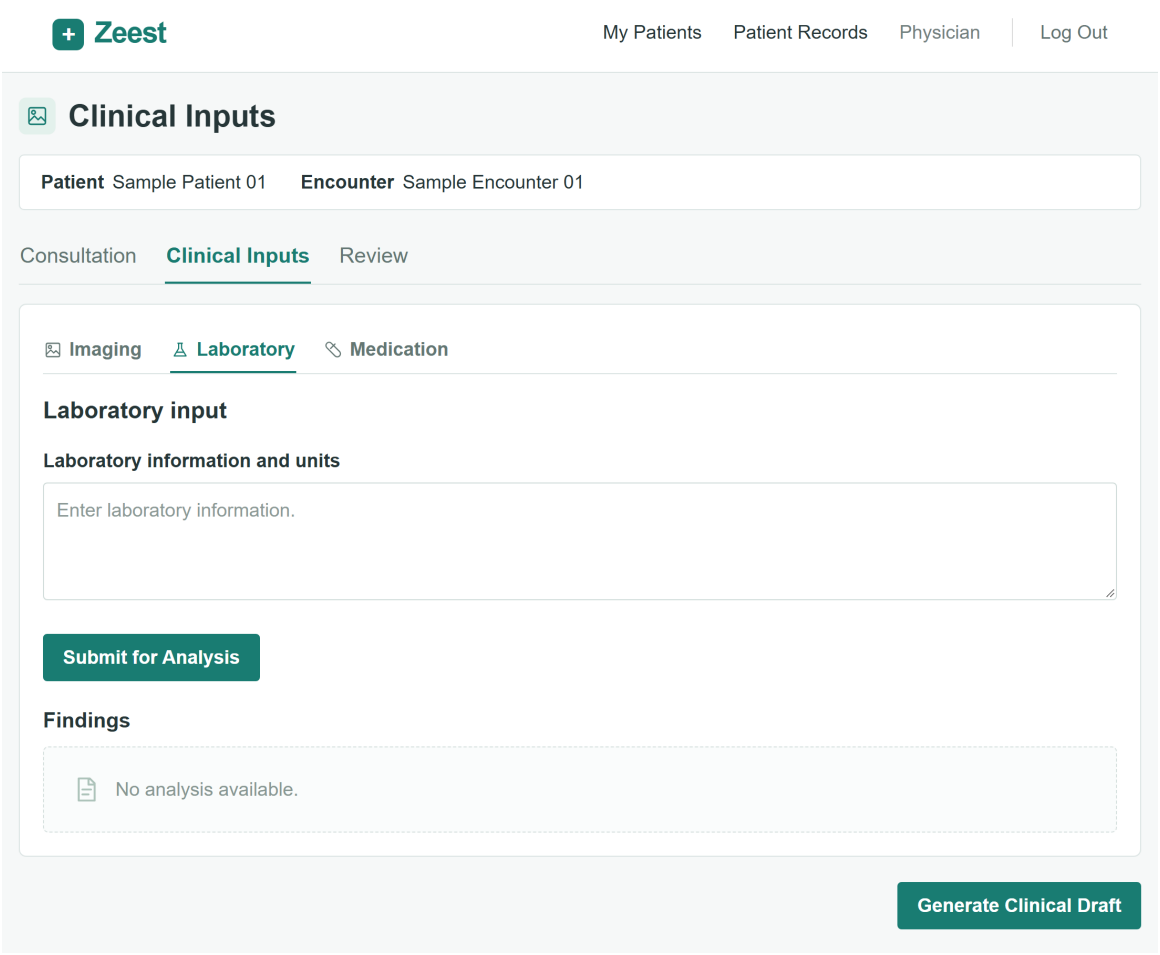


Figure 4.11: Laboratory input and findings area (Physician, UC10, UC12).

Table 4.37: Encounter attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`encounter_id`	UUID	PK; not null	Unique encounter.
`patient_id`	UUID	FK; not null	Patient associated with this encounter.
`recorded_at`	timestamptz	Required; not null	Encounter date and time.

Table 4.38: ClinicalInput attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`input_id`	UUID	PK; not null	Stable identifier of the supplied input.
`encounter_id`	UUID	FK; not null	Encounter where the input originates.
`input_kind`	text	Check; not null	Notes, imaging, laboratory or medication.

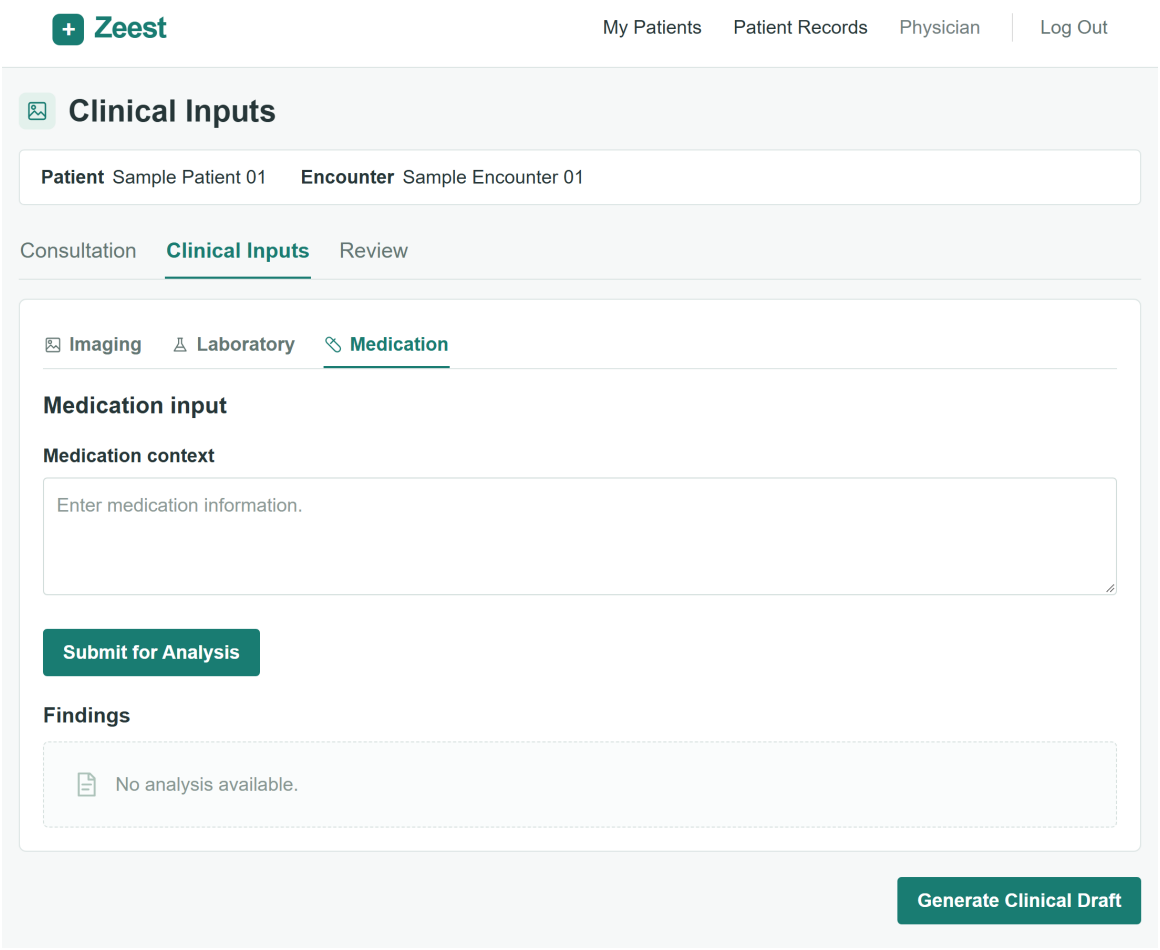


Figure 4.12: Medication input and findings area (Physician, UC11, UC12).

Table 4.39: InputRevision attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`input_id`	UUID	PK/FK; not null	Clinical input being versioned.
`revision_no`	integer	PK; >0; not null	Saved revision of this input.
`content_text`	text	Check; not null	Text content where applicable.
`file_ref`	text	Check; not null	Stored file reference where applicable.
`recorded_at`	timestamptz	Required; not null	Revision recording time.

Table 4.40: SpecialistFinding attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`finding_id`	UUID	PK; not null	Unique analysis finding or processing record.
`input_id`	UUID	FK pair; not null	Source clinical input.
`input_revision`	integer	FK pair; not null	Exact analysed input revision.
`agent_kind`	text	Check; not null	Imaging, laboratory or medication agent.
`model_ref`	text	Optional; nullable	Model/resource identifier where applicable.
`finding_text`	text	Check; not null	Available specialist findings.
`processing_status`	text	Check; not null	Pending, complete or failed.
`processed_at`	timestamptz	Optional; nullable	Completion or failure time.

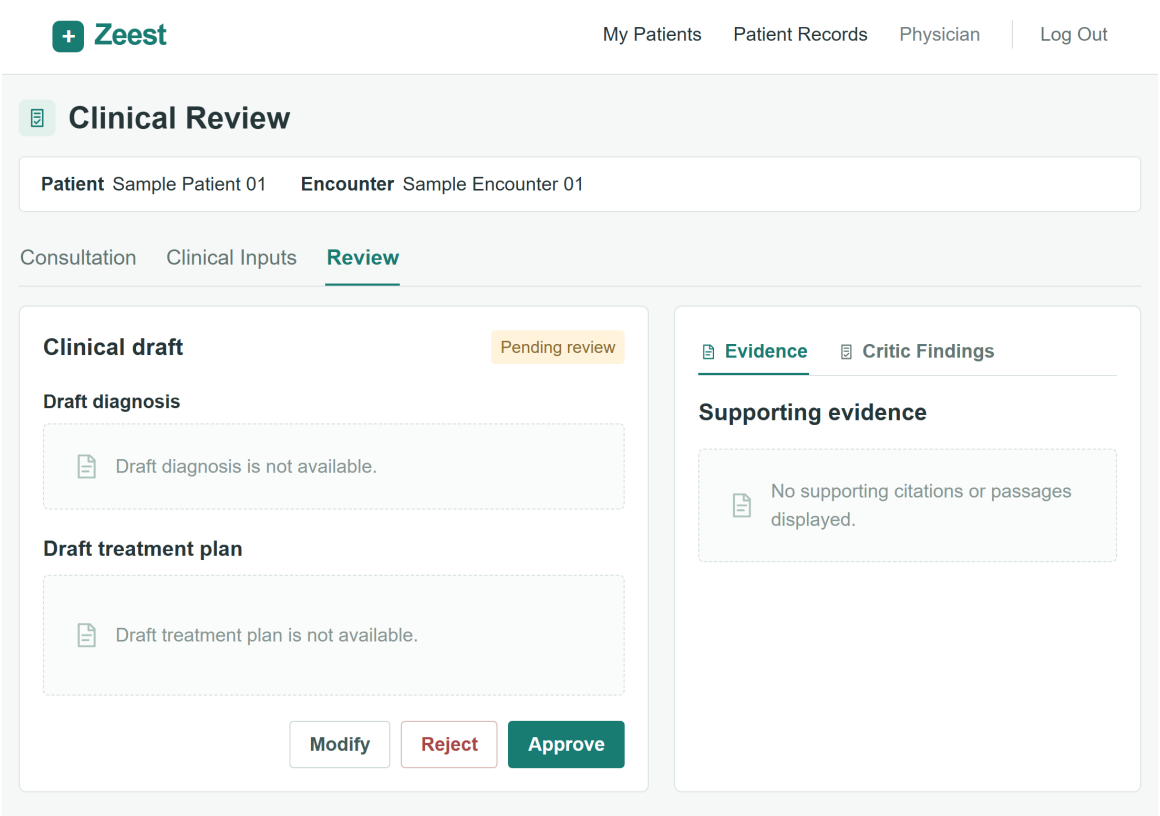


Figure 4.13: Clinical draft and supporting evidence tabs (Physician, UC13–UC14).

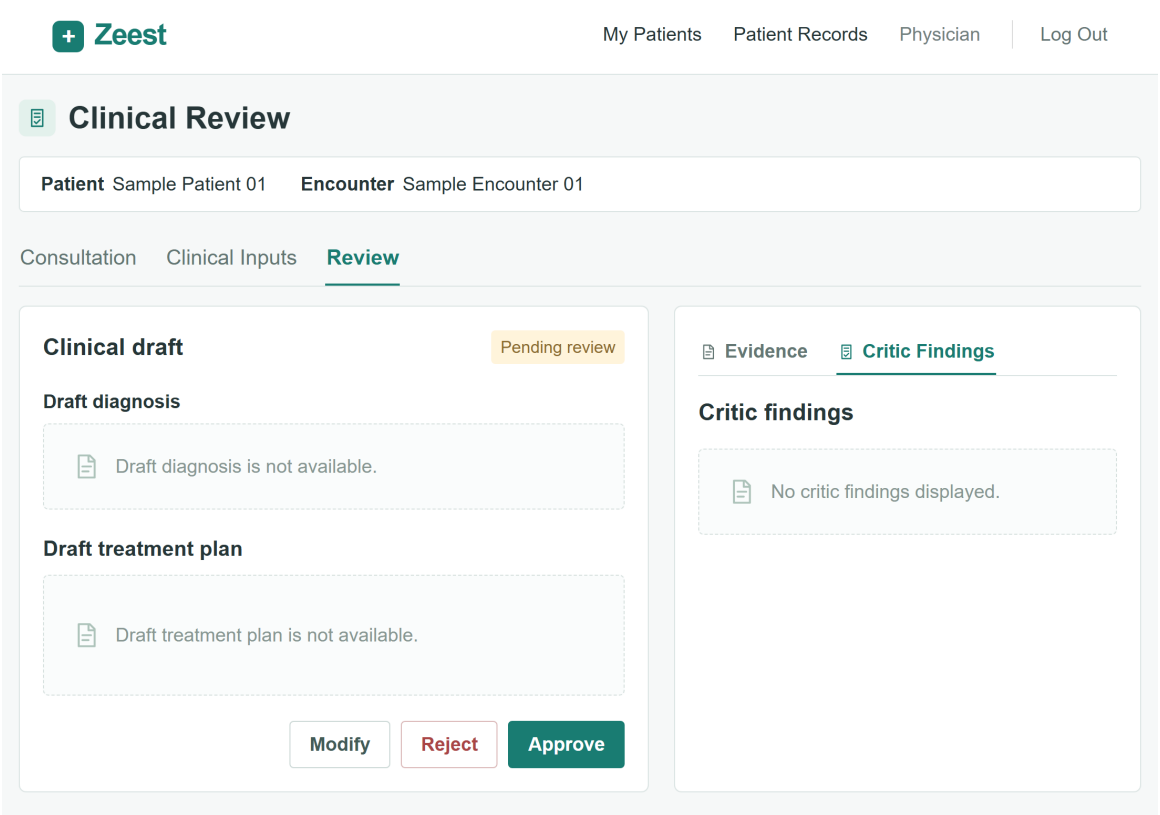


Figure 4.14: Critic findings beside the clinical draft (Physician, UC15).

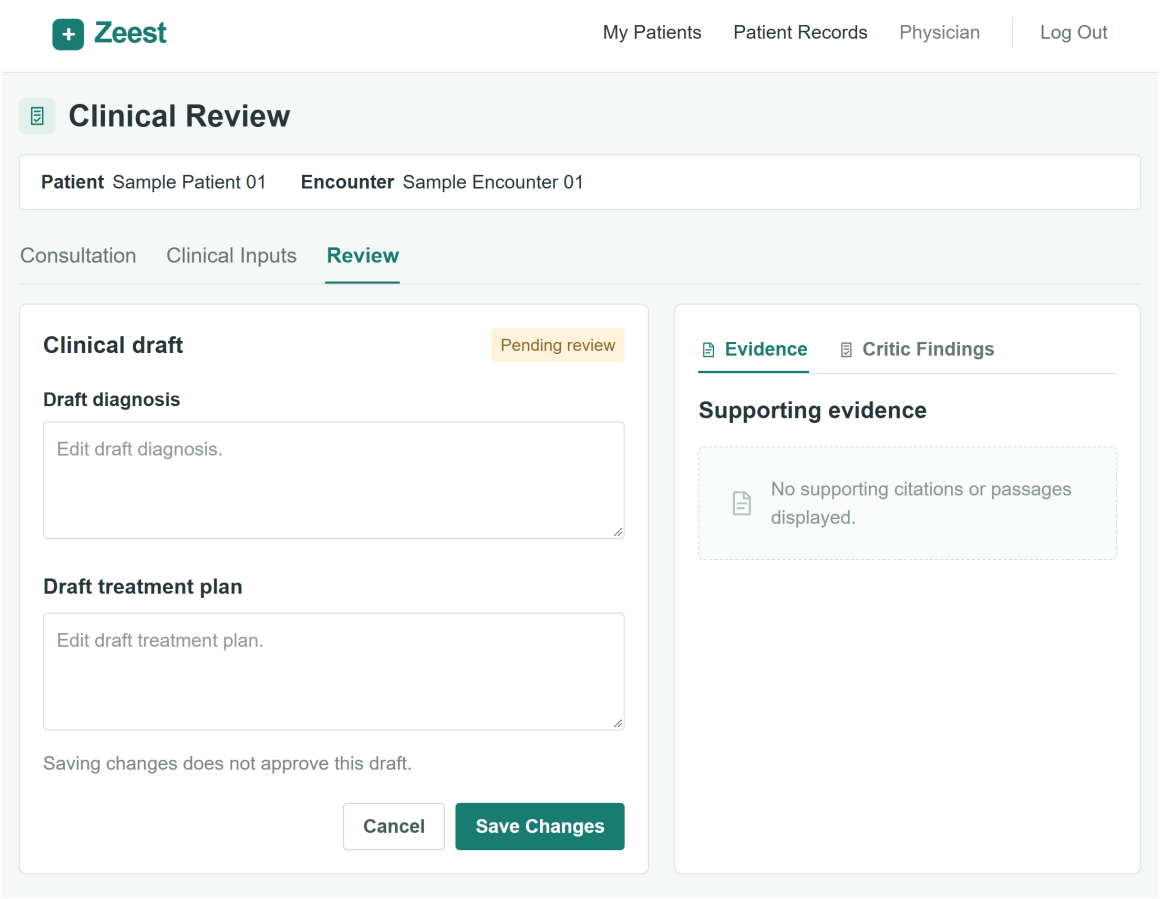


Figure 4.15: Draft revision editor and save controls (Physician, UC16).

Table 4.41: Proposal attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`proposal_id`	UUID	PK; not null	Stable identifier of a clinical draft.
`encounter_id`	UUID	FK; not null	Encounter for which the draft is generated.
`created_at`	timestamptz	Required; not null	Draft creation time.

Table 4.42: ProposalRevision attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`proposal_id`	UUID	PK/FK; not null	Draft being versioned.
`revision_no`	integer	PK; >0; not null	Saved draft revision.
`diagnosis_text`	text	Required; not null	Generated or physician-edited diagnosis draft.
`treatment_text`	text	Required; not null	Generated or physician-edited treatment draft.
`review_status`	text	Check; not null	Pending, approved or rejected.
`created_at`	timestamptz	Required; not null	Time this revision is saved.

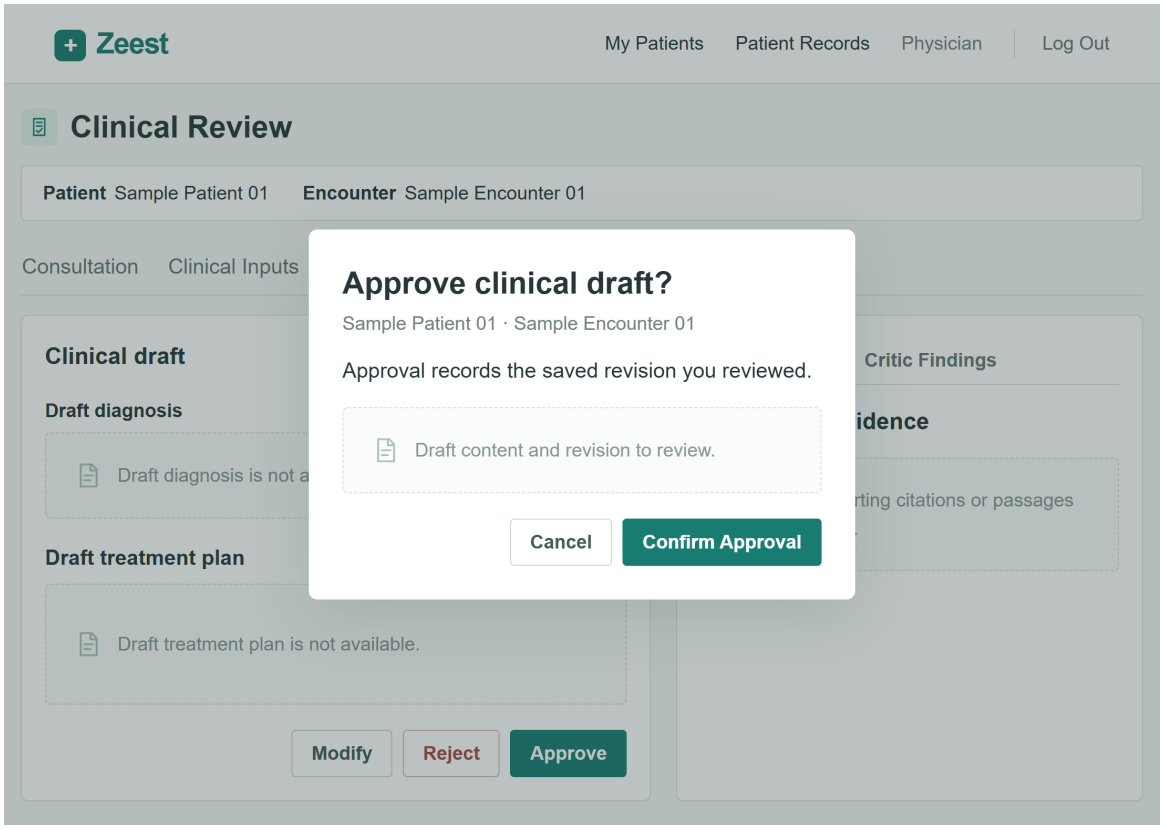


Figure 4.16: Confirmation dialog for explicit draft approval (Physician, UC17).

Table 4.43: ProposalInput attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`proposal_id`	UUID	PK/FK pair; not null	Draft using the input.
`proposal_revision`	integer	PK/FK pair; not null	Exact draft revision.
`input_id`	UUID	PK/FK pair; not null	Current or historical source input.
`input_revision`	integer	PK/FK pair; not null	Exact source input revision used.

Table 4.44: CriticReview attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`critic_id`	UUID	PK; not null	Unique critic-review operation.
`proposal_id`	UUID	FK pair; not null	Assessed clinical draft.
`proposal_revision`	integer	FK pair; not null	Exact assessed revision.
`review_status`	text	Check; not null	Pending, complete or failed.
`findings_text`	text	Check; not null	Safety and consistency findings when complete.
`reviewed_at`	timestamptz	Optional; nullable	Time the review completes or fails.

Table 4.45: EvidenceItem attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`evidence_id`	UUID	PK; not null	Unique retrieved evidence item.
`source_ref`	text	Required; not null	Citation or source reference.
`passage_text`	text	Required; not null	Retrieved supporting passage.
`retrieved_at`	timestamptz	Required; not null	Evidence retrieval time.

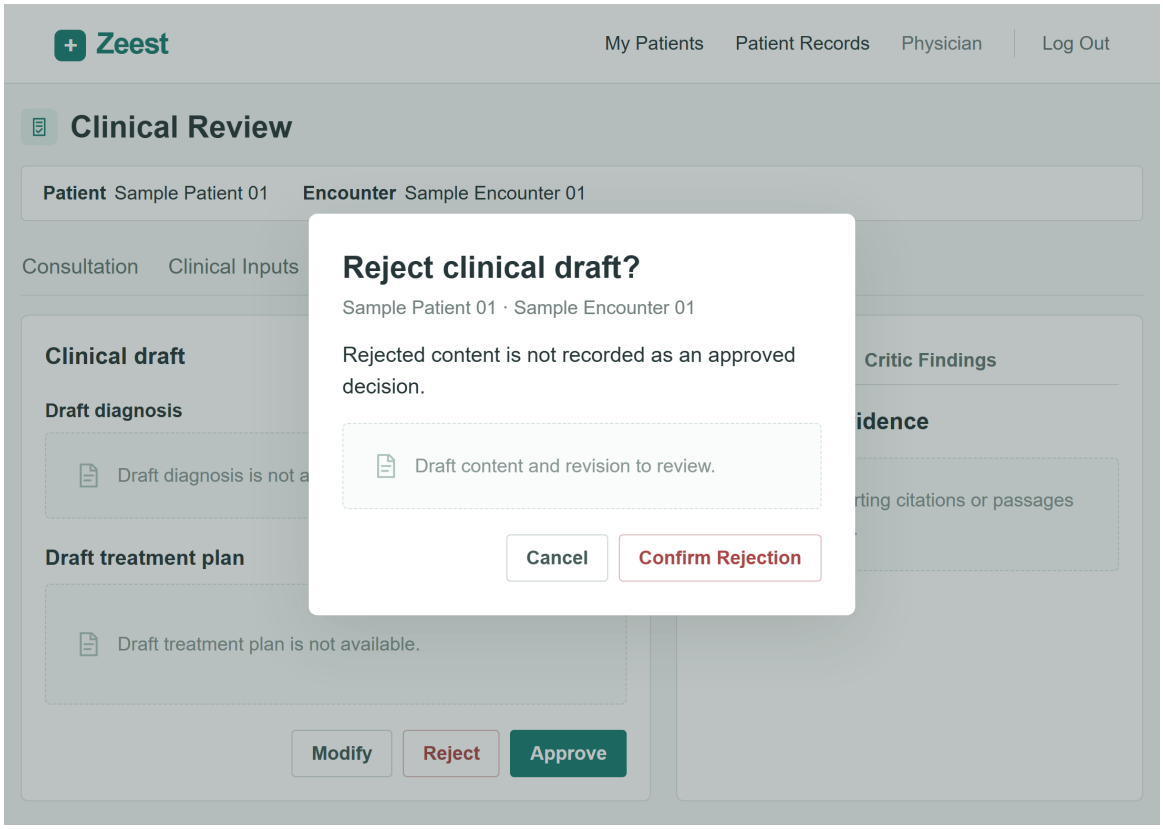


Figure 4.17: Confirmation dialog for draft rejection (Physician, UC18).

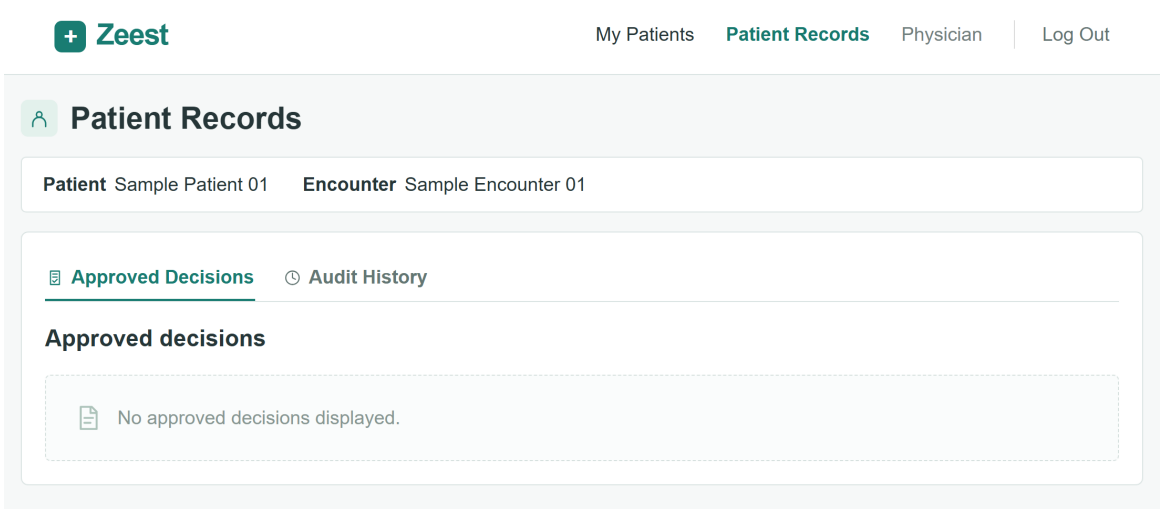


Figure 4.18: Selected encounter approved-decision view (Physician, UC19).

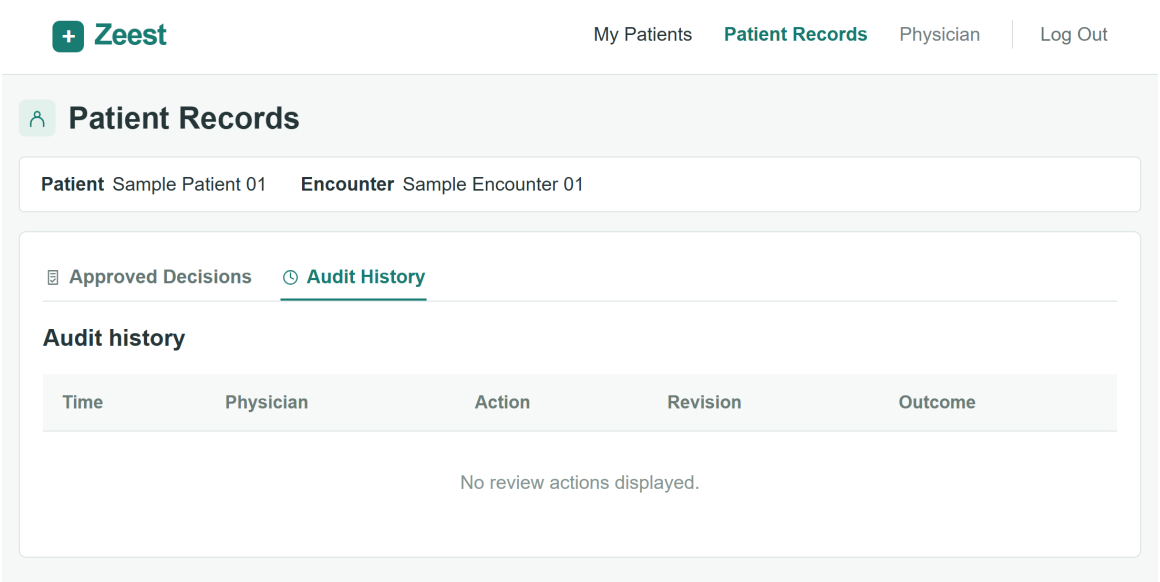


Figure 4.19: Encounter review-action history table (Physician, UC20).

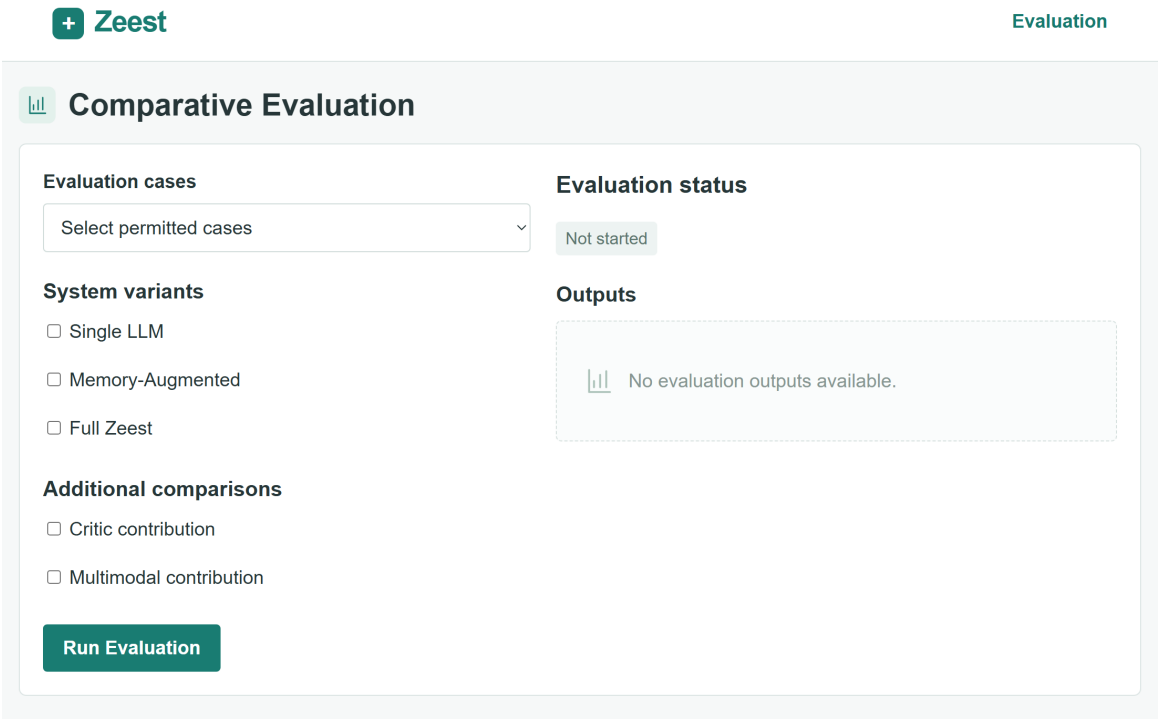


Figure 4.20: Case, variant and run controls for comparative evaluation (Research team, UC22).

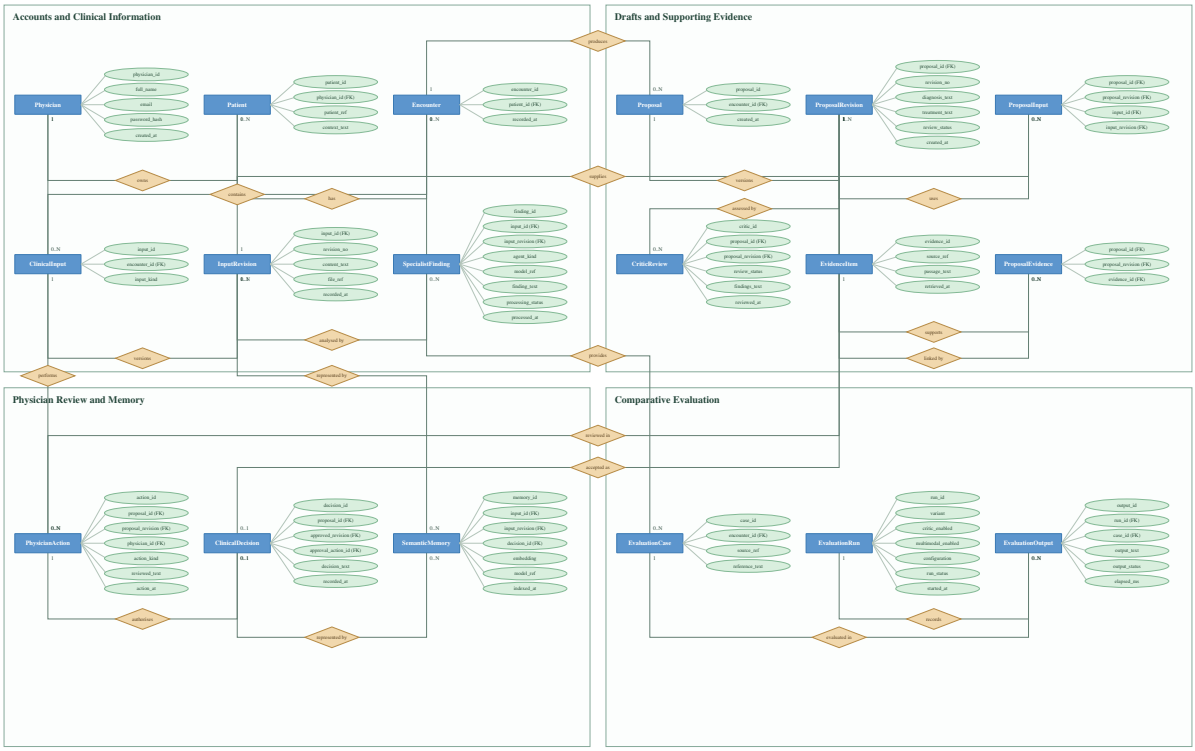


Figure 4.21: Complete grouped conceptual ER model with entity, attribute and relationship symbols.

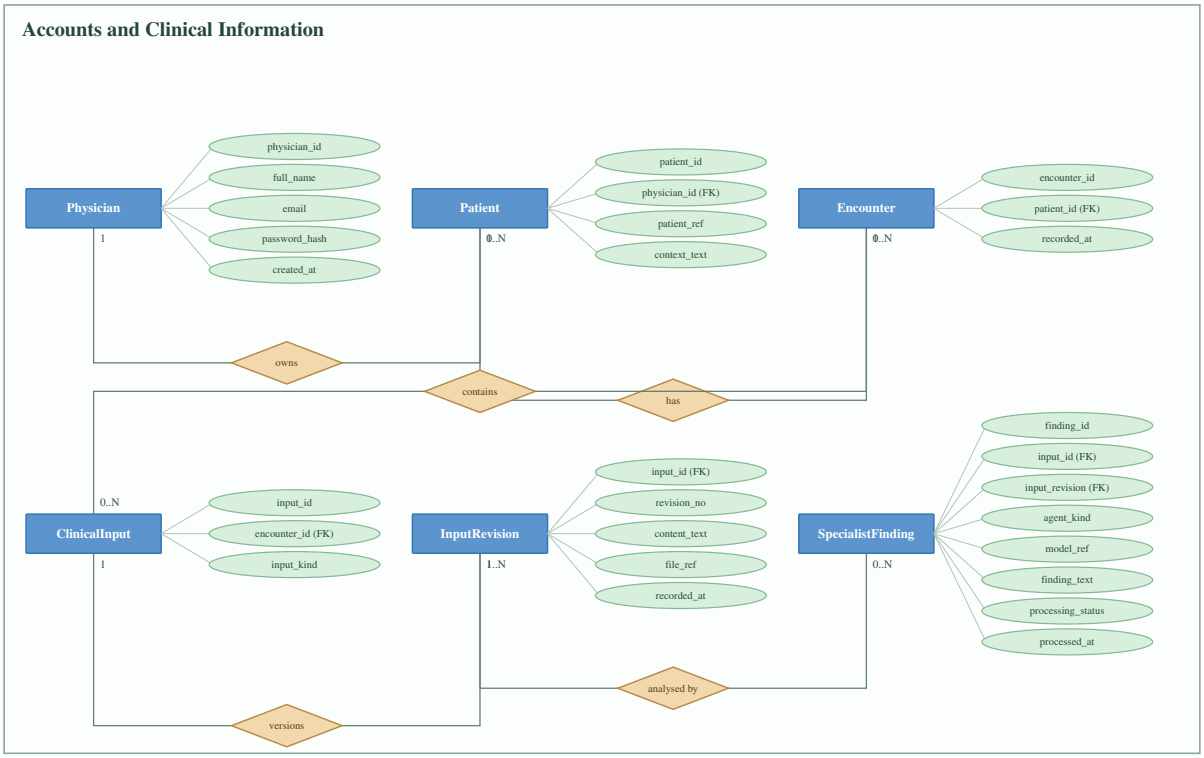


Figure 4.22: Accounts and clinical sources with attributes and relationship cardinalities.

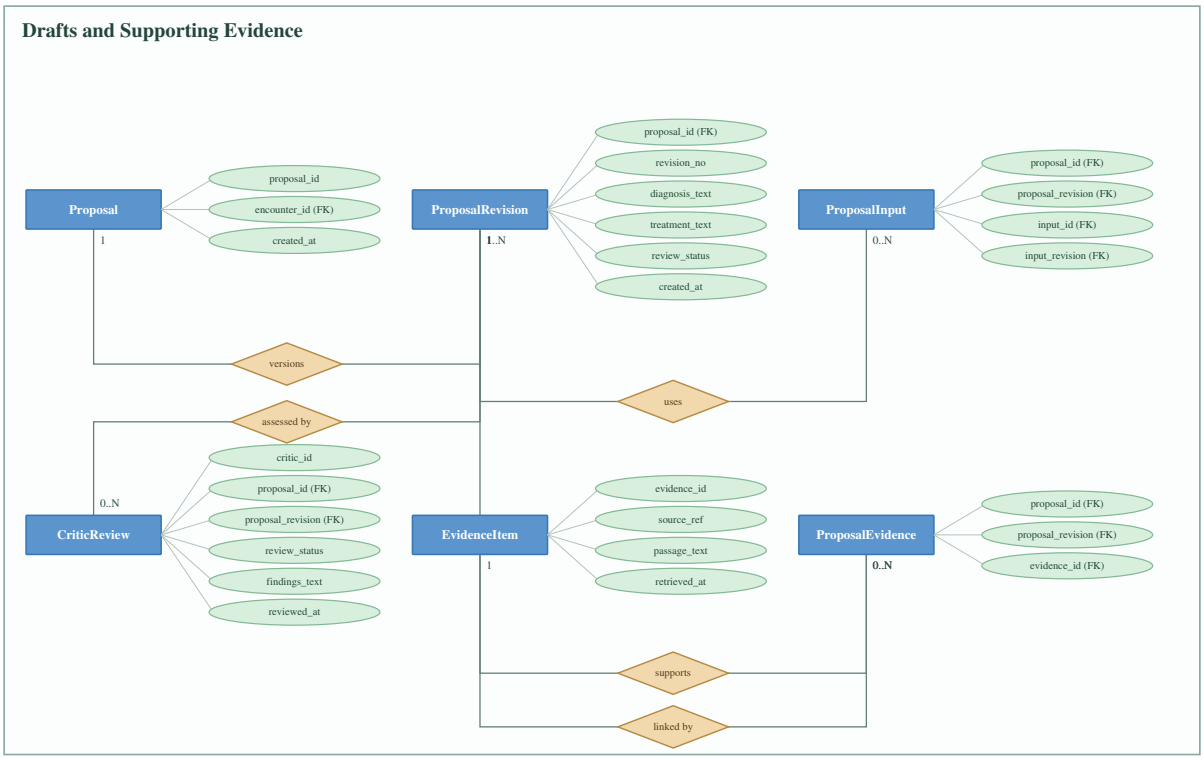


Figure 4.23: Draft revisions, evidence and critic review with attributes and relationships.

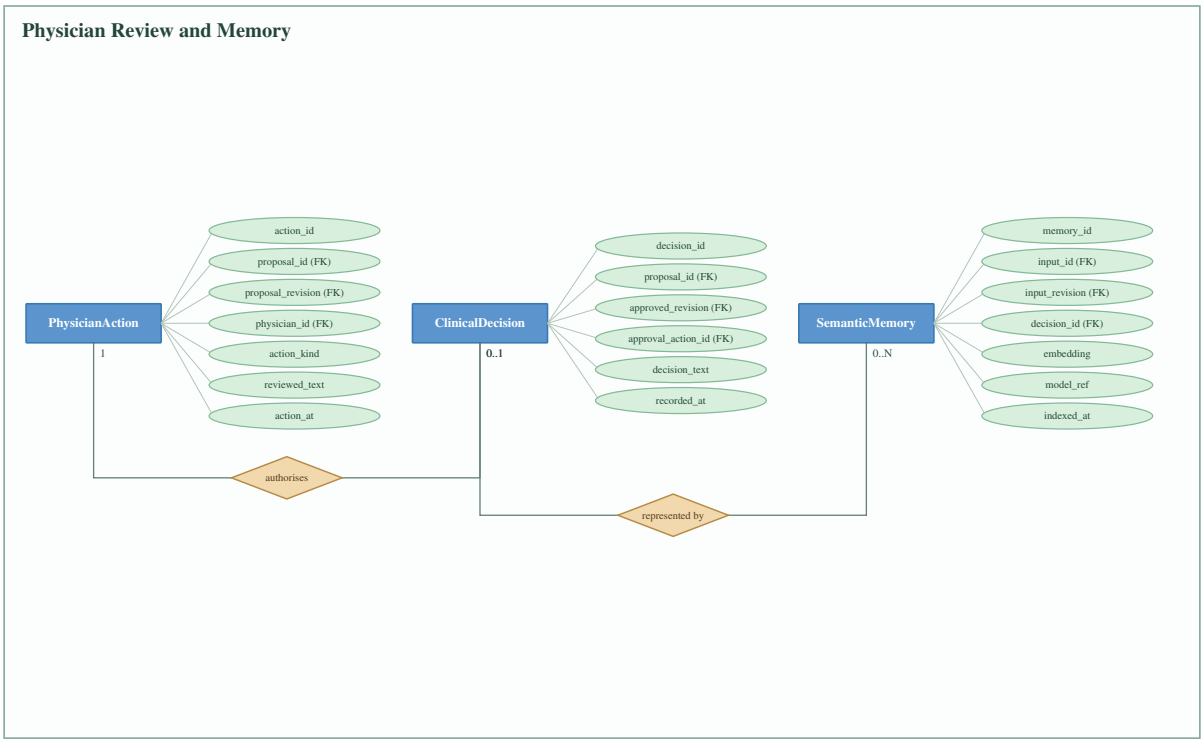


Figure 4.24: Physician actions, accepted decisions and semantic memory relationships.

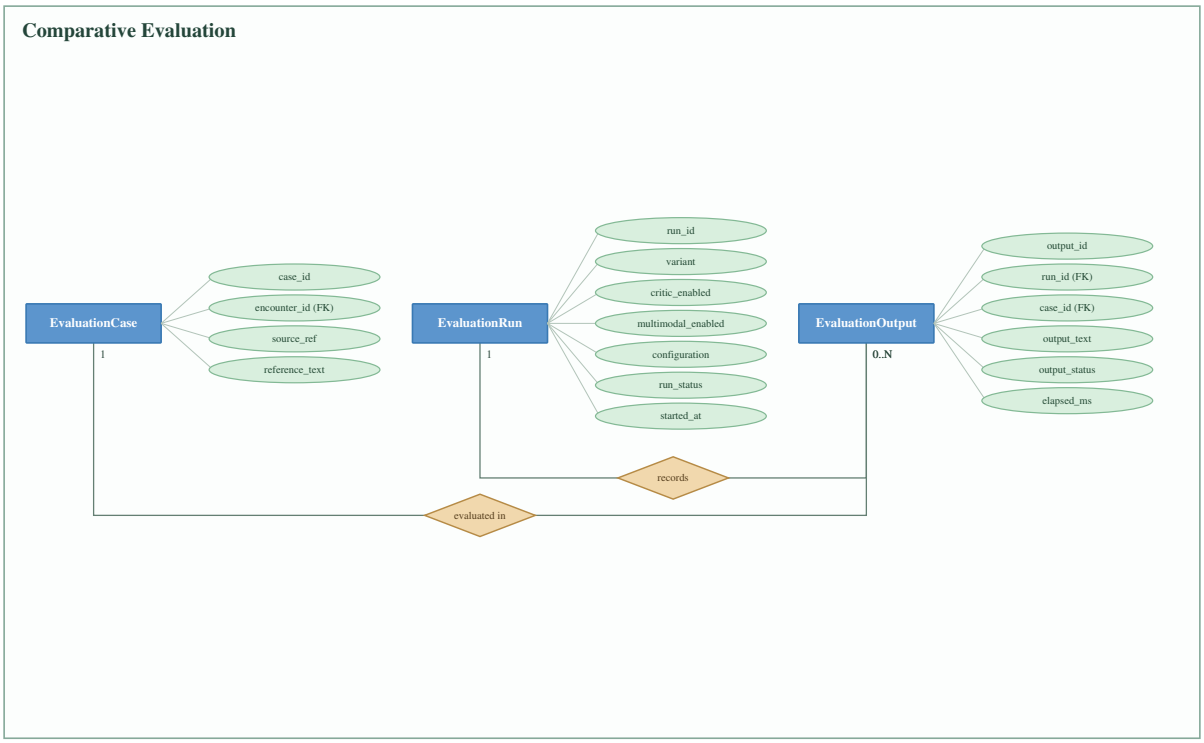


Figure 4.25: Evaluation cases, configurations and recorded output relationships.

Table 4.46: ProposalEvidence attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`proposal_id`	UUID	PK/FK pair; not null	Draft supported by evidence.
`proposal_revision`	integer	PK/FK pair; not null	Exact supported revision.
`evidence_id`	UUID	PK/FK; not null	Retrieved evidence item.

Table 4.47: PhysicianAction attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`action_id`	UUID	PK; not null	Unique physician review action.
`proposal_id`	UUID	FK pair; not null	Clinical draft being reviewed.
`proposal_revision`	integer	FK pair; not null	Exact reviewed revision.
`physician_id`	UUID	FK; not null	Physician performing the action.
`action_kind`	text	Check; not null	Modify, approve or reject.
`reviewed_text`	text	Required; not null	Snapshot of the content associated with the action.
`action_at`	timestamptz	Required; not null	Action recording time.

Table 4.48: ClinicalDecision attributes, types and constraints.

Attribute	Type	Constraints	Meaning
`decision_id`	UUID	PK; not null	Unique approved clinical decision.
`proposal_id`	UUID	FK pair; not null	Originating clinical draft.
`approved_revision`	integer	FK pair; unique pair; not null	Explicitly accepted draft revision.
`approval_action_id`	UUID	Unique FK; not null	Matching physician approval action.
`decision_text`	text	Required; not null	Exact accepted diagnosis and treatment content.
`recorded_at`	timestamptz	Required; not null	Decision recording time.

Table 4.49: SemanticMemory attributes, types and constraints.

Attribute	Type	Constraints	Meaning
memory_id	UUID	PK; not null	Unique semantic representation.
input_id	UUID	FK pair; not null	Input source when applicable.
input_revision	integer	FK pair; not null	Exact input-source revision.
decision_id	UUID	FK; not null	Approved-decision source when applicable.
embedding	vector(TBD)	Required; not null	Vector using the selected embedding dimension.
model_ref	text	Required; not null	Embedding model identifier and version.
indexed_at	timestamptz	Required; not null	Indexing time.

Table 4.50: EvaluationCase attributes, types and constraints.

Attribute	Type	Constraints	Meaning
case_id	UUID	PK; not null	Unique permitted evaluation case.
encounter_id	UUID	FK; not null	Case encounter supplying clinical context.
source_ref	text	Required; not null	Dataset or synthetic-case provenance.
reference_text	text	Optional; nullable	Reference assessment when established.

Table 4.51: EvaluationRun attributes, types and constraints.

Attribute	Type	Constraints	Meaning
run_id	UUID	PK; not null	Unique configured research run.
variant	text	Check; not null	Single LLM, memory-augmented or full Zeest.
critic_enabled	boolean	Required; not null	Declared critic configuration.
multimodal_enabled	boolean	Required; not null	Declared multimodal configuration.
configuration	jsonb	Required; not null	Model/resource versions and comparison settings.
run_status	text	Check; not null	Configured, running, complete or failed.
started_at	timestamptz	Optional; nullable	Actual run start time, when executed.

Table 4.52: EvaluationOutput attributes, types and constraints.

Attribute	Type	Constraints	Meaning
output_id	UUID	PK; not null	Unique run/case output.
run_id	UUID	FK; unique pair; not null	Research run producing this output.
case_id	UUID	FK; unique pair; not null	Evaluation case used by the run.
output_text	text	Check; not null	Recorded system output after execution.
output_status	text	Check; not null	Pending, complete or failed.
elapsed_ms	numeric	Optional; >=0; nullable	Measured duration only when available.

4.10 Risk Analysis

Table 4.53: Project risks, consequences and intended responses.

ID	Risk	Consequence	Response
R01	Data/model access	The selected clinical domain cannot be supported.	Confirm permitted cases and validated models before integration.
R02	Transcription/input quality	Incomplete or incorrect case information.	Allow note correction and report unusable inputs.
R03	Insufficient evidence	Draft recommendations lack inspectable support.	Show retrieved sources and make missing evidence explicit.
R04	Unsupported reasoning	Recommendations contain inconsistent findings.	Use critic review and retain physician decision authority.
R05	Patient-context mismatch	History or inputs are assigned to the wrong case.	Validate patient/encounter keys across processing and storage.
R06	Approval/revision error	Recorded content differs from the physician-approved draft.	Bind approval to the exact revision and content snapshot.
R07	Integration/service failure	The processing workflow is incomplete or incompatible.	Define contracts and expose stage-specific failures.
R08	Evaluation resources	Variant comparisons cannot answer the research questions.	Agree references, common cases and comparison conditions.

Table 4.53 outlines risks concerning resource permissions, supported clinical coverage, incomplete evidence, incorrect patient context and revision-specific approval. Their responses establish verification activities rather than completed mitigation outcomes. Dataset and model selection must confirm permitted use and task suitability before cases enter processing or comparison. Source identifiers and ownership checks must remain consistent through retrieval, analysis and review, while unavailable resources require visible failure states. Evidence inspection and critic findings support physician judgement without guaranteeing correctness. Approval must accept the reviewed revision and preserve matching action and decision records, including when requests fail or repeat. Integration tests and controlled comparisons are needed to assess these responses after implementation. The requirements, use cases, interface designs and logical schema therefore define a clinical workflow, while the methodology and architecture specify how its behaviour can be investigated. No performance result, executed acceptance test or demonstrated clinical safety outcome is established by these design artefacts.

4.11 Conclusion

The requirements establish a connected workflow from account access and physician-specific patient selection to consultation capture, specialist findings and clinical review. Functional requirements identify the operations users need, while quality attributes and non-functional requirements define ownership, integrity, traceability and understandable processing states. Use-case tables describe normal and alternative paths, and interface mappings connect these interactions to the corresponding screens. The logical

schema preserves input revisions, draft sources, review actions and accepted decisions so corrections and approval refer to identifiable content. Registration supports account access without introducing additional clinical roles, and each physician remains restricted to their own patients. Together, these specifications make the intended behaviour assessable through functional and integrity checks. Numerical performance limits and clinical resource configurations require an agreed environment before measurement. The requirements provide the contract for implementation and evaluation; they do not establish completed acceptance tests or clinical effectiveness. Subsequent design must preserve these relationships when connecting application services, agents and storage.

Chapter 5 Proposed Approach and Methodology

An integration methodology establishes how inputs, processing stages and evaluation procedures connect. The following sections describe case preparation, memory retrieval, specialist analysis, evidence handling and controlled comparisons, together with the dependencies needed to assess their contributions.

5.1 Case Preparation and Input Handling

The first methodological step is to create a consistent case package for the encounter. The physician opens a patient record, reviews prior encounters and records consultation audio for transcription according to the workflow in Chapter 4. Consultation audio is prepared as structured notes, which the physician can inspect and correct before downstream use. Imaging, laboratory information and medication context remain separate because they have different representations, source formats and analysis needs. Each input must retain its patient identifier, encounter identifier, type, source reference and revision. This prevents stages from mixing old notes with the encounter or treating unsupported input as a finding. The prepared package describes what information is available, where it came from and whether it is ready for analysis. This keeps failed or missing inputs visible of hiding them inside the draft generation process. The package must also distinguish authorised historical sources from inputs captured during the encounter.

5.2 Hybrid Memory and Case Organisation

Hybrid memory connects the encounter to historical context. Relational records preserve patient, encounter, input and approved-decision relationships, while semantic retrieval selects history related to the consultation or question. The Patient Memory Agent returns source-aware context, not a final diagnosis. The Clinical Case Agent then organises corrected notes, retrieved history and specialist inputs into a case representation for reasoning. This package identifies what is current, what is historical, what was retrieved and what is missing. If an input is absent or fails validation, that status is carried forward of creating a placeholder finding. This approach aligns with the database design in Chapter 4, where inputs, drafts, evidence, audit actions and approved decisions are separate records. It supports the comparison with a single-LLM baseline, because the baseline receives less patient-specific structure than the memory and full-system variants. Retrieved observations retain their source revisions and do not become accepted decisions through retrieval alone.

5.3 Proposed Agent Processing Pipeline

Figure 5.1 shows the processing pipeline used to coordinate the Zeest workflow. The physician-facing interface produces case inputs, while hybrid memory supplies patient history. The Clinical Case Agent combines those inputs and sends modality-specific information to the Imaging, Laboratory and Medication Safety agents. The Imaging Agent uses a validated pre-trained model only for imaging tasks. The Laboratory Agent analyses supplied laboratory information, and the Medication Safety Agent checks medication context against a resource still to be selected. The Evidence and RAG Agent retrieves external material with source references. The Clinical Reasoning Agent drafts the diagnosis and treatment proposal, and the Safety and Critic Agent checks it before physician review. The pipeline separates data movement from decision authority. Agents may prepare, analyse, retrieve or review information, but they do not record a final clinical decision. The physician sees the generated draft with evidence and critic findings, then approves, modifies or rejects it.

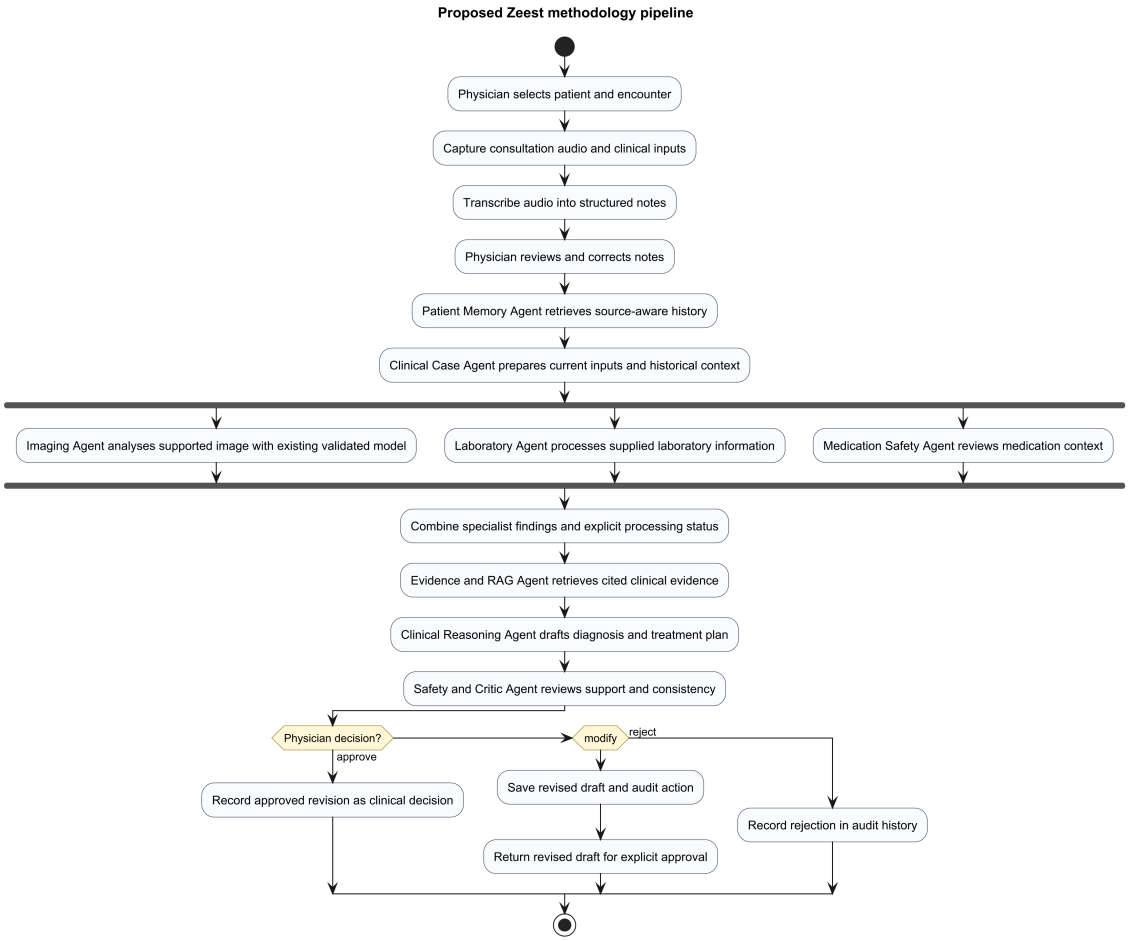


Figure 5.1: Proposed Zeest methodology pipeline connecting consultation inputs, hybrid memory, specialist agents, evidence retrieval, clinical drafting, critic review and physician decision controls.

5.4 Dataset, Model and Resource Register

The candidate data and model resources are identified in the reviewed literature. MIMIC-IV supports de-identified longitudinal EHR cases, while MIMIC-CXR and CheXpert support chest radiograph tasks. PubMed or guideline sources may support evidence retrieval, and Synthea can support synthetic patient cases. [17, 18, 10, 19] These resources are not interchangeable. Longitudinal memory needs encounter context, while imaging cases must match the selected model's supported labels. Evidence retrieval needs a permitted corpus and citation preservation. Medication review requires a knowledge resource with suitable coverage, and semantic memory requires an embedding model compatible with the vector store. Table 5.1 records each candidate's role, status and unresolved dependency so final selection is not assumed before implementation. a candidate resource becomes usable only after access, permitted use, preprocessing needs and clinical-domain fit are confirmed. It gives later evaluation work a checklist for deciding which cases and modalities can be included in the ablation study. Resource choices remain documented dependencies.

Table 5.1: Proposed dataset, model and resource register for Zeest methodology.

Candidate	Planned role	Current status	Dependency
MIMIC-IV [17]	Longitudinal EHR-style case context.	Candidate source.	Access, permitted use and case selection.
MIMIC-CXR [18]	Chest radiograph and report context.	Candidate source.	Access, pairing and supported task.
CheXpert [10]	Imaging benchmark/model candidate domain.	Candidate source.	Final model and label compatibility.
Existing imaging classifier	Supported imaging finding generation.	Not selected.	Validated model, inputs and labels.
PubMed/guideline corpus	External evidence retrieval.	Not finalised.	Corpus choice, citation format and retrieval policy.
Synthea [19]	Synthetic workflow cases.	Candidate source.	Scenario realism and mapping to Zeest fields.
Medication knowledge resource	Medication safety checking.	Not selected.	Resource availability and interaction coverage.
Embedding model	Semantic memory and retrieval.	Not selected.	Dimension, domain suitability and pgvector storage.

5.5 Data Preprocessing

Preprocessing depends on the chosen resources, but Zeest's workflow already defines the preparation rules. Patient and encounter identifiers must be de-identified or synthetic. Consultation notes, reports and other text inputs should be structured while remaining linked to their source. Imaging files should follow only the selected model's required input format, because changing image preparation without matching the model can invalidate the finding. Laboratory and medication information should be normalised enough for the agent to interpret it, but missing values should not be invented. Evidence passages should retain source references and retrieval timestamps. Ablation cases should be frozen before comparison so each variant receives the same allowed inputs for its configuration. This prevents a memory-augmented

variant or full-system variant from benefiting from accidental case changes. The preprocessing stage supports both system correctness and evaluation fairness. The resulting case definition must remain traceable to its permitted sources throughout preparation, comparison and review.

5.6 Evidence Retrieval Method

The evidence method separates patient-specific context from external clinical knowledge. Patient context comes from the encounter, corrected notes, retrieved history and specialist findings. External knowledge comes from guideline, literature or medical reference material selected for retrieval. The Evidence and RAG Agent forms a query from the selected case context, retrieves candidate passages, filters or ranks them by relevance and returns supporting text with citations. Figure 5.2 shows this flow. The Clinical Reasoning Agent may use retrieved evidence when drafting a diagnosis and treatment plan, but the evidence must remain visible to the responsible physician. This prevents a black box answer and lets the critic identify unsupported claims, missing evidence or contradictions. The procedure supports traceability because a draft should identify which retrieved passages were used and where they came from. The physician can then judge whether the support fits the patient and whether the draft should be approved, edited or rejected.

5.7 Ablation and Evaluation Protocol

The ablation protocol answers Zeest's evaluation questions. The comparison includes a single LLM baseline, a memory-augmented variant and the full Zeest system. The baseline receives the case prompt only. The memory variant adds retrieved patient history, while the full system adds specialist findings, evidence retrieval and critic review. Figure 5.3 shows how the same case set flows through the variants and into one assessment process. Additional contrasts examine critic and multimodal contribution where the cases support them. The same reference judgement process must be used for every variant. Table 5.2 defines the variants and comparison controls. Outputs should be assessed for case relevance, patient-history use, evidence support, consistency with specialist findings, critic handling and physician reviewability. Metrics, sample counts, rubrics and reviewer roles remain unresolved. Possible measures include agreement with a reference judgement, citation relevance, unsupported-claim count, error categories and physician review effort. Measured outcomes must remain separate from this evaluation procedure.

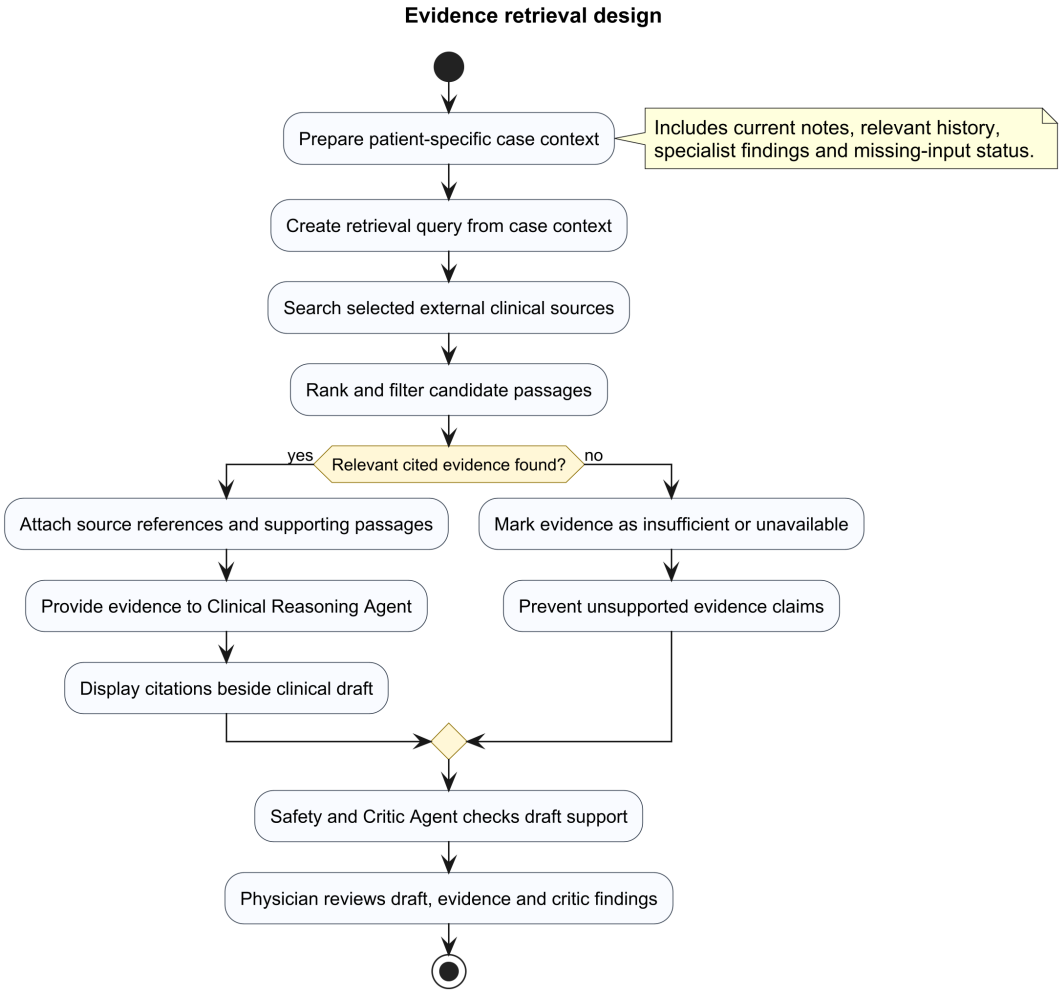


Figure 5.2: Patient context and external sources supplying cited evidence to the clinical draft.

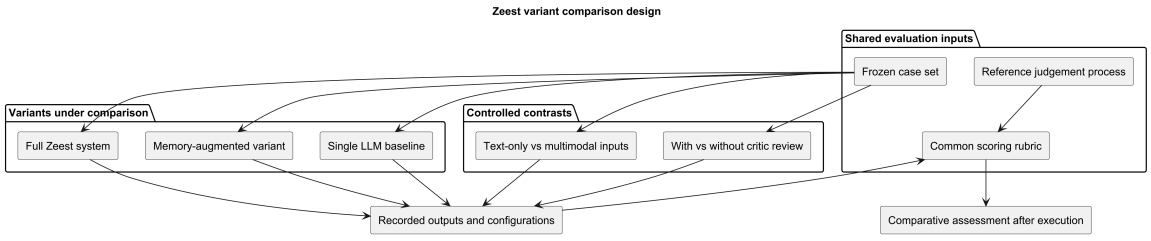


Figure 5.3: Single LLM, memory-augmented and full-system comparisons using shared cases and assessment.

Table 5.2: Planned ablation and evaluation protocol for Zeest variants.

Variant or contrast	Inputs allowed	Planned assessment focus	Status
Single LLM baseline	Current case prompt only.	Draft relevance and unsupported claims.	Not executed.
Memory-augmented	Current case plus retrieved longitudinal history.	Contribution of patient memory.	Not executed.
Full Zeest	Case, memory, specialist findings, evidence and critic review.	Integrated workflow performance and reviewability.	Not executed.
Critic contrast	Full pipeline with and without critic findings.	Unsupported or contradictory output handling.	Planned control.
Multimodal contrast	Text-only case versus supported multimodal case.	Contribution of imaging/lab/medication findings.	Planned control.

5.8 Methodological Traceability

The methodology traces to the objectives and requirements in Chapters 2 and 4. Hybrid memory supports O1, FR23 and FR24. Transcription and note correction support O2 and FR14 through FR18. Specialist agents support O3 and FR25 through FR30. Imaging integration supports O4 and FR26, while evidence retrieval supports O5, FR32 and FR36. Clinical draft generation supports O6, FR33 and FR34. Critic review supports O7, FR35 and FR37. Physician approval, modification, rejection and audit support O8 and FR38 through FR43. The ablation protocol supports O9 and RR01 through RR04. Table 5.3 summarises this mapping so design and evaluation work can remain aligned with the requirements. This traceability matters because the methodology should evaluate the same workflow described in the requirements, not a different or simplified system. Changes to resources or agent responsibilities must preserve these verification relationships. Patient ownership and accepted-decision views remain required even when an evaluation variant omits analytical components.

Table 5.3: Traceability between methodology stages and current requirement areas.

Method stage	Related Chapter 4 IDs	Planned verification
Case preparation	FR09–FR22	Check patient, encounter and source associations.
Hybrid memory retrieval	FR23–FR25	Compare retrieved history against selected cases.
Specialist analysis	FR26–FR30	Check input-specific findings and failures.
Evidence retrieval	FR32, FR36	Assess citation relevance and passage support.
Clinical draft and critic	FR33–FR37	Review draft consistency and critic findings.
Physician decision workflow	FR38–FR43	Verify revision-specific approval and audit records.
Ablation framework	RR01–RR04	Compare variants using shared cases and rubric.

5.9 Dependencies and Open Method Decisions

The methodology has unresolved dependencies before implementation and evaluation. The clinical domain must match available data, permitted use and a validated model. The transcription component, embedding model, evidence corpus, medication resource and LLM configuration remain to be selected. The LangGraph and Google ADK responsibility split remains a Chapter 6 design decision. Evaluation needs reference judgements, reviewer roles, sample size, rubric and thresholds. These dependencies are important because they determine whether a case can be used, which modalities can be compared and how outputs should be judged. They prevent the design from presenting design assumptions as completed results. The methodology establishes a structured procedure without pretending those choices are complete. The open decisions should be closed before implementation claims or numerical evaluation results are documented with their limitations, and any selected resource should be recorded with its permitted use and limitations. Unselected resources must not be represented as working integration dependencies.

5.10 Conclusion

Zeest's methodology combines case preparation, hybrid memory retrieval, specialist analysis, evidence-grounded drafting, critic review and physician decision control. It supports an R&D investigation of whether memory, multimodal inputs, agent decomposition, evidence retrieval and critic review improve clinical draft reviewability. The figures and tables define input flow, unresolved resources and ablation comparisons. The procedure keeps the generated output as a draft until the responsible physician approves, modifies or rejects it, preserving the approval boundary established in the functional requirements. Because the prototype is unbuilt, the chapter separates procedure from outcomes. Future work must confirm resources, implement the pipeline, execute comparisons and report measured findings with limitations. This structure gives the team a method for moving from the requirements toward a testable system while avoiding unsupported claims about clinical performance, dataset access or completed integration. It helps later chapters and reviews check whether the integrated design remains consistent with the evaluation.

Chapter 6 High-Level and Low-Level Design

System design translates requirements into component responsibilities, interfaces and information relationships. The following sections present architectural decisions, subsystem interactions, structural models and behavioural flows, showing how data and control move between the application and its supporting services.

6.1 System Overview

The physician workspace presents patient history, editable consultation notes, specialist findings and clinical drafts in a connected encounter context. Application services validate account access and check that the patient belongs to the requesting physician before retrieving or processing related records. Behind these services, eight agents divide patient-memory retrieval, case preparation, imaging, laboratory processing, medication review, evidence retrieval, clinical reasoning and critic review. Their outputs retain source identifiers and processing states so incomplete analysis does not become an apparent finding. PostgreSQL stores account, encounter, revision and decision relationships, while pgvector supports semantic retrieval over authorised sources. Generated content remains a draft until the physician accepts its reviewed revision. Modification creates a pending revision, and rejection retains the action without creating an accepted decision. Audit records connect each review action to its actor, time and encounter. Research outputs remain separate from accepted decisions even when an evaluation uses the same encounter context.

6.2 Design Considerations

Component boundaries must accommodate different clinical representations without losing patient ownership or source context. Consultation audio needs an incremental capture path, imaging requires model-compatible preparation, and laboratory and medication information require interpretable structured inputs. External model or knowledge services can fail independently, so their status and errors must remain visible to the application. Historical retrieval must distinguish source observations from accepted decisions and prevent another physician's records from entering the case. Draft preparation needs revision-specific evidence and critic findings, while approval requires the accepted content to match the saved revision. These concerns guide the division between interface, application gateway, orchestration, analytical services and persistence. Design choices need to support controlled research comparisons without treating components as automatically beneficial. Numerical latency limits, hardware capacity, model selection and resource coverage depend on the eventual environment. Consequently, the design specifies

responsibilities and failure boundaries while leaving configuration-dependent values to documented selection and measurement.

6.2.1 Assumptions and Dependencies

- Physicians can interpret patient information and assess clinical drafts.
- A supported browser and required service connectivity are available.
- Consultation recording requires microphone access and capture permission.
- Cases are de-identified or synthetic, with one responsible physician per patient.
- Imaging models match the selected domain and supported input formats.
- Transcription, embeddings and clinical resources require permitted use.
- PostgreSQL and pgvector support relational and semantic storage.
- LangGraph and Google ADK responsibilities are defined during integration.
- Resource versions, configurations and limitations are recorded for evaluation.

6.2.2 General Constraints

- Zeest supports clinical review; it does not treat or prescribe autonomously.
- Generated diagnoses and treatment plans remain drafts until approved.
- Authentication and patient ownership govern every clinical operation.
- Development uses permitted de-identified or synthetic cases.
- Unsupported inputs are rejected or marked unavailable.
- Performance claims require measurements in the selected environment.
- Input and draft revisions preserve the content used and reviewed.
- Evaluation records case permissions, resources and reference procedures.

6.2.3 Goals and Guidelines

- Keep patient, encounter and source context visible throughout the workflow.
- Distinguish clinical inputs, findings, evidence and generated drafts.
- Keep approval, modification and rejection outcomes clear.
- Use consistent component interfaces with source references and error status.

- Preserve agreed input and output contracts when replacing analytical models.
- Show incomplete or failed processing without inventing clinical findings.
- Use common case definitions and assessment conditions across comparisons.

6.2.4 Development Methods

Development follows the requirements, use-case definitions and logical data relationships established in Chapter 4. The initial work defines component contracts, account ownership checks and input and draft revisions before connecting model services. Small integration stages then establish patient navigation, consultation notes, specialist processing, evidence retrieval and the review boundary. Structural and behavioural diagrams describe responsibilities, messages and permitted state transitions, while test cases examine normal operations and rejected requests. Changes to an analytical component must preserve patient, encounter and source associations and expose any new input restrictions. Methodological comparisons use configurations so architecture changes can be distinguished from changes to cases or assessment conditions. The process keeps resource selection explicit: candidate datasets, knowledge sources and models require permission and domain checks before use. Implementation claims must follow working integration, and performance claims must follow measurements. The saved configuration identifies which resources and case restrictions governed each integration or evaluation run.

6.3 System Architecture

Figure 6.1 separates the physician interface, application gateway, agent orchestration and supporting services. Next.js provides the web interface, and FastAPI provides the Python application boundary. [21, 22] The gateway handles authentication, patient ownership, validated requests and revision-aware decision operations before dispatching analytical work. LangGraph and Google ADK inform the orchestration design; their exact division remains an implementation choice. [23, 16] Model adapters connect consultation transcription, existing imaging classifiers and Gemini-based reasoning without exposing internal services directly to the browser. [24, 25] PostgreSQL preserves relational records, and pgvector supports semantic retrieval. [26] Streaming communication serves consultation capture, while transactional requests serve saved case operations and physician decisions. Failures return status rather than implied findings. Draft preparation and accepted-decision recording follow different paths, so analytical assistance cannot bypass explicit approval. The architecture describes intended service responsibilities; deployment capacity, latency and runtime configuration require implementation and measurement. Analytical services return source-aware outputs without acquiring authority to accept a clinical decision.

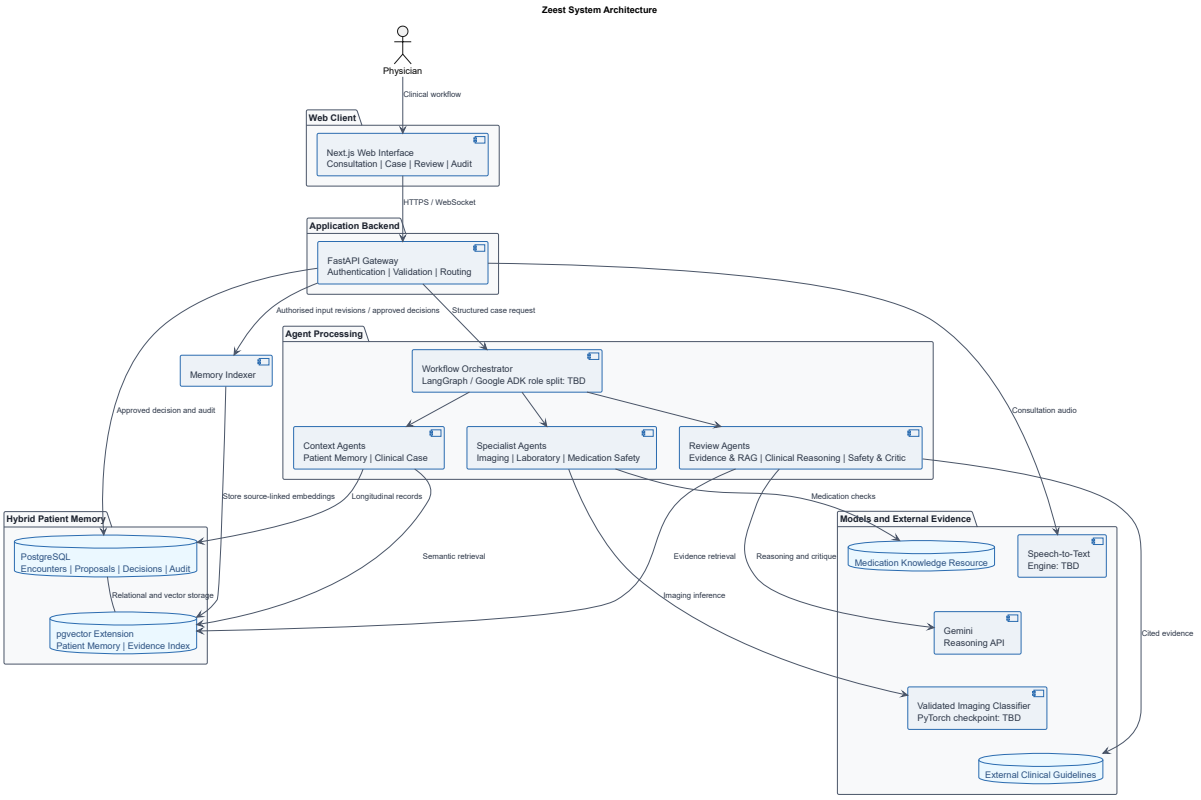


Figure 6.1: Zeest interface, gateway, agents, analytical services and hybrid memory.

6.3.1 Subsystem Architecture

Figure 6.2 groups the eight agents by their contributions to case preparation, specialised analysis and draft review. The Patient Memory Agent retrieves source-aware history, and the Clinical Case Agent combines it with current saved inputs. Imaging, Laboratory and Medication Safety agents return findings or explicit failure states for supported inputs. The Evidence and RAG Agent supplies cited passages, the Clinical Reasoning Agent creates a diagnosis and treatment draft, and the Safety and Critic Agent identifies concerns before presentation. Table 6.1 defines these handoffs. Figure 6.3 follows consultation capture through note correction, analysis and physician presentation; Figure 6.4 follows modification, approval and rejection through persistence. Independent analyses may run concurrently where their dependencies permit, but downstream reasoning must identify incomplete stages. Authorised input revisions and approved decisions can supply semantic memory; pending drafts cannot enter it as accepted clinical facts. These contracts preserve the distinction between preparation, recommendations and recorded decisions. Their outputs require verification.

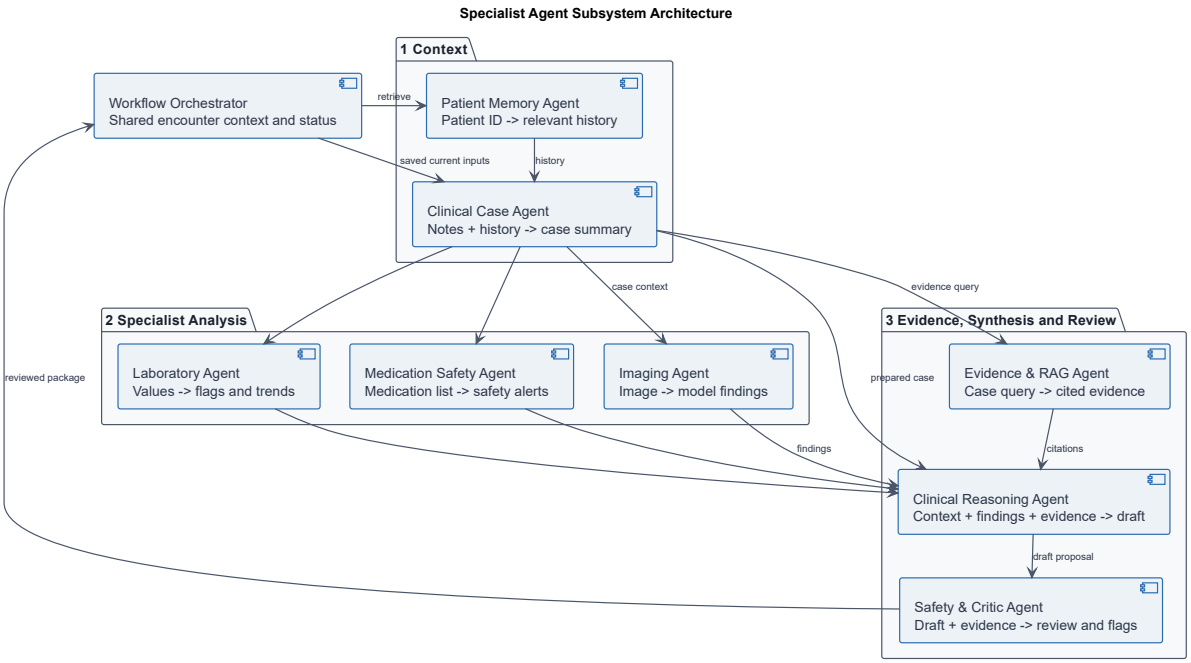


Figure 6.2: Eight specialist agents and their case-processing handoffs.

Table 6.1: Proposed specialist agent input, output, persistence and recipient contracts .

Agent Name	Primary Inputs	Primary Outputs	Storage & Recipient
Patient Memory Agent	`patient_id`, encounter context	Relational history, top-k semantic memories	PostgreSQL/pgvector; Orchestrator
Clinical Case Agent	Notes, multimodal data, retrieved history	Consolidated case representation	Working memory cache; Specialist agents
Imaging Agent	Radiological image tensor / file reference	Predicted pathology classes, confidence scores	Working cache; Reasoning Agent
Laboratory Agent	Laboratory panel values, reference ranges	Abnormal biomarker flags, trajectory trends	Working cache; Reasoning Agent
Medication Safety Agent	Active prescriptions, candidate drugs	Drug-drug interactions, contraindication alerts	Knowledge base; Reasoning Agent
Evidence & RAG Agent	Diagnostic keywords, clinical hypotheses	Cited guideline passages, source identifiers	pgvector vector index; Reasoning Agent
Clinical Reasoning Agent	Consolidated case, specialist findings, evidence	Differential diagnosis draft, treatment plan	Gemini API; Safety & Critic Agent
Safety & Critic Agent	Generated draft, case context, cited evidence	Consistency evaluation, risk alerts, critique	Working cache; Presentation tier

6.4 Architectural Strategies

The architectural strategies address patient continuity, processing dependencies and accountable review rather than assuming diagnostic improvement from added components. Hybrid persistence combines explicit account, patient and encounter relationships with semantic retrieval over identifiable sources. Orchestration keeps agent responsibilities separate and coordinates their inputs before drafting. Model adapters isolate service-specific formats and errors, while transactional application services govern ap-

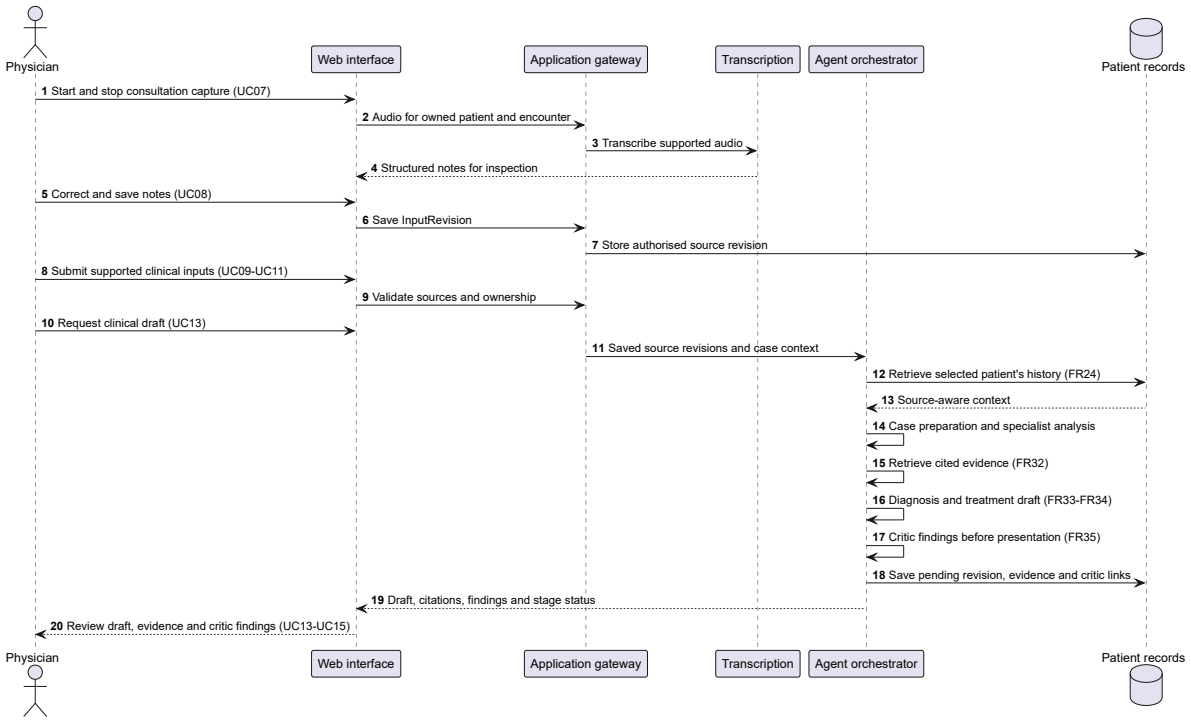


Figure 6.3: Consultation capture, source-aware analysis and draft presentation sequence.

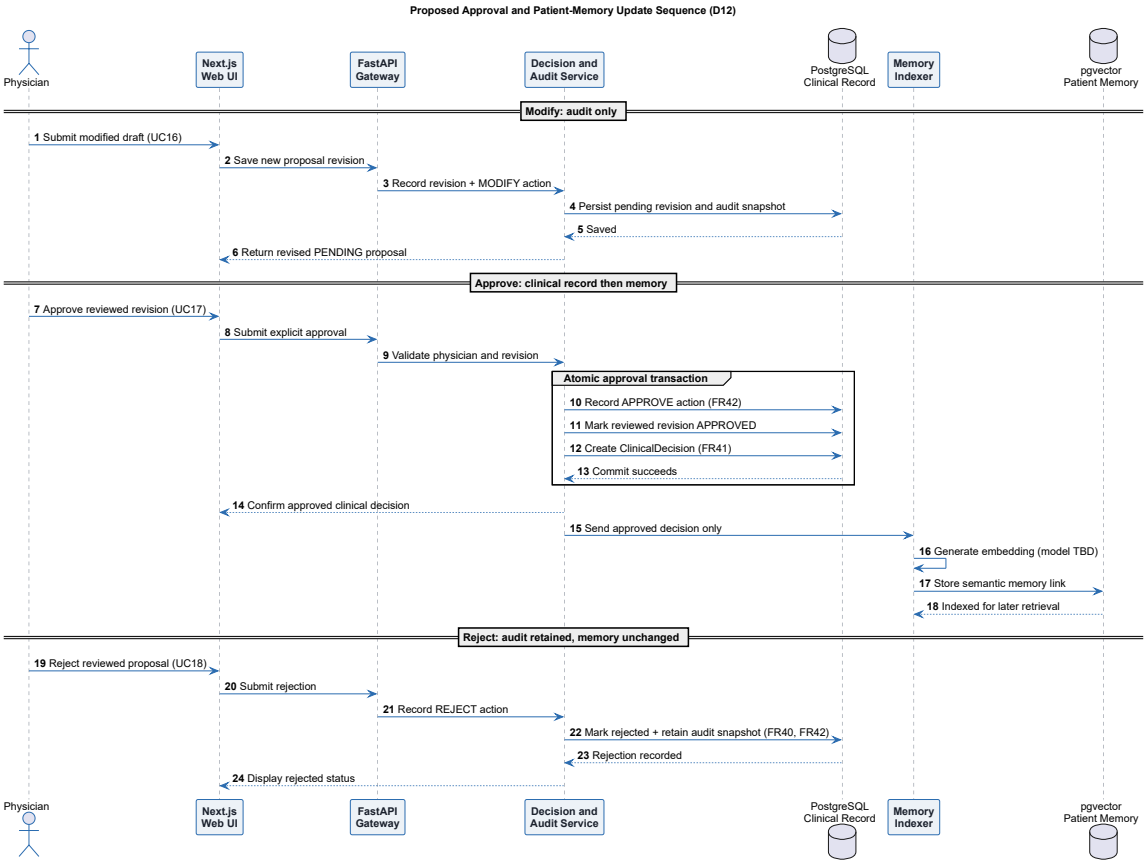


Figure 6.4: Revision-specific physician actions, decision recording and memory indexing.

proval and audit recording. Table 6.2 records the supported technology choices, trade-offs and configuration decisions that remain open. These strategies support controlled evaluation: a simpler variant can withhold a component while retaining the same case definition and reference process. Independent processing may use asynchronous execution, but dependencies must be respected and failed analyses must remain visible. No strategy establishes guaranteed throughput, clinical safety or regulatory compliance. Its value must be assessed through implementation checks and comparative runs. The design connects engineering choices to testable behaviour while preserving the physician's responsibility for every accepted revision.

6.4.1 Hybrid Relational and Vector Memory Persistence

PostgreSQL preserves physician ownership, patient and encounter associations, input revisions, draft revisions, evidence links, review actions and accepted decisions. Foreign keys and transaction boundaries support checks that related records belong to the same case and reviewed revision. Semantic memory uses pgvector representations linked to an authorised input revision or an approved decision. This distinction permits retrieval of source observations while preventing a pending or rejected draft from being treated as an accepted clinical conclusion. Retrieval returns the originating record and its context so similarity alone does not establish clinical relevance. Historical information must remain within the selected patient's ownership boundary, and observations must remain distinguishable from earlier encounters. Embedding model, dimension and indexing configuration require selection before storage and search can be evaluated. The hybrid design retains explicit relationships alongside semantic search, but its relevance and latency must be measured rather than inferred from the presence of a vector index.

6.4.2 Asynchronous Agent Orchestration and Model Runtime Segregation

The application separates case operations from analytical work that may depend on local inference or remote services. Consultation capture, saved note revisions and physician review should retain understandable states while an analysis is pending. The orchestrator starts independent specialist tasks only after the required case context is available, then combines returned findings and failure information before drafting. Dependencies determine whether a task can run concurrently; evidence retrieval must receive the information needed to form a query. Service adapters validate payloads and distinguish timeouts, rejected inputs and unavailable resources from completed results. A partial case can remain available for inspection, but missing analysis must not be silently represented as normal findings. LangGraph and Google ADK provide candidate orchestration mechanisms, with their final division during implementation. Asynchronous execution supports this separation without establishing a measured speed improvement. Stage status should remain visible while other services finish, fail or await required case

information.

Table 6.2: Architectural design decisions, supported choices, trade-offs and unresolved options .

Design Dimension	Proposed Choice	Architectural Trade-off	Unresolved Status
Backend Framework	FastAPI (Python)	High-speed async I/O and native Python ML ecosystem versus separate microservices	Production server concurrency tuning TBD
Frontend Framework	Next.js / TypeScript	Rich React component ecosystem and streaming support versus heavier bundle size	View layout refactoring planned
Multi-Agent Orchestrator	LangGraph / Google ADK StateGraph	Cyclic state-graph flexibility versus increased execution graph complexity	Formal division of roles between frameworks TBD
Primary Reasoning Model	Google Gemini API	Advanced multimodal context window versus reliance on cloud API latency	Production model version and quotas TBD
Radiology Classifier	PyTorch Pre-trained Model	Validated local inference speed versus fixed diagnostic pathology label sets	Exact pre-trained model checkpoint TBD
Memory Storage	PostgreSQL with pgvector extension	Unified relational ACID guarantees and vector search versus dedicated vector database	Vector dimension and HNSW index tuning TBD

6.5 Domain Model/Class Diagram

Figure 6.5 presents the clinical domain classes and their associations; the complete persistence attributes are defined in Chapter 4. Physician, Patient and Encounter maintain account access, ownership and patient encounter context. ClinicalInput and InputRevision distinguish an input from its saved versions, while SpecialistFinding identifies the revision analysed. Proposal and ProposalRevision separate a draft's identity from reviewed content; ProposalInput, EvidenceItem, ProposalEvidence and CriticReview preserve its sources and assessment. PhysicianAction records the actor, revision and physician review event, and ClinicalDecision stores the explicitly accepted content. SemanticMemory links authorised source revisions or approved decisions to their representations. EvaluationCase, EvaluationRun and EvaluationOutput keep research case inputs, configurations and outputs identifiable. Application services coordinate retrieval, analysis and transactional approval rather than assigning unrestricted database writes to agents. The model aligns with the eighteen-entity dictionary while distinguishing service behaviour from stored records. Class associations specify domain entity relationships; their integrity must be verified when persistence is implemented.

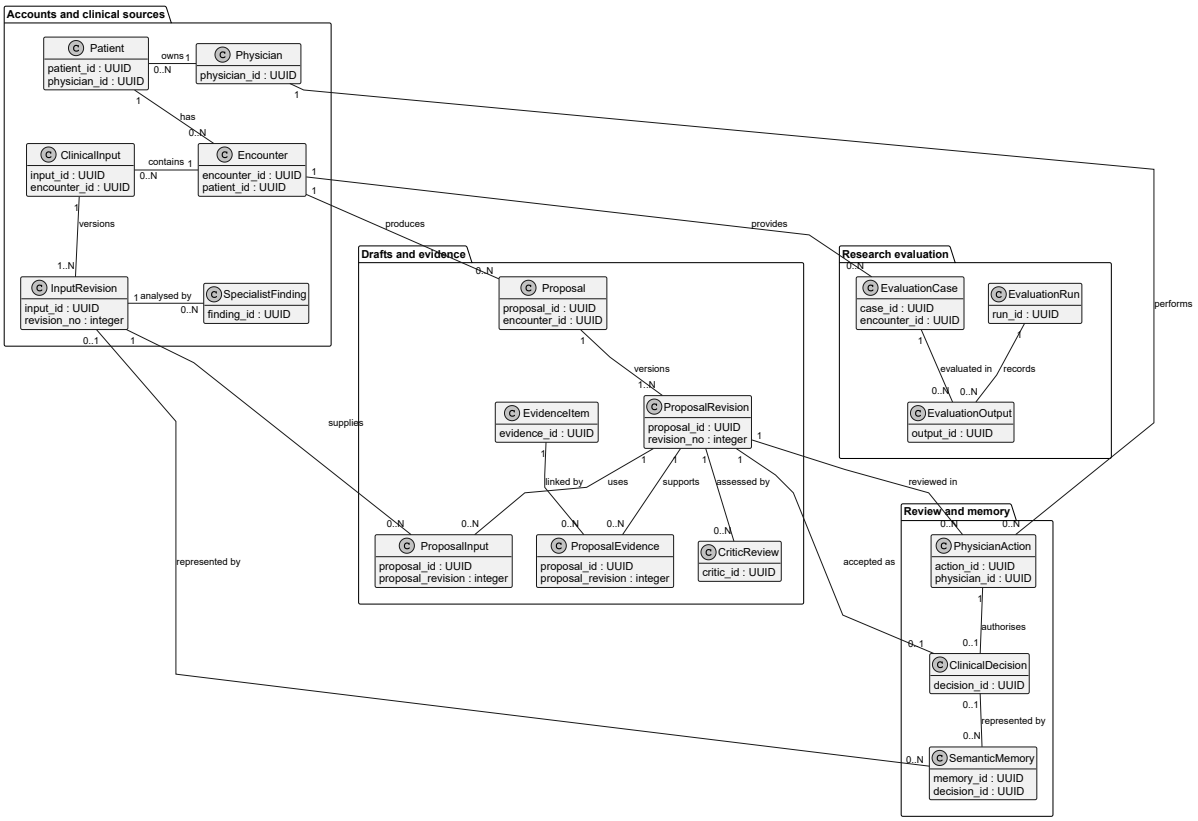


Figure 6.5: Domain classes with ownership, revision, evidence and decision associations.

6.6 Policies and Tactics

Local policies govern account access, source handling and state transitions within the architecture. The physician's authenticated identity is checked against the selected patient before any related record is read or processed. Saved revisions preserve the clinical content used for analysis and review, while source links identify the findings and passages supporting each draft. Review actions retain their actor, time and revision; approval records matching accepted content in a transaction, and modification or rejection does not create a decision. Evidence and critic findings remain inspectable even when they identify insufficient support. Analytical errors must be returned with their stage and source context so an incomplete operation cannot appear successful. Semantic retrieval distinguishes authorised observations from approved conclusions and excludes unaccepted draft content from accepted-decision memory. These policies provide verification targets for implementation and research cases. Checks must cover incorrect ownership, stale revisions and missing evidence as well as successful workflow operations.

6.6.1 Physician Review Boundary and Audit Immutability Policy

Figure 6.6 distinguishes pending, modified, approved and rejected draft states. Every generated draft begins pending physician review. Modification creates a saved revision and records a MODIFY action

while leaving the draft unapproved. Approval requires the authenticated physician to own the patient and review the exact saved revision. A transaction records the APPROVE action, accepted revision and matching ClinicalDecision, preventing an action from accepting different content. Rejection records a REJECT action and retains the review history without creating an accepted decision. Audit entries preserve the actor, time, encounter and revision rather than replacing earlier review events. Pending and rejected revisions remain available as drafts and history, but cannot enter accepted-decision memory. Authorised clinical input revisions can still provide historical context through their separate source links. These controls define the decision boundary; concurrency, failed transactions and repeated requests must be checked during implementation to confirm that their recorded outcomes preserve the required relationships.

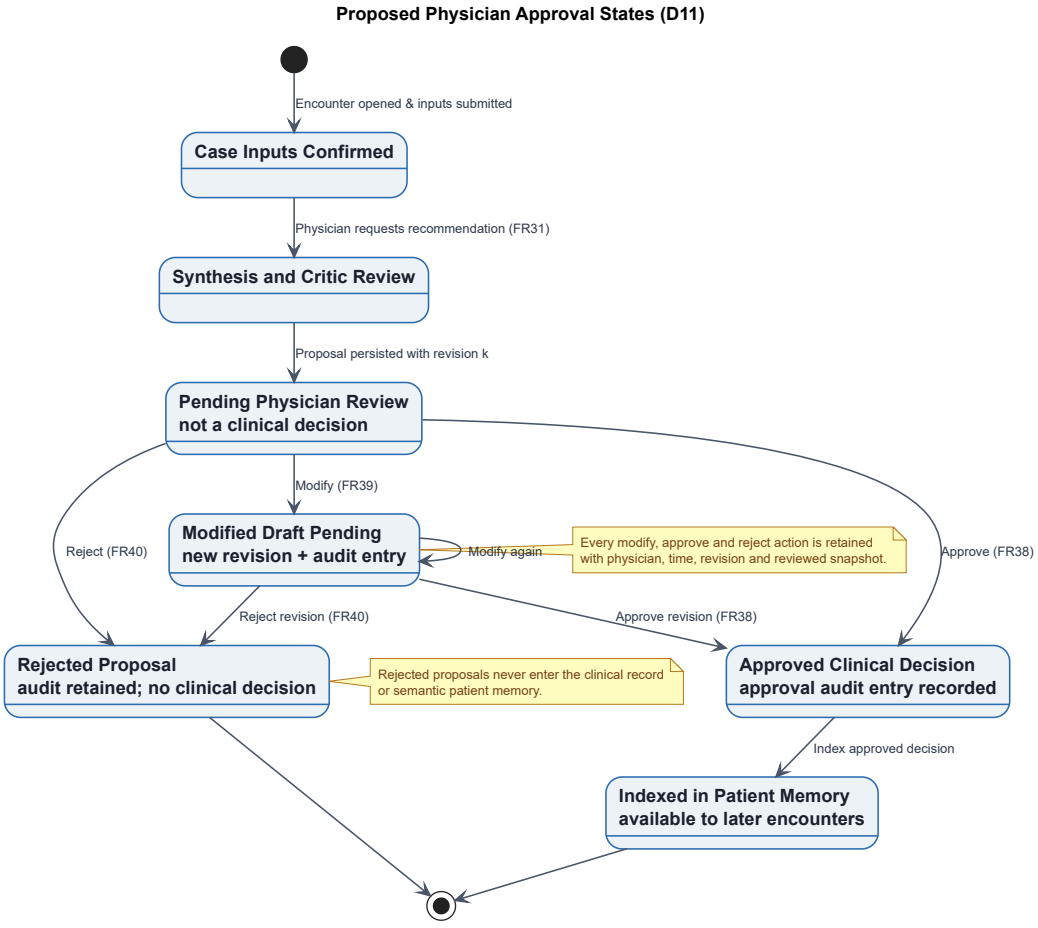


Figure 6.6: Physician-controlled draft states and explicit approval transitions.

6.6.2 Grounded Clinical Evidence Citation and Critic Verification Tactic

The Evidence and RAG Agent returns supporting passages with source identifiers and citations rather than only an explanation. The Clinical Reasoning Agent uses these passages alongside patient context and specialist findings when preparing a diagnosis and treatment draft. Evidence links identify the saved

revision they support, allowing the physician to inspect the relationship between a recommendation and its sources. The Safety and Critic Agent examines the draft for insufficient support, inconsistency and safety concerns before presentation. Critic findings retain their assessed revision and appear alongside the draft, including concerns that remain unresolved. Missing evidence must be shown as missing rather than replaced with fabricated citations, and critic review must not be described as proof of correctness. Physician approval remains necessary regardless of the critic outcome. These tactics complete the architecture's review pathway and establish comparison targets for evidence support and critic contribution; their effectiveness requires implemented workflows and measured evaluation.

6.7 Conclusion

The design translates Zeest's requirements into a physician interface, application gateway, coordinated agents, supporting analytical services and hybrid memory. Defined boundaries preserve patient ownership and distinguish source observations, specialist findings, generated drafts and accepted decisions. Revision relationships connect the information used for analysis with the evidence and critic findings presented during review. Application services govern approval and audit recording, while analytical agents remain responsible for preparation, retrieval and recommendations. The domain model and behavioural diagrams provide complementary views of these responsibilities and their permitted transitions. Component adapters and visible failure states support integration without implying that every candidate resource is already available. The architecture also accommodates controlled research comparisons by retaining case definitions, configurations and outputs independently of clinical decision records. Implementation must validate these contracts, transaction boundaries and failure paths before performance or clinical-value claims can be made. The resulting design provides a coherent development foundation while preserving physician authority and the limits of the current research scope.

Bibliography

[1] United Nations Department of Economic and Social Affairs, "Goal 3: Ensure healthy lives and promote well-being for all at all ages." https://sdgs.un.org/goals/goal3. [Accessed: October 4, 2026].

[2] Isabel Healthcare, "Isabel DDx Differential Diagnosis System." https://www.isabelhealthcare.com. [Accessed: October 5, 2026].

[3] Wolters Kluwer, "UpToDate: Evidence-based clinical decision support." https://www.uptodate.com. [Accessed: October 5, 2026].

[4] B. W. Mamlin, P. G. Biondich, B. A. Wolfe, H. Fraser, D. Jazayeri, C. Allen, J. Miranda, and W. M. Tierney, "Cooking up an open source EMR for developing countries: OpenMRS," in *AMIA Annual Symposium Proceedings*, pp. 529–533, 2006.

[5] K. Singhal, T. Tu, J. Gottweis, R. Sayres, E. Wulczyn, L. Hou, K. Clark, S. Pfohl, H. Cole-Lewis, D. Neal, M. Schaekermann, A. Wang, M. Amin, S. Lachgar, P. Mansfield, S. Prakash, B. Green, E. Dominowska, B. A. y Arcas, N. Tomasev, Y. Liu, R. Wong, C. Semturs, S. S. Mahdavi, J. Barral, D. Webster, G. S. Corrado, Y. Matias, S. Azizi, A. Karthikesalingam, and V. Natarajan, "Toward expert-level medical question answering with large language models," *Nature Medicine*, vol. 31, pp. 943–950, 2025.

[6] H. Nori, N. King, S. M. McKinney, D. Carignan, and E. Horvitz, "Capabilities of GPT-4 on medical challenge problems." arXiv:2303.13375, 2023. https://arxiv.org/abs/2303.13375.

[7] E. Alsentzer, J. R. Murphy, W. Boag, W.-H. Weng, D. Jindi, T. Naumann, and M. B. A. McDermott, "Publicly available clinical BERT embeddings," in *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, pp. 72–78, 2019.

[8] R. Luo, L. Sun, Y. Xia, T. Qin, S. Zhang, H. Poon, and T.-Y. Liu, "BioGPT: Generative pre-trained transformer for biomedical text generation and mining," *Briefings in Bioinformatics*, vol. 23, no. 6, p. bbac409, 2022.

[9] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng, "CheXNet: Radiologist-level pneumonia detection on chest x-rays with deep learning." arXiv:1711.05225, 2017. https://arxiv.org/abs/1711.05225.

[10] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya, J. Seekins, D. A. Mong, S. S. Halabi, J. K. Sandberg, R. Jones, D. B. Larson, C. P. Langlotz, B. N. Patel, M. P. Lungren, and A. Y. Ng, "CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, pp. 590–597, 2019.

[11] S. Bannur, S. Hyland, Q. Liu, F. Perez-Garcia, M. Ilse, D. C. Castro, B. Boecking, H. Sharma, K. Bouzid, A. Thieme, M. P. Lungren, M. Wetscherek, J. Alvarez-Valle, and O. Oktay, "Learning to exploit temporal structure for biomedical vision-language processing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15016–15027, 2023.

[12] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.

[13] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking retrieval-augmented generation for medicine," in *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 6233–6251, 2024.

[14] Q. Jin, W. Kim, Q. Chen, D. C. Comeau, L. Yeganova, W. J. Wilbur, and Z. Lu, "MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs for zero-shot biomedical information retrieval," *Bioinformatics*, vol. 39, no. 11, p. btad651, 2023.

[15] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations*, 2023.

[16] Google, "Agent development kit." https://google.github.io/adk-docs/. [Accessed: October 5, 2026].

[17] A. E. W. Johnson, L. Bulgarelli, L. Shen, *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Scientific Data*, vol. 10, p. 1, 2023.

[18] A. E. W. Johnson, T. J. Pollard, S. J. Berkowitz, N. R. Greenbaum, M. P. Lungren, C. ying Deng,

R. G. Mark, and S. Horng, "MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports," *Scientific Data*, vol. 6, p. 317, 2019.

[19] J. Walonoski, M. Kramer, J. Nichols, A. Quina, C. Moesel, D. Hall, C. Duffett, K. Dube, T. Gallagher, and S. McLachlan, "Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record," *Journal of the American Medical Informatics Association*, vol. 25, no. 3, pp. 230–238, 2018.

[20] OWASP Foundation, "Password storage cheat sheet." https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html. [Accessed: October 5, 2026].

[21] Vercel Inc., "Next.js: The React framework for the web." https://nextjs.org/. [Accessed: October 8, 2026].

[22] S. Ramírez, "FastAPI: Modern, fast (high-performance), web framework for building APIs with Python." https://fastapi.tiangolo.com/. [Accessed: October 8, 2026].

[23] LangChain Inc., "LangGraph." https://langchain-ai.github.io/langgraph/. [Accessed: October 5, 2026].

[24] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, pp. 8024–8035, 2019.

[25] Gemini Team, Google, "Gemini: A family of highly capable multimodal models." arXiv:2312.11805, 2023. https://arxiv.org/abs/2312.11805.

[26] A. Kane, "pgvector: Open-source vector similarity search for PostgreSQL." https://github.com/pgvector/pgvector. [Accessed: October 8, 2026].

Chapter A Traceability and Design Decisions

The following tables provide supplementary objective traceability and configuration decisions for development and evaluation.

A.1 Objective Traceability

Table A.1 connects the nine objectives with their requirements and intended verification.

Table A.1: Objectives, requirement locations and intended verification.

Objective	Requirement location	Verification
O1 Hybrid memory	FR23–FR24	Check patient-scoped relational and semantic retrieval.
O2 Consultation notes	FR14–FR18	Check capture, transcription and saved corrections.
O3 Specialist agents	FR25–FR30	Check source-aware findings and stage failures.
O4 Existing imaging model	FR19, FR26	Check supported inputs against the selected model.
O5 Cited evidence	FR32, FR36	Assess retrieved source relevance and passage support.
O6 Clinical draft	FR31, FR33–FR34	Assess diagnosis and treatment drafts against reference cases.
O7 Critic review	FR35, FR37	Compare concerns and outputs with and without critic review.
O8 Physician control	FR38–FR44	Check revision-specific actions, decisions and audit records.
O9 Comparative evaluation	RR01–RR04	Compare the baseline, memory variant and full system under common conditions.

A.2 Configuration Decisions

Patient access is restricted to the responsible physician. Clinical decisions require approval of the exact saved revision; pending drafts are excluded from accepted memory. Before evaluation, select permitted cases, analytical models, evidence sources and orchestration responsibilities. Record the case count, reference process, rubric, workload and thresholds with each comparison configuration.